



AWS CLF-C02 EXAM PREP — CHEAT SHEET (CLF-C02)

45 Cheat Sheets & Diagrams to Master the AWS Cloud



Tip: Review one section at a time, practice with scenarios, and reinforce your learning.



1. AWS FOUNDATIONS & ARCHITECTURE

#	Topic	Key Concepts
1	AWS Global Infrastructure	Regions, AZs, Edge Locations, Local Zones, Regional Edge Cache
2	AWS Shared Responsibility Model	Security IN the cloud (customer) vs. OF the cloud (AWS); IaaS / PaaS / SaaS / Serverless breakdown
3	AWS Shared Responsibility Model — Deep Dive	EC2 vs RDS vs Lambda responsibility breakdown; exam traps; who patches the OS
4	AWS Deployment Models	Cloud (Public), Private Cloud, On-Premises, Hybrid Cloud, Multi-Cloud — control, cost, security trade-offs
5	AWS Well-Architected Framework	6 Pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability
6	AWS Cloud Economics & Value Proposition	CapEx vs OpEx, TCO comparison, 6 core cloud economics principles, Pricing Calculator
7	AWS Cloud Adoption Framework (CAF)	6 Perspectives: Business, People, Governance, Platform, Security, Operations — who owns each



3. STORAGE

#	Topic	Key Concepts
14	Amazon S3 Storage Classes Comparison	Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant, Glacier Flexible, Glacier Deep Archive, Express One Zone
15	AWS Storage Services Comparison	S3, EBS, EFS, FSx, Storage Gateway — storage types, protocols, access patterns, hybrid support



5. NETWORKING & CONTENT DELIVERY

#	Topic	Key Concepts
17	AWS Networking Cheat Sheet	VPC, subnets, IGW, NAT Gateway, Route Tables, Security Groups, NACLs, CIDR, Load Balancers, Route 53
18	Security Groups vs NACLs	Stateful vs Stateless, allow-only vs allow/deny, instance level vs subnet level, rule evaluation
19	AWS VPC Connection Options	IGW, NAT GW, VPC Endpoints, VPC Peering, Transit Gateway, Site-to-Site VPN, Direct Connect
20	AWS Transit Gateway	Hub-and-spoke model, transitive routing, TGW vs VPC Peering at scale (10 VPCs = 45 vs 10 connections)
21	AWS Load Balancers Comparison	ALB (L7), NLB (L4), GWLB (L3), CLB (Legacy) — protocols, routing, cross-zone LB, use cases
22	Amazon Route 53 Routing Policies	Simple, Weighted, Latency, Failover, Geolocation, Geoproximity, Multi-value; Health Checks; Record Types
23	CloudFront vs Global Accelerator	CDN (HTTP/HTTPS caching) vs TCP/UDP performance routing, edge features, architecture, exam traps



7. EVENT-DRIVEN & INTEGRATION ARCHITECTURE

#	Topic	Key Concepts
32	Event-Driven AWS Architecture	EventBridge (route), SNS (fan-out), SQS (queue/decouple), Lambda (compute), Step Functions (orchestrate)



8. ANALYTICS & DATA

#	Topic	Key Concepts
33	AWS Analytics Query Services	Athena + Glue + S3 power trio; Redshift (data warehouse); EMR (big data); Kinesis (streaming); QuickSight (BI)



10. MIGRATION & HYBRID

#	Topic	Key Concepts
35	AWS Migration Services Comparison	Migration Hub, MGN, DMS, SMS, Transfer Family, Snow Family — migration type, downtime, best for
36	AWS Migration & Hybrid Services Architecture	7 R's (Rehost, Replatform, Refactor, Repurchase, Retire, Relocate); hybrid architecture overview
37	AWS Snow Family	Snowball Edge Storage Optimized vs Compute Optimized; discontinued devices; OpsHub; offline data transfer



12. EXAM PREPARATION

#	Topic	Key Concepts
43	High-Frequency Exam Traps	SG vs NACL, Read Replica vs Multi-AZ, CloudFront vs Global Accelerator, storage/DB/AI/analytics traps
44	Most Common CLF-C02 Scenario Questions Map	Need → Keyword → Best AWS service, organized by 9 categories
45	AWS CLF-C02 Keywords Cheat Sheet	Keyword triggers mapped to the right AWS service across all 9 domains



2. COMPUTE

#	Topic	Key Concepts
8	AWS Compute Services Comparison	EC2, Lambda, ECS, EKS, Fargate, Lightsail, Batch — management level, workload type, pricing model
9	EC2 Pricing Options Comparison	On-Demand, Compute Savings Plans, EC2 Instance Savings Plans, Reserved Instances, Spot, Dedicated Hosts, Dedicated Instances
10	Auto Scaling + High Availability Architecture	ASG components, scaling policies (Target Tracking, Step, Simple, Scheduled), multi-AZ design, self-healing
11	AWS Container Services Comparison	ECS, EKS, Fargate, App Runner — orchestration vs serverless containers, infrastructure management
12	AWS Serverless Services Comparison	Lambda, API Gateway, SNS, SQS, EventBridge, Step Functions, DynamoDB — event-driven architecture
13	AWS Edge & Hybrid Infrastructure	Local Zones (city-level latency), Wavelength (5G/mobile edge), Outposts (on-premises AWS)



4. DATABASES

#	Topic	Key Concepts
16	AWS Database Services Comparison	RDS, Aurora, DynamoDB, ElastiCache, Redshift, Neptune — data models, scalability, availability, pricing



6. SECURITY & IDENTITY

#	Topic	Key Concepts
24	IAM Best Practices	Least privilege, groups vs users, MFA, roles, permissions boundaries, access keys, root user, policy best practices
25	AWS Security Services Comparison	GuardDuty, Inspector, Security Hub, Macie, Shield, WAF, IAM Access Analyzer — purpose, targets, integrations
26	Amazon GuardDuty vs Amazon Inspector	Threat detection (active threats / logs & events) vs Vulnerability scanning (CVEs / agents & scans)
27	AWS Governance & Multi-Account Services	Organizations, SCPs, Control Tower, IAM Identity Center (SSO), Config, Security Hub — how they work together
28	AWS KMS vs Secrets Manager vs Parameter Store	Encryption keys vs rotating secrets vs configuration values — purpose, auto-rotation, cost, exam keywords
29	AWS Compliance & Security Standards	HIPAA, PCI DSS, SOC 1/2/3, ISO 27001, GDPR; AWS Artifact vs Audit Manager; compliance lifecycle
30	AWS Identity Services	Cognito User Pools, Cognito Identity Pools, Directory Service, IAM Identity Center (SSO) — who uses each
31	AWS Systems Manager (SSM)	Session Manager (no SSH/RDP), Parameter Store (config/secrets), Patch Manager (OS patching), Run Command



9. AI & MACHINE LEARNING

#	Topic	Key Concepts
34	AWS AI Services Comparison	Bedrock, SageMaker, Rekognition, Textract, Comprehend, Lex, Polly, Transcribe — input/output, use cases, exam traps



11. MONITORING, COST & MANAGEMENT

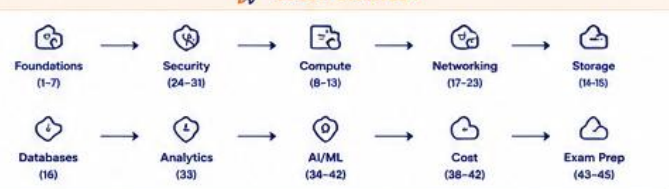
#	Topic	Key Concepts
38	AWS Monitoring & Logging Services Comparison	CloudWatch, CloudTrail, VPC Flow Logs, CloudWatch Logs, X-Ray, AWS Health, Config — purpose, data source
39	AWS Disaster Recovery Strategies	Backup & Restore, Pilot Light, Warm Standby, Multi-Site Active/Active — RTO, RPO, cost, complexity
40	AWS Billing & Pricing Tools	Pricing Calculator, Cost Explorer, Budgets, CUR, Cost Categories, Savings Plans, Trusted Advisor
41	AWS Cost Optimization Services	Cost Explorer, Budgets, Compute Optimizer, Trusted Advisor, Savings Plans, Reserved Instances, Spot
42	AWS Support Plans Comparison	Basic, Developer, Business, Enterprise On-Ramp, Enterprise — response times, TAM, Trusted Advisor, Well-Architected Review

QUICK-REFERENCE SUMMARY

Cloud Concepts	Foundations, Cloud Economics, Deployment Models, CAF, Well-Architected
Security & Compliance	IAM, Shared Responsibility, Security Services, Governance, GuardDuty vs Inspector, KMS/Secrets/SSM, Compliance Standards, Identity Services
Technology	Compute, Storage, Databases, Networking, Analytics, AI/ML, Serverless, Containers, Event-Driven
Billing & Pricing	Cost Tools, Support Plans, Cost Optimization, EC2 Pricing Models
Migration & Hybrid	Migration Services, 7 R's, Hybrid Architecture, Snow Family
Exam Prep	Exam Traps, Scenario Map, Keywords Cheat Sheet



STUDY STRATEGY



Remember: Understand the concepts → Practice scenarios → Review regularly → Focus on weak areas → You'll pass CLF-C02!

AWS GLOBAL INFRASTRUCTURE – CHEAT SHEET (CLF-C02)

AWS has a globally distributed network of data centers to provide high availability, fault tolerance, and low latency.

AWS INFRASTRUCTURE OVERVIEW

REGION
A Region is a physical location in the world where AWS has multiple, isolated Locations.

AVAILABILITY ZONE (AZ)
Each Region consists of multiple AZs that are isolated from each other.

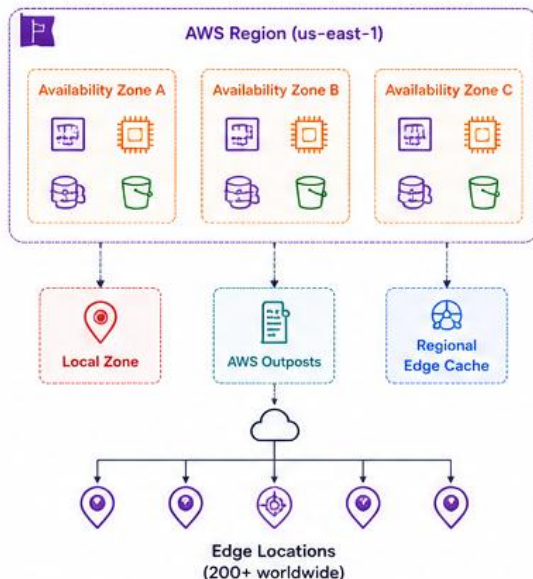
EDGE LOCATION
Used by CloudFront to cache content closer to end users (200+ worldwide).

REGIONAL EDGE CACHE
Used by CloudFront for caching content at the edge in a Region (13 worldwide).

AWS LOCAL ZONE
Extends AWS infrastructure, services, and capabilities closer to large population, industrial, and IT centers.

AWS OUTPOSTS
Fully managed AWS infrastructure on-premises for a consistent hybrid experience.

Example: AWS Region



AWS REGIONS AROUND THE WORLD (as of May 2024)

Regions

- US East (N. Virginia)
- US West (Oregon)
- Europe (Ireland)
- Europe (Frankfurt)
- Asia Pacific (Tokyo)
- Asia Pacific (Singapore)
- Asia Pacific (Sydney)
- Canada (Central)
- South America (São Paulo)
- AWS GovCloud (US)
- Middle East (UAE)
- Africa (Cape Town)



35 Regions
111+ Availability Zones
13 Regional Edge Caches
200+ Edge Locations

Note: AWS continues to expand its global infrastructure. Check the AWS Global Infrastructure page for the latest updates.

KEY POINTS

- ✓ Resources in different AZs are isolated from failures in other AZs.
- ✓ Choose the Region closest to your customers for lower latency.
- ✓ Use multiple AZs within a Region for high availability.
- ✓ Use CloudFront Edge Locations to deliver content with low latency.
- ✓ Use Local Zones for ultra-low latency workloads at the edge.
- ✓ Use Outposts for on-premises integration with AWS services.

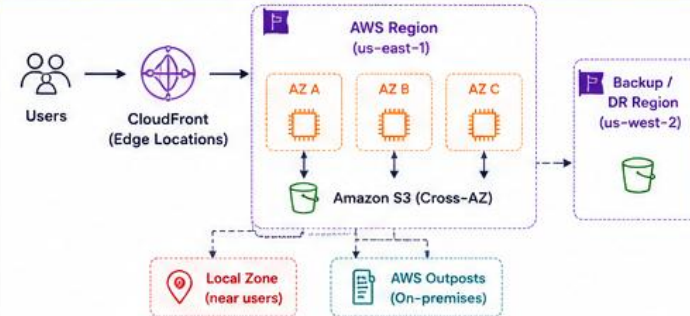
REGION vs AZ

Aspect	Region	Availability Zone (AZ)
Scope	Multiple AZs in a geographic area	One or more data centers within a Region
Isolation	Isolated from other Regions	Isolated from other AZs within the same Region
Latency	Higher (between Regions)	Low (within a Region)
Use Case	Disaster recovery, data residency, compliance	High availability, fault tolerance
Example	us-east-1 (N. Virginia)	us-east-1a, us-east-1b, us-east-1c

WHEN TO USE WHAT?

Multiple AZs	High availability, fault tolerance for applications and databases
Multiple Regions	Disaster recovery, business continuity, data residency
CloudFront (Edge Locations)	Low latency content delivery (static & dynamic)
Local Zones	Low latency for applications close to users
Outposts	Hybrid cloud, data residency, on-premises workloads

EXAMPLE ARCHITECTURE



QUICK REFERENCE

Region	Physical location in the world (e.g., us-east-1)
AZ	Isolated data center within a Region (e.g., us-east-1a)
Edge Location	CloudFront edge sites (200+ worldwide)
Regional Edge Cache	CloudFront cache in Region (13 worldwide)
Local Zone	Extend AWS closer to users for low latency
Outposts	AWS infrastructure on-premises

BENEFITS

- ✓ **High Availability:** AZs are isolated from each other.
- ✓ **Fault Tolerance:** Design apps across multiple AZs.
- ✓ **Low Latency:** Use Regions and Edge Locations close to users.
- ✓ **Scalability:** Global infrastructure scales with your needs.
- ✓ **Compliance:** Choose Regions for data residency.
- ✓ **Hybrid Capability:** Use Outposts for seamless integration.

EXAM TIPS

- 💡 A Region is a physical location, not a data center.
- 💡 AZs are isolated from each other, even within the same Region.
- 💡 Use multiple AZs for HA; multiple Regions for DR.
- 💡 Edge Locations are part of CloudFront.
- 💡 Local Zones = low latency; Outposts = on-premises.
- 💡 Always check the latest Region list on AWS website.

REMEMBER

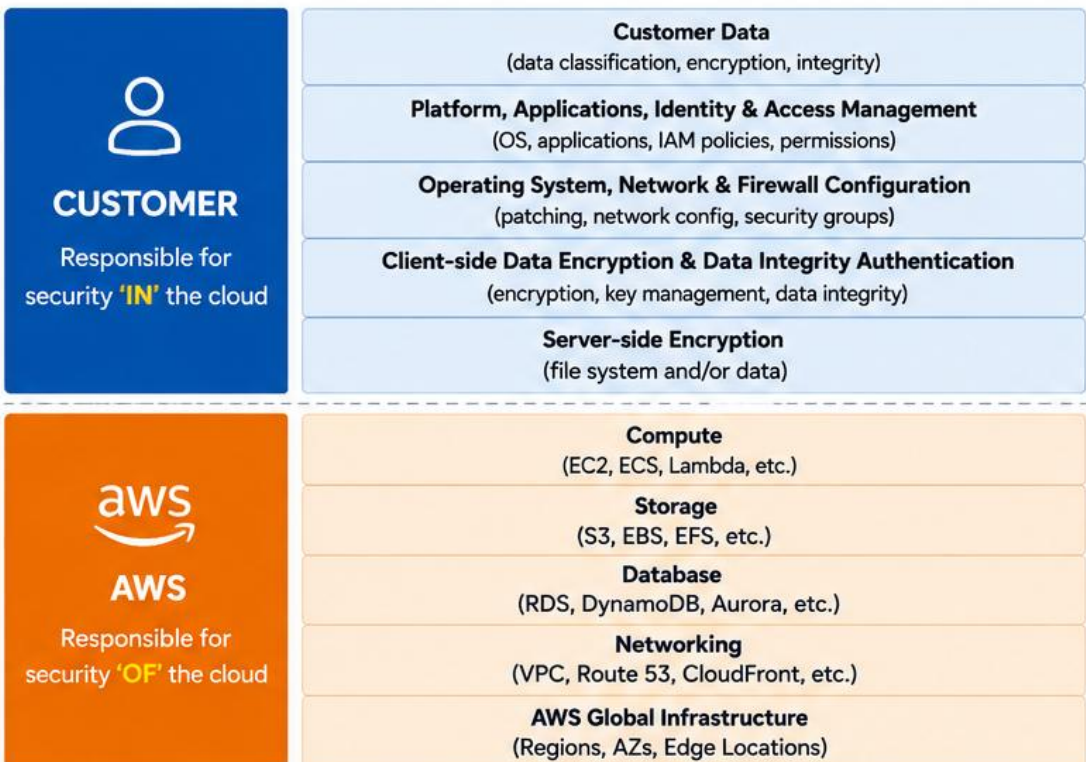


Think:
Global (Region) → Resilient (AZ) → Fast (Edge)
→ Hybrid (Outposts)



AWS SHARED RESPONSIBILITY MODEL

Security and compliance are a **shared effort** between AWS and the customer.



Responsibility Area		On-Premises (You manage)	AWS Cloud (AWS manages)
	Data Your data, backups, encryption		
	Identity & Access Management Users, permissions, roles		
	Operating System Patching, configuration		
	Network & Firewall Configuration		
	Client-side Encryption & Data Integrity		
	Server-side Encryption (File System / Data)		
	Compute		
	Storage		
	Database		
	Networking		
	AWS Global Infrastructure Regions, AZs, Edge Locations		

You manage AWS manages

EXAMPLES BY SERVICE MODEL

IaaS (e.g., EC2)

You manage:

- Guest OS, applications, and data
- Security groups, IAM, encryption
- Network configuration

You

OS / Apps / Data
Network Config

AWS

AWS manages:

- Compute, storage, database
- Networking, infrastructure

Compute / Storage
Database / Network
Infrastructure

PaaS (e.g., RDS)

You manage:

- Data
- IAM, users, permissions
- Network access, encryption

You

Data / Users /
Permissions

AWS

AWS manages:

- OS, patching
- Database engine
- Infrastructure

OS / DB Engine /
Infrastructure

SaaS (e.g., S3, DynamoDB)

You manage:

- Data
- IAM, access policies
- Encryption (optional)

You

Data / Access
Policies

AWS

AWS manages:

- Application
- OS, infrastructure
- Scaling, availability

Application /
Infrastructure

Serverless (e.g., Lambda)

You manage:

- Code
- IAM, permissions
- Event source config

You

Code / Config

AWS

AWS manages:

- Runtime
- Infrastructure
- Scaling, availability

Runtime /
Infrastructure



Key Takeaway: The more managed services you use (PaaS, SaaS, Serverless), the more security responsibilities AWS takes on.

AWS SHARED RESPONSIBILITY MODEL (CLF-C02)

Security and compliance in the cloud is a shared responsibility between AWS and the customer.



KEY TAKEAWAY

Security "in" the cloud is AWS's responsibility.
Security "of" the cloud is your responsibility.

CUSTOMER RESPONSIBILITY

Security "in" the cloud (everything you put in, configure, and manage)

	Data	Classify, protect, and manage your data.
	Applications	Secure your applications and APIs.
	Identity & Access Management	Manage users, groups, roles, permissions, and MFA.
	Operating System, Network & Firewall Configuration	Configure OS, patch (if applicable), manage network, security groups, NACLs, firewalls.
	Client-side Data Encryption & Data Integrity	Encrypt data in transit and at rest (customer-managed keys if used).
	Logging, Monitoring & Alerting	Enable and manage CloudTrail, CloudWatch, VPC Flow Logs, GuardDuty, etc.
	Backup & Recovery	Implement backup policies and test recovery.

CUSTOMER "IN" THE CLOUD



AWS "OF" THE CLOUD

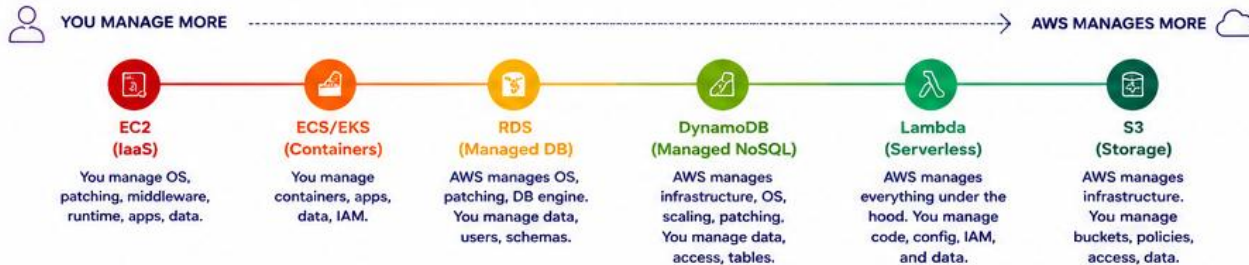


AWS RESPONSIBILITY

Security "of" the cloud (the cloud itself)

	Compute	Physical servers, host hardware, virtualization layer, and Hypervisor.
	Storage	Storage devices, durability, and data centers.
	Database	Physical database servers, storage, replication (for managed services like RDS, DynamoDB).
	Networking	Infrastructure for VPC, AWS Global Network, Route 53, Edge locations, and connectivity.
	AWS Global Infrastructure	Regions, Availability Zones, Edge Locations, and hardware facilities.

MANAGED SERVICES NUANCE (MORE MANAGED = MORE AWS RESPONSIBILITY)



EC2 vs RDS vs LAMBDA – WHO MANAGES WHAT?

RESPONSIBILITY AREA	EC2 (IaaS)	RDS (Managed DB)	Lambda (Serverless)
Physical Servers	AWS	AWS	AWS
Networking (VPC, Route 53)	AWS	AWS	AWS
OS (Operating System)	Customer	AWS	AWS
Patching OS	Customer	AWS	AWS
Database Engine	Customer	AWS	N/A
Middleware / Runtime	Customer	Customer	AWS
Application Code	Customer	Customer	Customer
Data	Customer	Customer	Customer
Backups	Customer	Customer (AWS provides tools)	Customer
Scaling Infrastructure	Customer	AWS (Auto Scaling for storage/compute)	AWS (Automatic)

EXAM TRAPS: WHO PATCHES THE OS?

SERVICE	WHO PATCHES OS?
EC2	Customer
RDS	AWS
Lambda	AWS
ECS (EC2 Launch Type)	Customer
ECS (Fargate)	AWS
EKS (Worker Nodes)	Customer
DynamoDB	AWS
S3	AWS



EXAM TRAP ALERTS

- ✗ If you see EC2, assume YOU patch and manage the OS.
- ✗ If it's a managed service (RDS, DynamoDB, Lambda, S3), AWS patches the OS.
- ✗ Serverless (Lambda) = AWS manages everything underlying.
- ✗ Don't confuse application patching vs OS patching!

KEY TAKEAWAYS

- ✓ You are always responsible for your data, access, and applications.
- ✓ The more managed the service, the more AWS takes care of.
- ✓ EC2 = you manage the OS. RDS & Lambda = AWS manages the OS.
- ✓ Know the boundary for each service – this is heavily tested!



EXAMPLES IN REAL LIFE



You launch EC2:
You install OS, patch it, install Apache, run your app.



You use RDS:
AWS handles OS, patching, backups, replication. You focus on schemas and data.



You use Lambda:
AWS runs, patches, scales, and maintains the platform. You just upload code and set permissions.



REMEMBER: Security "in" the cloud = Your job. Security "of" the cloud = AWS's job.



AWS DEPLOYMENT MODELS – CHEAT SHEET (CLF-C02)

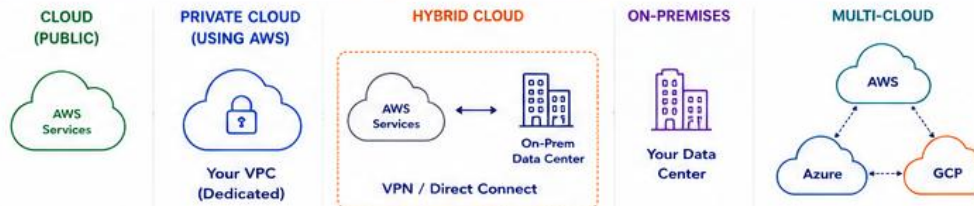
Choose the right deployment model to meet your business, security, and compliance needs.

COMPARE	1. CLOUD (PUBLIC CLOUD)	2. PRIVATE CLOUD (Using AWS)	3. ON-PREMISES (Traditional)	4. HYBRID CLOUD	5. MULTI-CLOUD
Description	Resources are owned and operated by AWS, delivered over the internet.	Resources are dedicated to a single organization on AWS in a virtual environment.	Resources are hosted in your own data center and managed by your organization.	Use a mix of on-premises infrastructure and AWS cloud services.	Use services from multiple different cloud providers (AWS + others).
Infrastructure Ownership	AWS	AWS (dedicated to your organization)	Your Organization	Both (You + AWS)	Multiple Cloud Providers (Shared ownership)
Management Responsibility	AWS manages the infrastructure. You manage your data, applications, and access.	AWS manages the infrastructure. You manage your data, applications, and access.	You manage everything (servers, storage, networking, applications, security, etc.)	You manage on-premises. AWS manages cloud resources. You manage applications and data across both.	You manage resources across multiple clouds. Each provider manages their own infrastructure.
Access	Over the Internet	Over the Internet or AWS Direct Connect/ VPN	Local network	Internet + Private network (VPN/Direct Connect)	Over the Internet or provider-specific connections
Scalability	Virtually unlimited (on-demand)	High (within your private environment limits)	Limited by your hardware capacity	High (cloud side) + limited (on-prem side)	High (across providers, but more complex)
Cost Model	Pay-as-you-go	Pay-as-you-go (dedicated resources may cost more)	High upfront CAPEX + ongoing OPEX	Pay-as-you-go for cloud + CAPEX/OPEX for on-prem	Pay-as-you-go across providers (egress & complexity costs)
Control & Customization	Lower control over infrastructure	More control than public cloud, less than on-prem	Full control	Control over on-prem + flexibility in cloud	High complexity to standardize and control
Security & Compliance	AWS is compliant with many standards. You secure your apps and data.	Greater isolation and control. Easier to meet strict compliance requirements.	You are fully responsible for security and compliance.	Keep sensitive data on-prem. Use cloud for scalability and less critical workloads.	More complex to maintain consistent security and compliance.
Best For	Startups, web apps, test/dev, HR systems, variable workloads, rapid innovation.	Regulated workloads needing dedicated environment, high compliance, predictable workloads.	Legacy systems, no internet access, strict control, specialized hardware needs.	Organizations modernizing gradually, data sovereignty needs, burst capacity.	Avoid vendor lock-in, best-of-breed services, global enterprises with diverse needs.
Examples	Amazon EC2, S3, RDS, Lambda, DynamoDB	Amazon VPC (single-tenant), Dedicated Hosts, Amazon Outposts (connected to AWS)	VMware on local servers, Windows Server, Storage Arrays	On-premises datacenter + AWS (via VPN/Direct Connect) using services like EC2, S3	AWS + Microsoft Azure + Google Cloud Platform (e.g., EC2 + Azure SQL + GCP BigQuery)

HYBRID CLOUD CONNECTION OPTIONS

- VPN (Site-to-Site): Encrypted connection over the internet.
- AWS Direct Connect: Dedicated network connection to AWS for higher bandwidth and lower latency.
- AWS Transit Gateway: Connects VPCs and on-premises networks across multiple Regions.

HOW THEY WORK (AT A GLANCE)



KEY TAKEAWAYS

- Start with Public Cloud for speed and cost efficiency.
- Use Private Cloud (AWS) for greater isolation and compliance.
- Choose On-Premises when you need full control or have limited connectivity.
- Adopt Hybrid Cloud for flexibility and phased migration.
- Use Multi-Cloud for best-of-breed solutions and resilience.



CLF-C02 TIP

Understand the trade-offs between cost, control, security, and complexity. Most organizations use a hybrid approach.



EXAM POINTERS

- Know the 5 AWS deployment models and their characteristics.
- Understand when to use each model based on business and technical requirements.
- Be familiar with connection options for Hybrid Cloud.
- Remember shared responsibility in cloud vs. full responsibility on-prem.



AWS WELL-ARCHITECTED FRAMEWORK – CHEAT SHEET (CLF-C02)

Build secure, high-performing, resilient, and efficient systems in the cloud.

SUSTAINABILITY = MINIMIZE ENVIRONMENTAL IMPACT
Key services: Graviton processors, S3 Intelligent-Tiering, AWS Compute Optimizer, lower-carbon regions.

THE 6 PILLARS OF THE AWS WELL-ARCHITECTED FRAMEWORK

1. OPERATIONAL EXCELLENCE	2. SECURITY	3. RELIABILITY	4. PERFORMANCE EFFICIENCY	5. COST OPTIMIZATION	6. SUSTAINABILITY
Run and monitor systems to deliver business value and continuously improve supporting processes and procedures.	Protect information, systems, and assets while delivering business value through risk assessments and mitigation.	Ensure systems perform their intended function correctly and consistently when expected.	Use computing resources efficiently to meet system requirements and maintain performance as demand changes.	Run systems to deliver business value at the lowest price point.	Minimize the environmental impacts of running cloud workloads.
✓ Perform operations as code ✓ Make frequent, small, reversible changes ✓ Refine operations procedures ✓ Learn from all failures	✓ Implement strong identity controls ✓ Apply least privilege access ✓ Enable traceability ✓ Protect data in transit and at rest	✓ Design for failure ✓ Automate recovery ✓ Test recovery procedures ✓ Scale horizontally to increase availability	✓ Select the right resource types ✓ Monitor and maintain performance ✓ Scale to match demand ✓ Optimize over the lifecycle	✓ Implement cloud financial management ✓ Adopt appropriate purchasing models ✓ Eliminate waste ✓ Analyze and attribute expenditure	✓ Understand your impact ✓ Establish sustainability goals ✓ Maximize utilization ✓ Use lower-carbon services and regions
Example AWS Services 	Example AWS Services 	Example AWS Services 	Example AWS Services 	Example AWS Services 	KEY SERVICES & PRACTICES Graviton processors S3 Intelligent-Tiering AWS Compute Optimizer Lower-carbon regions Example AWS Services

</

KEY TAKEAWAYS	EXAM TIPS (CLF-C02)	POPULAR AWS SERVICES BY PILLAR (AT A GLANCE)	
✓ Use the 6 pillars as a guide to build better cloud architectures. ✓ Continuously review and improve workloads. ✓ Make data-driven decisions based on metrics. ✓ Balance trade-offs based on your business priorities.	✓ Know the purpose and key focus of each pillar. ✓ Understand trade-offs between pillars. ✓ Be able to identify pillar improvements in scenario-based questions. ✓ Use AWS services that align with the pillar's goals.	Operational Excellence Security Reliability Performance Efficiency Cost Optimization Sustainability	CloudWatch, Systems Manager, CloudTrail, AWS Config IAM, KMS, GuardDuty, AWS WAF, Shield EC2 Auto Scaling, ELB, Route 53, RDS Multi-AZ, Backup EC2, ElastiCache, CloudFront, S3, Lambda, Amazon MQ AWS Cost Explorer, Budgets, Savings Plans, Trusted Advisor AWS Regions, Compute Optimizer, S3 Intelligent-Tiering, Graviton



REMEMBER: The Well-Architected Framework is not prescriptive. Use it to make informed trade-offs and build workloads that meet your business needs.



EXAM TIP: Sustainability is a tested pillar! Don't skip it. Know the goal and key services that support it.



LEARN MORE: aws.amazon.com/well-architected



AWS CLOUD ECONOMICS & VALUE PROPOSITION – CHEAT SHEET (CLF-C02)

Understand the economic benefits of cloud computing and how AWS helps you save money.

CAPEX vs OPEX: TRADITIONAL IT vs AWS CLOUD

Aspect	Traditional IT (On-Premises) CapEx (Capital Expense)	AWS Cloud OpEx (Operational Expense)
Definition	Upfront investment in physical infrastructure and assets.	Pay only for what you use, when you use it.
Payment Model	Large upfront cost.	Pay-as-you-go, no upfront commitment.
Cost Type	Fixed expense (regardless of usage).	Variable expense (scales with usage).
Cash Flow Impact	High initial impact.	Low initial impact, predictable operating costs.
Scalability	Slow to scale, requires new hardware purchases.	Instant scale up or down.
Utilization	Often underutilized (10–30% average utilization).	Pay for actual usage, reduce waste.
Risk	High risk of over-provisioning or obsolescence.	Lower risk, no upfront commitment.
Maintenance	You manage hardware, software, facilities, and upgrades.	AWS manages infrastructure, reducing your overhead.
Accounting	Shown as assets on balance sheet, depreciated over time.	Expensed as operating cost, improves agility.
Example	Buying servers, networking gear, racks, data center space.	EC2, S3, RDS, Lambda – only pay for what you use.
Bottom Line	Trade agility for control.	Trade fixed expense for variable expense and agility.

6 CORE CLOUD ECONOMICS PRINCIPLES

- 1 PAY-AS-YOU-GO MODEL**
 - Pay only for the resources you use.
 - No upfront costs or long-term contracts required.
- 2 TRADE FIXED EXPENSE FOR VARIABLE EXPENSE**
 - Eliminate large capital investments.
 - Convert CapEx to OpEx for better cash flow and flexibility.
- 3 ELASTICITY & SCALABILITY**
 - Scale resources up or down in minutes.
 - Respond quickly to business demands and avoid over-provisioning.
- 4 BENEFIT FROM MASSIVE ECONOMIES OF SCALE**
 - AWS spreads infrastructure costs across millions of customers.
 - You benefit from lower prices than you could achieve on your own.
- 5 STOP GUESSING CAPACITY**
 - No need to predict peak loads years in advance.
 - Scale as needed and pay for actual usage.
- 6 FOCUS ON INNOVATION, NOT INFRASTRUCTURE**
 - AWS manages the undifferentiated heavy lifting.
 - You focus on building and delivering business value.

TOTAL COST OF OWNERSHIP (TCO) COMPARISON

Traditional IT (On-Premises) High TCO	AWS Cloud Lower TCO
Hardware (Servers, Storage, Networking) + Software Licensing + Power & Cooling + Data Center / Facilities (Rent, Space, Racks) + IT Staff & Operations (Manage, Monitor, Maintain) + Maintenance & Upgrades (Spare parts, Refresh cycles) + Security & Compliance + Downtime & Risk	✓ No hardware to buy ✓ No data center to build or run ✓ No power or cooling costs ✓ No hardware maintenance ✓ No software patching ✓ No capacity planning ✓ Built-in security & compliance ✓ High availability by design ✓ Pay only for what you use RESULT: Lower costs, less risk, more agility, faster innovation.
TCO includes all direct and indirect costs of running and maintaining an application over its lifecycle.	

AWS PRICING CALCULATOR: ESTIMATE YOUR TCO



- ✓ Build and compare cost estimates for AWS services.
- ✓ Compare AWS vs on-premises costs.
- ✓ Adjust usage, instance types, storage, and regions.
- ✓ Save and share your estimates.
- ✓ Export detailed reports for planning.

Access it at: calculator.aws/#/

WHEN TO USE CLOUD (THE ECONOMIC CASE)

- Use AWS when you need:**
 - ✓ Variable workloads or unpredictable demand
 - ✓ Fast scalability and rapid time-to-market
 - ✓ Lower upfront costs and improved cash flow
 - ✓ Global infrastructure without large investment
 - ✓ Focus on innovation, not infrastructure
 - ✓ Built-in security, compliance, and reliability
- Not ideal when you have:**
 - ✗ Extremely steady, high, fixed utilization (over many years)
 - ✗ Strict data sovereignty with no cloud option
 - ✗ Applications that cannot run in the cloud
 - ✗ Regulatory requirements that prohibit use of cloud providers

EXAM TIPS (CLF-C02)

- ✓ Identify whether the question is asking about cost model (CapEx vs OpEx) or cost benefits.
- ✓ Look for keywords: pay-as-you-go, scale, upfront cost, TCO, economies of scale.
- ✓ AWS saves money by reducing TCO, not just instance pricing.
- ✓ Use AWS Pricing Calculator for TCO comparison questions.
- ✓ Remember: Cloud = lower TCO, more agility, less risk.



KEY TAKEAWAY: AWS helps you spend less by eliminating upfront costs, reducing operational overhead, and letting you pay only for what you use while benefiting from massive economies of scale.



FURTHER READING:
AWS Well-Architected Framework – Cost Optimization Pillar | AWS Pricing Calculator User Guide



REMEMBER: Choose the right services and optimization strategies to maximize your savings.



AWS CLOUD ADOPTION FRAMEWORK (CAF) – CHEAT SHEET (CLF-C02)

A framework of organizational capabilities to help you successfully adopt and scale AWS.



CLF-C02 EXAM NOTE

CAF helps organizations align people, process, and technology for successful cloud adoption.

THE 6 PERSPECTIVES OF THE AWS CLOUD ADOPTION FRAMEWORK

1 BUSINESS

 Who Owns It?	Business leaders, finance leaders
 Focus (What It Drives)	ROI, business outcomes, value realization
 Key Responsibilities (Examples)	- Define business strategy and objectives - Build the business case and track ROI - Prioritize investments based on value - Measure outcomes and benefits

PEOPLE

 Who Owns It?	HR leaders, staff managers
 Focus (What It Drives)	Training, change management, culture
 Key Responsibilities (Examples)	- Develop skills and training plans - Drive change management - Communicate and engage teams - Build a culture of innovation

GOVERNANCE

 Who Owns It?	CIO, program managers, policy owners
 Focus (What It Drives)	Policies, risk management, compliance
 Key Responsibilities (Examples)	- Define cloud policies and standards - Manage risk, compliance, and audit - Set guardrails and decision rights - Ensure regulatory requirements are met

PLATFORM

 Who Owns It?	Cloud architects, platform engineers
 Focus (What It Drives)	Infrastructure, architecture patterns, automation
 Key Responsibilities (Examples)	- Design and build scalable architectures - Provide reusable platform services - Automate provisioning and deployment - Manage landing zones and guardrails

SECURITY

 Who Owns It?	CISO, security team, security architects
 Focus (What It Drives)	IAM, controls, data protection
 Key Responsibilities (Examples)	- Define security policies and controls - Manage IAM and access governance - Protect data and ensure privacy - Monitor security and respond to threats

OPERATIONS

 Who Owns It?	IT operations managers, operations team
 Focus (What It Drives)	Monitoring, reliability, incident management
 Key Responsibilities (Examples)	- Operate and monitor workloads - Manage incidents and problems - Optimize performance and cost - Ensure backup, DR, and continuity

HOW THE 6 PERSPECTIVES WORK TOGETHER



- ✓ All 6 perspectives are interdependent.
- ✓ Strong collaboration across perspectives leads to faster, safer, and more successful cloud adoption.
- ✓ Weakness in one area can impact overall success.
- ✓ Review and mature each perspective continuously.

COMMON EXAM SCENARIOS (WHO OWNS WHAT?)

Scenario	Which Perspective?
Define ROI and track business value	Business
Plan training and drive culture change	People
Create cloud policies and manage compliance	Governance
Design scalable architectures and landing zones	Platform
Manage IAM, encryption, and data protection	Security
Monitor systems and handle incidents	Operations

KEY TAKEAWAYS

- ✓ CAF provides a holistic approach to cloud adoption.
- ✓ Know which team/role owns each perspective.
- ✓ Focus on business value while maintaining security and compliance.
- ✓ Continuous improvement across all 6 perspectives drives success.



EXAM TIP If the question asks "which team is responsible for...?" map it to one of the 6 perspectives.
Example: "Who defines cloud policies?" → Governance. "Who manages incidents?" → Operations.



FURTHER READING AWS Cloud Adoption Framework Overview | CAF Whitepaper | AWS Well-Architected Framework



AWS COMPUTE SERVICES COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right compute service based on your control needs, scalability, pricing model, and use case.

FEATURE	AMAZON EC2 (Elastic Compute Cloud)	AWS LAMBDA (Serverless)	AMAZON ECS (Elastic Container Service)	AMAZON EKS (Elastic Kubernetes Service)	AWS FARGATE (Serverless for Containers)	AWS LIGHTSAIL (Virtual Private Servers)	AWS BATCH (Batch Processing)
What it is	Resizable virtual servers in the cloud.	Run code without provisioning or managing servers.	Managed container orchestration service (Docker containers).	Managed Kubernetes service to run containers at scale.	Run containers without managing servers (works with ECS & EKS).	Easy-to-use virtual private servers with predefined plans.	Fully managed service for batch computing at any scale.
Management	You manage OS, applications, scaling.	AWS manages everything (servers, scaling, OS).	AWS manages the service; you manage containers and tasks.	AWS manages the Kubernetes control plane; you manage nodes/workloads.	AWS manages the infrastructure; you manage containers.	You manage OS and applications.	AWS manages the infrastructure; you manage jobs and applications.
Server Management	You manage servers.	No servers to manage.	You manage EC2 instances (or use Fargate).	You manage worker nodes (or use Fargate).	No servers to manage.	You manage servers.	No servers to manage.
Workload Type	General purpose (any workload).	Event-driven, short-running code, microservices.	Containerized applications.	Containerized applications (Kubernetes workloads).	Containerized applications without managing servers.	Simple apps, websites, development, testing.	Batch jobs, simulations, ETL, data processing.
Scaling	Manual, Auto Scaling (you define).	Automatic scaling (concurrency based).	Manual, Service Auto Scaling.	Cluster Auto Scaling (you configure).	Automatic scaling.	Manual (vertical scaling by plan upgrade).	Automatic scaling (compute environments).
Pricing Model	Pay for instance (by the second/hour).	Pay per request + duration (ms).	Pay for underlying EC2 instances.	Pay for control plane + worker nodes (EC2/Fargate).	Pay per vCPU, memory + per second.	Monthly fixed price bundled plans.	Pay for vCPU/Memory allocated per second.
Control	Full control over OS, networking, storage.	No control over infrastructure, only code.	Full control over containers, tasks, networking.	Full control over Kubernetes resources and workloads.	Focus on containers; no infrastructure control.	Moderate control (over OS & apps).	Focus on jobs; no infrastructure control.
Startup Time	Minutes	Milliseconds	Minutes	Minutes	Seconds	Minutes	Minutes
Best For	• Lift-and-shift applications • Custom software • Full control needs	• Event-driven apps • APIs, backends • Data processing • IoT, real-time	• Microservices • Web applications • Container workloads	• Complex container apps • Kubernetes workloads • Multi-cloud/hybrid	• Container apps without managing servers • Variable workloads	• Small websites • Dev/Test environments • Simple applications	• Batch processing • Data analysis • HPC workloads
Integrates With	All AWS services	API Gateway, S3, DynamoDB, SNS, SQS, EventBridge, etc.	ALB, CloudWatch, ECR, IAM, VPC, etc.	VPC, Load Balancers, CloudWatch, IAM, etc.	ECS, EKS, ALB, CloudWatch, ECR, IAM, etc.	DNS, Load Balancers, RDS, S3, etc.	S3, EFS, CloudWatch, IAM, Step Functions, etc.
OS Support	Windows, Linux, etc.	Provided runtimes only (Node.js, Python, Java, etc.)	Based on container image	Based on container image	Based on container image	Linux/Unix, Windows	Based on job container or AMI (Linux/Windows)
Use Your Own Infrastructure?	No (AWS managed)	No	No	No	No	No	No
Ideal For (CLF-C02 Tip)	When you need maximum control and customization.	When you want to run code without managing servers.	When you run Docker containers on a managed platform.	When you need Kubernetes features and portability.	When you want containers without managing infrastructure.	When you need simple, low-cost virtual servers.	When you run large batch jobs or parallel processing.

QUICK DECISION GUIDE

- Need full control over OS & environment? → Use **EC2**
- Running event-driven code or APIs? → Use **Lambda**
- Using containers (Docker) and want AWS to manage it? → Use **ECS**
- Need Kubernetes for containers? → Use **EKS**
- Want containers but don't want to manage servers? → Use **Fargate**
- Need simple VPS for small projects? → Use **Lightsail**
- Running batch jobs or large-scale processing? → Use **Batch**

COMMON EXAM SCENARIOS

- ✓ Run a web application with full control → **EC2**
- ✓ Run backend code in response to S3 upload → **Lambda**
- ✓ Deploy a containerized app with microservices → **ECS**
- ✓ Run Kubernetes-based app → **EKS**
- ✓ Run containers without managing servers → **Fargate**
- ✓ Host a small website quickly → **Lightsail**
- ✓ Process large datasets overnight → **Batch**

KEY TAKEAWAYS

- ★ EC2 = Virtual servers (full control)
- ★ Lambda = Serverless functions (no servers)
- ★ ECS = Containers on AWS (easy)
- ★ EKS = Kubernetes on AWS (powerful)
- ★ Fargate = Serverless for containers
- ★ Lightsail = Simple VPS (easy & cheap)
- ★ Batch = Batch computing at scale



CLF-C02 TIP

Understand the level of control, management, and pricing model for each compute service—this is key for many exam questions.

Think:

- Full control? → **EC2**
- No servers? → **Lambda** or **Fargate**
- Containers? → **ECS** or **EKS**
- Simple VPS? → **Lightsail**
- Batch jobs? → **Batch**

Remember: All these services integrate with other AWS services like VPC, IAM, CloudWatch, S3, RDS, and more.

EC2 PRICING OPTIONS COMPARISON (CLF-C02)

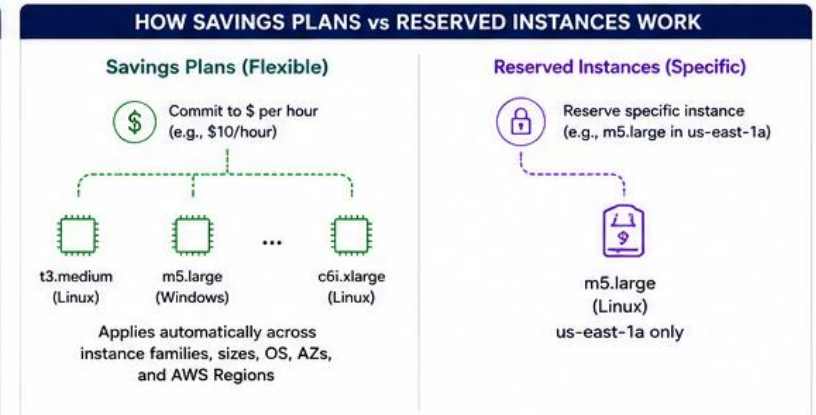
Choose the right pricing model to optimize cost for your workload

FEATURE	ON-DEMAND	SAVINGS PLANS		RESERVED INSTANCES (RI)	SPOT INSTANCES	DEDICATED HOSTS	DEDICATED INSTANCES
		Compute Savings Plans	EC2 Instance Savings Plans				
Description	Pay for compute capacity by the hour or second with no long-term commitment.	Commit to a consistent amount of compute (USD/hour) in exchange for a lower price.	Commit to a specific instance family in a specific region (USD/hour) for a lower price.	Reserve specific instances for 1 or 3 years for a significant discount.	Bid for unused EC2 capacity at a discounted price.	Physical server dedicated to your use. Most control and visibility.	Instance dedicated to you on shared hardware.
Applies To	Any instance in any region	EC2, Fargate, Lambda (any family, any region)	EC2 instances only (specific family, specific region)	EC2 instances only (specific instance type, size, OS, AZ – can modify some)	Any instance, any AZ (availability varies)	EC2 instances only	EC2 instances only
Payment Option	No upfront payment	No upfront payment	No upfront payment	All Upfront / Partial Upfront / No Upfront	No upfront payment	Hourly	Hourly
Discount (Up to)	None	1 or 3 years	1 or 3 years	1 or 3 years	None	None	None
Flexibility	0%	Up to 66%	Up to 72%	Up to 72%	Up to 90%	0% (billed per host)	Up to 15% vs On-Demand
Best For	High – scale up/down as needed	Highest – automatically applies across instance families, sizes, OS, and regions	Lower – locked to instance family and region (size flexible within family)	Low – specific instance type, size, OS, AZ (can modify some)	Low – capacity can be interrupted	Low – physical host level	Medium – instance level
Interruptions	Short-term, unpredictable, burst workloads	Steady-state usage across multiple services and workloads	Steady usage of the same instance family in the same region	Steady, predictable usage for specific instances	Fault-tolerant, flexible, interruption-tolerant workloads	Compliance, licensing, socket/host-level control	Compliance, reduced noisy neighbor risk
Capacity Reservation	No	No	No	No	Yes – can be interrupted with 2-minute notice	No	No
Scope	Any instance, any AZ	EC2, Fargate, Lambda (any family, any region)	EC2 instances only (specific family, specific region)	EC2 instances only (specific configuration)	Any instance, any AZ (availability varies)	Yes (entire host) Host level (multiple instances per host)	No Instance level

 Note: Discounts vary by region and instance type. Savings Plans provide the most flexibility. Use the AWS Pricing Calculator for accurate estimates.

SUMMARY COMPARISON				
Option	Commitment	Payment	Best Savings	Flexibility
On-Demand	None	No upfront	0%	★★★★★
Compute SP	1 or 3 years	No upfront	Up to 66%	★★★★★
EC2 Instance SP	1 or 3 years	No upfront	Up to 72%	★★★★★
Reserved Instances	1 or 3 years	Upfront options	Up to 72%	★★★★★
Spot Instances	None	No upfront	Up to 90%	★★★★★
Dedicated Hosts	None	Hourly	0%	★★★★★
Dedicated Instances	None	Hourly	Up to 15%	★★★★★

WHEN TO USE WHAT?		
	Unpredictable traffic, development, testing	→ On-Demand
	Steady usage across multiple services (EC2, Fargate, Lambda) across families/regions	→ Compute Savings Plans
	Steady usage of the same instance family in the same region	→ EC2 Instance Savings Plans
	Steady, predictable usage of specific instances	→ Reserved Instances
	Fault-tolerant, interruption-tolerant workloads	→ Spot Instances
	Compliance, licensing, socket/host control	→ Dedicated Hosts
	Compliance, reduced noisy neighbor risk	→ Dedicated Instances



EXAM TIPS

- ✓ Compute Savings Plans = Most flexible (applies to EC2, Fargate, Lambda across any region).
- ✓ EC2 Instance Savings Plans = More savings but locked to instance family + region.
- ✓ Reserved Instances = Specific instance, AZ, OS (less flexible than Savings Plans).
- ✓ Spot Instances = Cheapest but can be interrupted.
- ✓ If no commitment / short term / unpredictable → On-Demand.



KEY TAKEAWAY

Use the right pricing model to match your workload pattern. Savings Plans for flexibility, RIs for maximum discount on predictable usage, and Spot Instances for highest savings on interruption-tolerant workloads.



CLF-C02 TIP

For most scenarios, think:

On-Demand (short term) → Savings Plans (most flexible) → Reserved Instances (max savings for steady workloads) → Spot Instances (cheapest for fault-tolerant workloads).

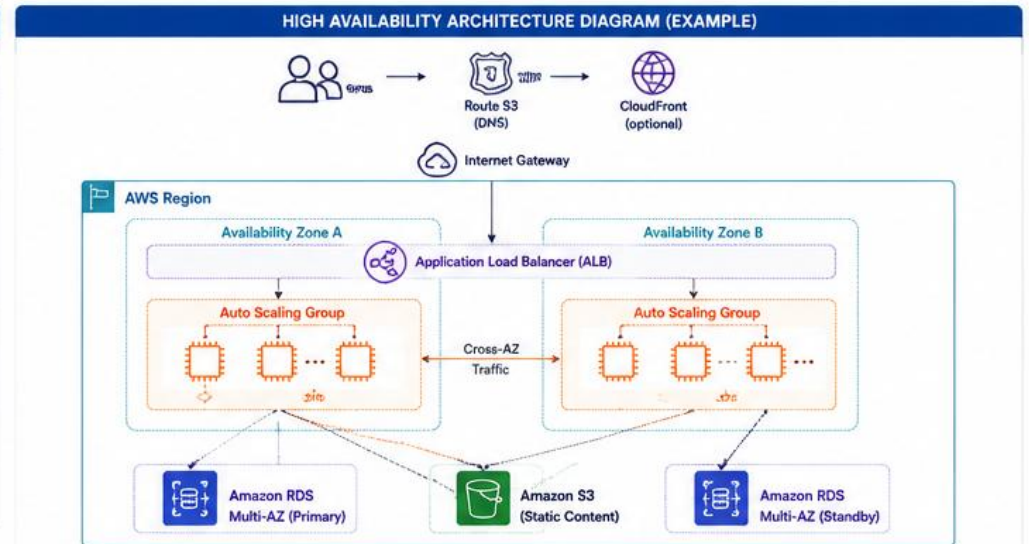


AWS AUTO SCALING + HIGH AVAILABILITY ARCHITECTURE – CHEAT SHEET (CLF-C02)

Design applications that automatically scale to handle demand and remain available even if components fail.

AUTO SCALING GROUP (ASG) – OVERVIEW		ASG COMPONENTS	HOW AUTO SCALING WORKS		SCALING POLICIES
Auto Scaling automatically adjusts the number of EC2 instances in your application to maintain performance and availability.		Launch Template (Recommended) Defines instance configuration (AMI, instance type, key pair, security groups, user data).			Target Tracking Scaling (Recommended) Maintains a specific metric at a target value (e.g., CPU utilization at 50%).
Scalability	Scale out (add instances) or scale in (remove instances) based on demand.	Auto Scaling Group Manages a group of EC2 instances with similar configurations.	Scale Out (Add Instances) - Metric exceeds threshold (e.g., CPU > 70%) - ASG launches new instances - Instances register with ELB - Traffic is distributed		Step Scaling Scales based on metric thresholds using step adjustments.
Availability	Automatically replaces unhealthy instances and balances across Availability Zones.	Scaling Policies Define when and how to scale (out/in) based on metrics.	Scale In (Remove Instances) - Metric drops below threshold (e.g., CPU < 30%) - ASG terminates instances - Remaining instances handle traffic		Simple Scaling Adds or removes a fixed number of instances when threshold is breached.
Cost Optimization	Scale in during low traffic to reduce costs.	Health Checks Monitors instance health and replaces unhealthy instances.			Scheduled Scaling Scales based on a recurring schedule (e.g., increase at 9 AM, decrease at 6 PM).
Fault Tolerance	Distributes instances across AZs and replaces unhealthy instances.				

HIGH AVAILABILITY ARCHITECTURE – BEST PRACTICES			HIGH AVAILABILITY AWS SERVICES	
Use Multiple AZs	Deploy resources across 2 or more Availability Zones.		SERVICE	ROLE IN HIGH AVAILABILITY
Eliminate Single Points of Failure	No critical component should be in a single AZ.		Amazon EC2 Auto Scaling	Automatically adjusts capacity and replaces unhealthy instances.
Use Load Balancing	Distribute traffic across multiple instances/AZs.		Elastic Load Balancing (ALB/NLB)	Distributes traffic across healthy instances in multiple AZs.
Health Checks	Use ELB and ASG health checks to detect failures.		Amazon RDS Multi-AZ	Automatic standby in another AZ with synchronous replication.
Automate Recovery	Use Auto Scaling to replace unhealthy instances.		Amazon ElastiCache (Multi-AZ)	Automatic failover to standby node.
Back Up Data	Use database backups and cross-AZ replication.		Amazon Route 53	DNS failover routing to healthy endpoints.
Decouple Components	Use AWS services (SQS, SNS, RDS, S3) to decouple.		Amazon S3	11 9s durability, data replicated across multiple AZs.
Test for Failure	Regularly test failover and recovery (Chaos Engineering).		Amazon CloudFront	Global edge locations for high availability and low latency.
			AWS Global Accelerator	Improves availability and performance using AWS global network.



SAMPLE SCALING POLICY (TARGET TRACKING)		COMMON ASG SETTINGS	COMMON EXAM SCENARIOS	KEY TAKEAWAYS
Metric:	Average CPU Utilization	Min Size: Minimum number of instances	✓ Application experiences unpredictable traffic → Use ASG + Target Tracking.	★ Auto Scaling = scalable, cost-effective, self-healing.
Target Value:	50%	Max Size: Maximum number of instances	✓ Ensure application is available even if an AZ fails → Deploy across multiple AZs.	★ High Availability = multi-AZ, no single points of failure.
Scale Out:	Add 1 instance if CPU > 50%	Desired Capacity: Initial/desired number of instances	✓ Replace unhealthy instances automatically → Use ASG with health checks.	★ Use managed AWS services to build resilient architectures.
Scale In:	Remove 1 instance if CPU < 50%	Health Check Type: EC2 / ELB	✓ Distribute traffic across instances → Use ALB in front of ASG.	★ Combine Route 53, ELB, ASG, and Multi-AZ services for high availability.
Cooldown:	300 seconds	Health Check Grace Period: Time for instance to initialize before health checks start	✓ Highly available database → Use RDS Multi-AZ.	
		Termination Policies: Order in which instances are terminated during scale in	✓ Static website needs high availability → Host on S3 + CloudFront.	



CLF-C02 TIP: Understand Auto Scaling concepts, scaling policies, and how to design highly available architectures using AWS services. Focus on Multi-AZ, Load Balancing, Failover, and Self-Healing.



Learn more: AWS Auto Scaling: aws.amazon.com/autoscaling/ | AWS Architecture Center: aws.amazon.com/architecture/

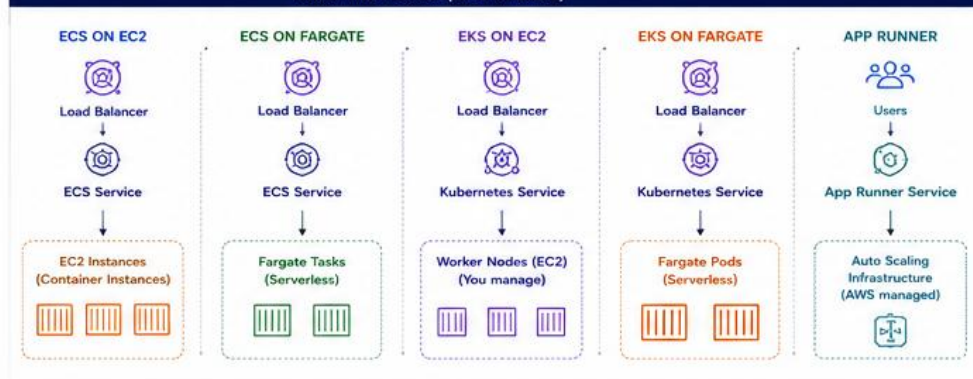


AWS CONTAINER SERVICES COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right container service to build, deploy, and run containerized applications on AWS.

FEATURE / SERVICE	 AMAZON ECS (Elastic Container Service)	 AMAZON EKS (Elastic Kubernetes Service)	 AWS FARGATE (Compute for Containers)	 AWS APP RUNNER (Fully Managed Service)
 What it is	Highly scalable, high-performance container orchestration service with support for Docker containers.	Managed Kubernetes service to run Kubernetes without needing to install and operate it.	Serverless compute engine for containers that works with both ECS and EKS (no server management).	Fully managed service to build and run containerized web apps and APIs with ease.
 Managed by AWS	ECS control plane (scheduling)	EKS control plane	Server provisioning, scaling, patching	Everything (build, deploy, scale, manage)
 Orchestration	AWS-native orchestration	Kubernetes (open source)	Works with ECS or EKS	AWS-native, abstracts containers
 Infrastructure Management	You manage EC2 instances (ECS on EC2) or use Fargate (serverless).	You manage worker nodes (EC2, Bottlerocket, or Fargate).	No servers to manage (serverless). Pay for vCPU and memory used.	No servers, clusters, or infrastructure to manage.
 Workload Type	Long-running containers, batch jobs, microservices	Complex apps, microservices, portable workloads	Any containerized workload (ECS/EKS)	Web apps, APIs, background workers
 Use Cases	Web applications, APIs, microservices, batch processing, CI/CD workloads	Cloud-native apps, multi-cloud K8s portability, stateful apps, service mesh	Workloads that need serverless compute and automatic scaling	Simple containerized web apps and APIs (mostly HTTP/HTTPS)
 Key Features	- Task definitions & services - Integration with AWS services - Service discovery, load balancing - Spot capacity support - Blue/green deployments	- Upstream Kubernetes compatibility - Managed control plane (HA) - IAM authentication & RBAC - Add-ons (CoreDNS, CNI, metrics server) - Works with Fargate, EC2, Bottlerocket	- Serverless compute for containers - Automatic scaling - Pay-per-use (per vCPU/memory-second) - No cluster or node management	- Automatic build from source or container image - Auto deploy, scale, and manage - Custom domains & HTTPS - Integrated CI/CD and rollbacks - Observability and logging built-in
 Scaling	Manual, Auto Scaling, or with Fargate	Cluster Autoscaler, Karpenter, or with Fargate	Automatic based on workload	Automatic (based on HTTP traffic)
 Pricing Model	Pay for EC2 (if using ECS on EC2) or Fargate usage (vCPU/memory/requests)	Pay for control plane (per cluster) + worker nodes (EC2/Fargate)	Pay per vCPU and memory per second (plus requests for ECR data transfer)	Pay per vCPU, memory, requests, and data transfer
 Best For	Teams that want AWS-native orchestration with flexibility in infrastructure.	Teams needing Kubernetes features, portability, and ecosystem.	Teams that want serverless containers without managing servers.	Teams that want the simplest way to run containerized web apps.
 Supports	Linux, Windows containers	Linux containers	Linux, Windows containers (with ECS on Fargate)	Linux containers

HOW THEY WORK (AT A GLANCE)



WHEN TO CHOOSE WHAT?

SCENARIO	BEST CHOICE
Need simple AWS-native container service with control over infrastructure	Amazon ECS (on EC2)
Want serverless containers without managing servers	ECS on Fargate or EKS on Fargate
Need Kubernetes features, portability, and ecosystem	Amazon EKS
Need the easiest way to run a web app or API in a container	AWS App Runner
Existing Kubernetes apps to migrate to AWS	Amazon EKS
Batch jobs or one-off tasks	ECS (with Fargate) or AWS Batch
Need hybrid capacity (on-demand + Spot) for cost savings	Amazon ECS or Amazon EKS

KEY TAKEAWAYS

- ✓ ECS = AWS-native, flexible, great integration.
- ✓ EKS = Kubernetes power and portability.
- ✓ Fargate = Serverless compute for containers (no servers to manage).
- ✓ App Runner = Simplest way to run containerized web apps.
- ✓ Use the right mix based on control, complexity, and ease of use.



CLF-C02 TIP

ECS and EKS are orchestration services.
Fargate is compute for containers (works with both).
App Runner is a fully managed platform for web apps.



EXAM POINTERS

- Know the differences between ECS, EKS, Fargate, and App Runner.
- Understand when to use serverless containers (Fargate/App Runner).
- Remember: Fargate removes the need to manage servers.
- EKS is Kubernetes; ECS is AWS-native.



FURTHER READING

AWS ECS User Guide | AWS EKS User Guide | AWS Fargate Overview
AWS App Runner Developer Guide | AWS Containers Documentation



AWS SERVERLESS SERVICES COMPARISON – CHEAT SHEET (CLF-C02)

Build and run applications without managing servers. Pay only for what you use.



CLF-C02 SCOPE NOTE

The CLF-C02 exam tests these 7 serverless services. Amazon Cognito is covered under IAM/Identity, not here.

FEATURE / SERVICE	AWS LAMBDA Run Code	AMAZON API GATEWAY Build APIs	AMAZON SNS Publish / Subscribe	AMAZON SQS Message Queueing	AMAZON EVENTBRIDGE Events & Scheduling	AWS STEP FUNCTIONS Orchestrate Workflows	AMAZON DYNAMODB Serverless NoSQL DB
What it is	Run code in response to events without provisioning servers.	Create, publish, maintain, monitor, and secure APIs at any scale.	Fully managed pub/sub messaging service for distributing messages.	Fully managed message queuing service for decoupling applications.	Event bus service to connect apps using events. Includes scheduling.	Visual workflows to coordinate multiple AWS services and Lambda.	Fully managed NoSQL database with single-digit millisecond performance.
Primary Use Cases	Backend logic, data processing, file processing, automation, microservices.	REST APIs, WebSocket APIs, HTTP APIs for web/mobile/backend applications.	Broadcast notifications, alerts, mobile push, fan-out to SQS, email, SMS.	Decouple components, buffer messages, load leveling, retry mechanisms.	Event-driven architectures, SaaS event integration, scheduled jobs (cron-like).	Orchestrate serverless workflows, ETL jobs, approval processes, microservice orchestration.	High scale apps, gaming, mobile/web apps, session store, IoT, caching.
How it Works	Event triggers Lambda function. Code runs in managed environment and scales automatically.	Receives API calls and routes them to backend integrations (Lambda, HTTP, AWS services).	Publisher sends message to a topic. SNS delivers to all subscribed endpoints.	Producer sends messages to queue. Consumers poll and process messages asynchronously.	Ingests events from AWS services, custom apps, or SaaS. Routes to targets.	Define state machine of steps. Executes and tracks workflow, handles retries and branching.	Data stored in tables. Automatic scaling, high availability, and replication managed by AWS.
Key Features	- No server management - Auto scaling - Support for many runtimes - Event source integrations	- REST, HTTP, WebSocket - Auth (Cognito, IAM, Lambda authorizers) - Throttling & caching - Usage plans	- Pub/Sub messaging - Multiple protocols (SQS, HTTP/S, Email, SMS, Lambda) - Message filtering	- Standard & FIFO queues - Message retention - Delay queues - Dead-letter queues - Long polling	- Default & custom event buses - Event filtering - Scheduling (cron/rate) - SaaS event sources	- Visual workflow - Parallel execution - Error handling & retries - Wait states, choices, Map states	- Key-value & document - Global tables - Auto scaling - Streams & TTL - Encryption
Integration	Works with almost all AWS services.	Integrates with Lambda, HTTP, AWS services, VPC Link.	Integrates with SQS, Lambda, HTTP/S, Email, SMS endpoints.	Integrates with Lambda, SNS, EventBridge, AWS services.	Integrates with 100+ AWS services, SaaS apps, custom targets.	Integrates with Lambda, AWS services, Activity workers, API Gateway.	Integrates with Lambda, DAX, Streams, S3, Kinesis, AppSync.
Scaling	Automatic, up to thousands of concurrent executions.	Automatic scaling to handle traffic.	Virtually unlimited publish/subscribe throughput.	Virtually unlimited message throughput.	High-throughput event routing at scale.	State transitions scale automatically.	Automatic scaling of read/write capacity.
Pricing Model	Pay per request + duration (ms) + memory allocated.	Pay per API call + data transfer.	Pay per publish request and data delivery.	Pay per request (API call).	Pay per event + rules + custom connections.	Pay per state transition.	Pay per read/write capacity unit or on-demand request.
Best For	Event-driven compute without servers.	Exposing and managing APIs securely.	Fan-out messaging and notifications.	Decoupling and buffering messages.	Event-driven architectures and scheduling.	Coordinating complex serverless workflows.	High-performance, serverless NoSQL applications.
Example	Image resize when file uploaded to S3.	Mobile app calls a REST API.	Send email alert to all subscribers.	Process orders asynchronously.	Run task every day at 8 AM or on event from SaaS.	Order fulfillment workflow (validate, pay, ship, notify).	Store user profiles for a social app.

SERVERLESS ARCHITECTURE EXAMPLE



WHEN TO USE WHAT?

- Lambda:** When you need to run code in response to events.
- API Gateway:** When you need to create secure APIs.
- SNS:** When you need to send messages to many subscribers.
- SQS:** When you need to decouple, buffer, and retry messages.
- EventBridge:** When you need event-driven rules or schedules.
- Step Functions:** When you need to coordinate multiple steps.
- DynamoDB:** When you need a fast, scalable NoSQL database.

KEY TAKEAWAYS

- ✓ Serverless = No server management + automatic scaling.
- ✓ Pay only for what you use.
- ✓ Build event-driven, scalable, and resilient applications.
- ✓ Combine these services to build powerful serverless solutions.



CLF-C02 TIP

Know the purpose, use cases, and how these services integrate. Focus on event-driven architectures and cost benefits.

EXAM POINTERS

- ✓ Understand the difference between SNS and SQS.
- ✓ Know when to use Step Functions vs. Lambda.

✓ Know how API Gateway integrates with Lambda.

✓ Understand EventBridge rules and scheduling.

✓ Know DynamoDB pricing and scaling.

OUT OF SCOPE (NOT TESTED AS SERVERLESS SERVICES)

- ✗ Amazon Cognito (Identity / Authentication) — studied in IAM/Identity domain.



AWS EDGE & HYBRID INFRASTRUCTURE COMPARISON (CLF-C02)

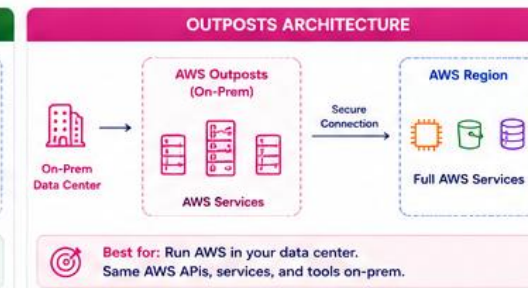
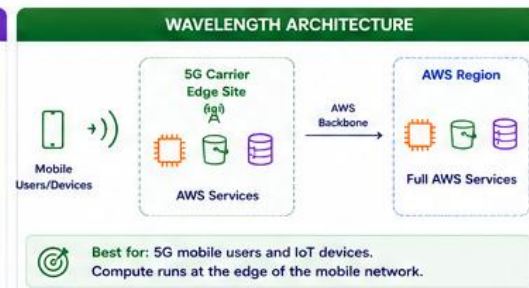
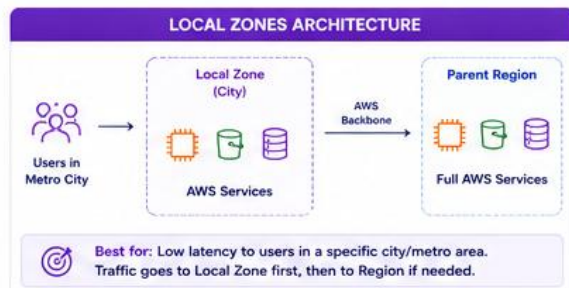
Choose the right AWS compute option closer to your users or on premises.



KEY TAKEAWAY

Local Zones = City-level low latency
Wavelength = 5G network edge
Outposts = AWS in your data center

SERVICE	AWS LOCAL ZONES Low Latency City Compute	AWS WAVELENGTH 5G/Mobile Edge Compute	AWS OUTPOSTS AWS On-Premises
WHAT IT IS	Extension of an AWS Region that places compute, storage, database, and other services closer to end users.	Infrastructure embedded in 5G carrier networks that brings AWS compute to within the mobile network.	AWS hardware and services that you install and run in your own data center.
BEST FOR	Low latency for users in a specific metro area.	Ultra-low latency for mobile users and IoT devices over 5G.	Run AWS services locally for data residency, latency, or operational requirements.
WHERE IT RUNS	In a city/metro area outside an AWS Region, connected to the parent Region.	Inside 5G carrier network locations (edge sites) worldwide.	In your on-premises data center or colocation facility.
HOW IT WORKS	Deploy supported AWS services in Local Zones. Services connect back to the parent Region over the AWS backbone.	AWS services run at the edge of the 5G network. Traffic stays on the mobile network for lower latency.	Outposts rack connects to your network. You run AWS infrastructure locally with the same AWS APIs, tools, and services.
KEY BENEFITS	• Millisecond latency improvement for local users • Built on AWS infrastructure and services • Consistent programming model with AWS • Ideal for local data processing and caching	• Single-digit millisecond latency for mobile users • Process data at the edge of the 5G network • Improve mobile experiences & real-time apps • Reduce backhaul costs	• Run AWS services on-prem with full control • Consistent AWS experience (APIs, tools, IAM, etc.) • Data residency & compliance • Works offline (limited operations)
KEY LIMITATIONS	• Limited number of AWS services • Not for full region isolation (depends on parent Region) • Data leaves Local Zone for some services	• Limited to specific 5G carrier locations • Limited AWS services available • Requires 5G network	• High upfront cost • You manage the physical environment • Limited to Outposts-capable services
USE CASE EXAMPLES	• Low latency gaming • Financial trading applications • Media & content delivery • Real-time analytics near users	• AR/VR on mobile devices • Real-time gaming on mobile • IoT & video analytics in the field • Mobile edge computing	• Manufacturing & industrial workloads • Healthcare & government data residency • Low/No connectivity environments • Lift-and-shift on-prem applications
INTEGRATES WITH	VPC RDS SQS Lambda (and more)	5G Carrier Network EC2 ECS Lambda (and more)	VPC RDS ECS EKS VMware Cloud on AWS (and more)
PRICING MODEL	Standard AWS pricing + Local Zone data transfer fees	Pay for compute + data transfer over 5G network	Upfront Outposts hardware cost + hourly usage fees



HOW TO CHOOSE?	
IF YOU NEED...	CHOOSE
Low latency for users in a specific city/metro	Local Zones
Ultra-low latency for mobile/5G users	Wavelength
Run AWS on-premises in your data center	Outposts
Simplest to get started quickly	Local Zones
Edge compute for mobile network	Wavelength
Data residency / on-prem control	Outposts

KEY DIFFERENCES AT A GLANCE			
Aspect	Local Zones	Wavelength	Outposts
Location	City/metro area	5G carrier edge sites	Your data center
Owned/Managed By	AWS	AWS (in carrier network)	You (AWS hardware)
Connection to Region	AWS backbone	AWS backbone (via carrier)	Private connection / Internet
Primary Users	Internet users in a metro area	Mobile users & IoT devices	On-prem applications & users
Latency	Low (ms)	Ultra-low (single-digit ms)	Very low / local
Offline Capable	No	No	Yes (limited operations)
Service Availability	Limited subset	Limited subset	Limited subset

COMMON EXAM GOTCHAS	
✗	Local Zones are not a separate AWS Region.
✗	Wavelength is only for 5G/mobile use cases.
✗	Outposts does not provide all AWS services.
✗	Data from Local Zone/Wavelength/Outposts may still go to the parent Region.
✗	Outposts requires hardware purchase and capacity planning.



CLF-C02 EXAM TIPS

- Local Zones = "Close to users in a city"
- Wavelength = "Inside the 5G network"
- Outposts = "AWS in your data center"
- Look for keywords: low latency, mobile/5G, on-prem, data residency.
- Match the requirement to the right option.

Amazon S3 Storage Classes (Updated)

● General Purpose
 ● Infrequent Access
 ● Archive
 ● Performance

Feature	 S3 Standard	 S3 Intelligent-Tiering	 S3 Standard-IA	 S3 One Zone-IA	 S3 Glacier Instant Retrieval	 S3 Glacier Flexible Retrieval	 S3 Glacier Deep Archive	 S3 Express One Zone
Designed for	Frequently accessed data	Unknown or changing access patterns (automatic tiering)	Infrequently accessed data	Infrequently accessed, re-creatable data in a single AZ	Long-lived data requiring millisecond retrieval	Long-lived data with minutes to hours retrieval	Long-lived data with hours retrieval, lowest cost	High performance workloads, low latency (single AZ)
Durability	99.99999999% (11 9's)							
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.9%	99.9%	99.95%
Availability SLA	99.9%	99%	99%	99%	99%	99%	99%	99.9%
Min. Storage Duration Charge	None	None	30 days	30 days	90 days	90 days	180 days	None
Min. Billable Object Size*	N/A	128 KB	128 KB	128 KB	128 KB	Varies*	Varies*	N/A
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	None
Retrieval Time	Instantaneous	Instantaneous (when in Frequent or Infrequent tier)	Instantaneous	Instantaneous	Instantaneous (milliseconds)	Expedited: 1–5 mins Standard: 3–5 hrs Bulk: 5–12 hrs	Standard: ~12 hrs Bulk: ~48 hrs	Single-digit milliseconds
Typical Use Cases	 - Big data analytics - Mobile & gaming - Content distribution	 - Data with changing access patterns - Unknown access patterns	 - Backup - Disaster recovery - Long-term data retention	 - Secondary backup - Re-creatable data - Lower-cost DR	 - Data lakes - Analytics - Long-term databases	 - Video archives - Medical imaging - Long-term retention with flexible access	 - Regulatory archiving - Digital preservation - Lowest-cost storage	 - AI/ML - Real-time analytics - High performance apps

*Objects smaller than 128 KB may be billed at the minimum billable size (128 KB) depending on the storage class and object metadata.

Key Takeaways

- ✓ Choose based on access frequency, retrieval speed, and cost.
- ✓ Glacier classes are for long-term archival with retrieval delays.
- ✓ One Zone-IA and Express One Zone store data in a single AZ.
- ✓ Intelligent-Tiering automatically moves data across 5 access tiers.

S3 Intelligent-Tiering Access Tiers (5 Tiers)



S3 Naming Rules

- ✓ 3–63 characters
- ✓ Lowercase letters, numbers, dots (.), hyphens (-)
- ✓ Must start and end with a letter or number
- ✓ No IP address format (e.g., 192.168.1.1)
- ✓ Bucket names must be globally unique



AWS STORAGE SERVICES COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right storage service based on your data type, access pattern, performance, and where it is used.

FEATURE	AMAZON S3 (Simple Storage Service)	AMAZON EBS (Elastic Block Store)	AMAZON EFS (Elastic File System)	AMAZON FSx (Fully Managed File Systems)	AWS STORAGE GATEWAY (Hybrid Storage)
Storage Type	Object Storage	Block Storage	File Storage (NFS)	File Storage (Managed) (Windows / Lustre / ONTAP / OpenZFS)	Hybrid Storage (Cloud + On-Prem)
Best For	Storing and retrieving any amount of data from anywhere	Persistent block storage for EC2 instances	Shared file storage for Linux workloads	High performance file systems for specialized workloads	Extending on-prem storage to AWS (Azure-like hybrid)
Data Access	Objects via API (HTTP/HTTPS)	Raw block level (iSCSI)	Files via NFS	Files via SMB, NFS, or custom protocols (Lustre, etc.)	Files (NFS/SMB) or Blocks (iSCSI) via Gateway
Access Protocols	REST API (HTTP/HTTPS)	iSCSI	NFSv4.1	- Windows File Server: SMB - Lustre: Lustre - ONTAP: NFS, SMB, iSCSI - OpenZFS: NFS, SMB, iSCSI	- File Gateway: NFS, SMB - Volume Gateway: iSCSI - Tape Gateway: iSCSI (VTL)
Attach to EC2	❌ No	✅ Yes (as a volume)	❌ No (mounted on instances)	❌ No (mounted on instances)	✅ Yes (via Gateway)
Scalability	Virtually unlimited	Scale by modifying volume size (up to 16 TiB per volume)	Elastic – scales automatically (Petabyte scale)	Scales to petabytes (depending on file system type)	Scales based on underlying on-prem storage
Performance	Optimized for high throughput, not low latency	Low latency, high IOPS (SSD-backed)	General purpose performance (~ Max I/O mode for higher perf.)	High performance for specialized workloads (HPC, ML, etc.)	Depends on network and on-prem hardware
Durability	99.999999999% (11 9's)	99.999% (within AZ)	99.999999999% (11 9's) across AZs	99.99% (Multi-AZ options available)	Data stored locally and in AWS (Depending on gateway type)
Availability	Highly available across multiple AZs	AZ scoped	Highly available across multiple AZs	Multi-AZ deployment options available	Depends on gateway deployment and configuration
Use Cases	- Backup & restore - Data lakes - Static website hosting - Media storage & distribution - Big data analytics	- OS volumes - Databases - Transaction logs - Applications - Scratch data	- Shared home directories - Content management - Web serving - Container workloads - Dev/Test environments	- Windows file shares - High performance computing (Lustre) - Enterprise apps (ONTAP) - Backup & restore (OpenZFS) - Machine learning workloads	- Hybrid cloud storage - Disaster recovery - Extend on-prem storage to cloud - Backup to AWS - Low latency local access
Key Features	- Versioning, Lifecycle policies - Replication, CORS - Object Lock, Glacier tiers	- Snapshots, AMIs - Encryption - Multi-Attach (io1/io2)	- POSIX-compliant - Access Points - Lifecycle Management	- Fully managed - Built-in data protection - Integration with AD, backup tools	- Local cache of data - Seamless cloud integration - Multiple gateway types
Hybrid / On-Prem Support	✅ Yes (via S3 API, Storage Gateway or Direct Connect/VPN)	❌ No	✅ Yes (via VPN/Direct Connect using NFS)	✅ Yes (via VPN/Direct Connect using SMB/NFS)	✅ Yes (Designed for hybrid cloud environments)
Pricing Model	Pay for storage used + requests + data transfer	Pay for provisioned storage (GB-month) + IOPS (if applicable) + snapshots	Pay for storage used (GB-month) + throughput/mode + requests	Pay for storage, throughput/capacity, and features	Pay for gateway + storage used in AWS + data transfer

QUICK DECISION GUIDE	
Need to store any type of data as objects?	Use S3
Need persistent block storage for EC2?	Use EBS
Need shared file storage for Linux?	Use EFS
Need high-performance file system?	Use FSx
Need hybrid access to on-prem storage?	Use Storage Gateway

STORAGE TYPES EXPLAINED	
	Object Storage – Stores data as objects in a flat namespace (Example: S3).
	Block Storage – Stores data as blocks, like a hard drive (Example: EBS).
	File Storage – Stores data as files in a hierarchical file system (Example: EFS, FSx).

COMMON EXAM SCENARIOS	
✅ Host a static website with images and videos.	– S3
✅ High IOPS for a database on EC2.	– EBS (io1/io2)
✅ Shared Linux filesystem across EC2 instances.	– EFS
✅ Windows file shares for corporate apps.	FSx for Windows File Server
✅ On-prem application needs low latency access to data in AWS.	Storage Gateway (File/Volume)

KEY TAKEAWAYS	
✅ S3 = Object storage, massive scale, accessed via API.	
✅ EBS = Block storage, attached to EC2, low latency.	
✅ EFS = Elastic file system for Linux, shared via NFS.	
✅ FSx = Managed file systems with high performance.	
✅ Storage Gateway = Bridge between on-prem and AWS.	



CLF-C02 TIP: Understand the difference between object, block, and file storage, and when each service should be used based on access patterns and workloads.



AWS DATABASE SERVICES COMPARISON

Choose the right database for your application

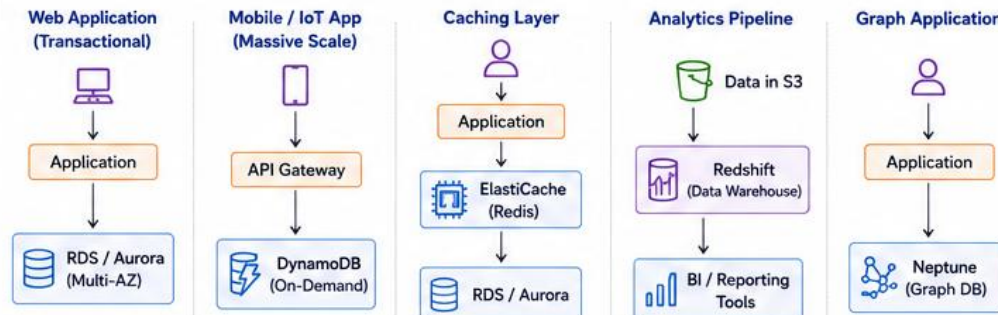
✓ Relational (SQL) 🗃️ Non-Relational (NoSQL) ⌚ Analytics

Feature	Amazon RDS (Relational Database Service)	Amazon Aurora (MySQL / PostgreSQL compatible)	Amazon DynamoDB (NoSQL Key-Value & Document)	Amazon ElastiCache (In-Memory Cache)	Amazon Redshift (Data Warehouse)	Amazon Neptune (Graph Database)
Database Type	Relational (SQL)	Relational (SQL)	NoSQL	In-Memory Cache	Data Warehouse	Graph (Property Graph)
Use Cases	OLTP applications, web apps, content management	High performance apps, web & mobile, SaaS, gaming	Mobile apps, gaming, IoT, sessions, catalog, carts	Caching, sessions, leaderboards, pub/sub, real-time apps	Analytics, reporting, BI, data warehousing	Fraud detection, social networks, recommendations, knowledge graphs
Data Model	Tables (rows & columns)	Tables (rows & columns)	Key-Value, Document	Key-Value, String, Hash, Lists, Sets, Sorted Sets	Columnar (tables)	Nodes, Edges, Properties
Engine Options	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server	Aurora MySQL, Aurora PostgreSQL	—	Redis, Memcached	Redshift (built on PostgreSQL)	Neptune (SPARQL support)
Scalability	Scale up (vertical)	Scale up to 128 TB, 6-way replication	Virtually unlimited (horizontal)	Scale out (read replicas & sharding)	Scale out (compute nodes)	Scale out (read replicas)
Availability	Multi-AZ	Multi-AZ (3 AZs), 6-way replication	Multi-AZ	Multi-AZ (Replication Groups)	Multi-AZ	Multi-AZ (3 AZs)
Durability	Automated backups, S3 storage	6-way replication across 3 AZs	11 9's durability across 3 AZs	Backups to S3 (Snapshots)	Snapshots to S3	Backups to S3, storage replicated across 3 AZs
Best For	Lift & shift relational databases to AWS	High performance with MySQL/PostgreSQL compatibility	Massive scale, low-latency NoSQL workloads	Improve performance of application data	Large-scale analytics & data warehousing	Highly connected data & complex relationships
Key Features	Automated backups, read replicas, Multi-AZ, encryption	High performance, auto-scaling storage, fast failover	Serverless, Single-digit millisecond latency, DAX (optional)	Microsecond latency, persistence (Redis), sub-millisecond performance	Columnar storage, MPP (Massively Parallel Processing), integration with S3	ACID transactions, highly available, supports SPARQL and Gremlin
Pricing Model	Per instance hour + storage + I/O requests + backups	Per instance hour + storage (10GB increments)	Per request (R/W capacity or on-demand) + storage	Per node hour + data transfer	Per node hour + storage (S3) + data transfer	Per instance hour + storage + I/O requests

When to Use Which?

RDS	Existing SQL applications, standard databases, consistent performance
Aurora	High performance, high availability, MySQL/PostgreSQL compatibility
DynamoDB	Serverless, massive scale, flexible schema, low-latency access
ElastiCache	Caching, session store, leaderboard, real-time analytics
Redshift	Business intelligence, reporting, data warehousing, large datasets
Neptune	Graph data, relationship-heavy applications

Common Use Case Architectures



Quick Notes

- ✓ RDS is easiest to manage for traditional databases.
- ✓ Aurora is faster and more available than RDS.
- ✓ DynamoDB is fully serverless and scales automatically.
- ✓ ElastiCache improves read performance and reduces load on databases.
- ✓ Redshift is optimized for analytical workloads, not for OLTP.
- ✓ Neptune is purpose-built for storing and querying graph data.

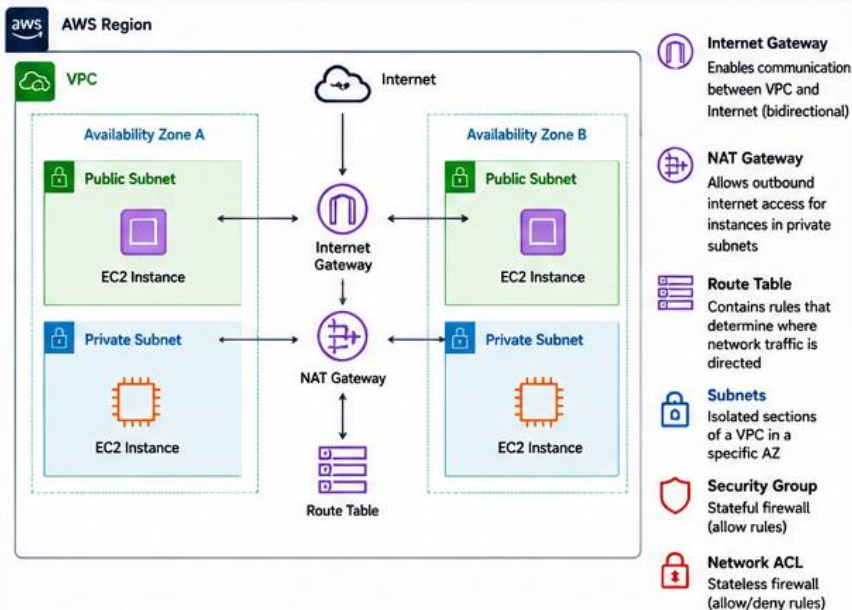


Tip: In the exam, identify the workload (OLTP vs OLAP, relational vs non-relational, real-time vs batch, graph relationships) to choose the right database service.

AWS NETWORKING CHEAT SHEET (CLF-C02)

Core networking services and concepts you must know

1. AMAZON VPC – HIGH LEVEL ARCHITECTURE



Best Practice: Use private subnets for backend resources. Use NAT Gateway (not NAT Instance) for outbound internet access.

2. VPC COMPONENTS

Component	Description
VPC	A logically isolated virtual network in AWS.
Subnet	A range of IP addresses in an AZ (public or private).
Internet Gateway	Allows communication between VPC and the Internet.
NAT Gateway	Enables instances in private subnets to access the Internet (outbound only).
Route Table	Contains rules (routes) that determine where traffic is sent.
Security Group	Stateful firewall at instance level (allow rules).
Network ACL	Stateless firewall at subnet level (allow/deny rules).
Elastic IP	Static IPv4 address for dynamic cloud resources.
VPC Peering	Connects two VPCs privately (using AWS backbone).
VPC Endpoint	Private connection to AWS services without Internet.

3. NETWORK TRAFFIC FLOW

Traffic Type	Source	Destination	Path
Internet-bound (from private subnet)	EC2 in Private Subnet	Internet	EC2 → Route Table → NAT Gateway → Internet Gateway → Internet
Internet-bound (from public subnet)	EC2 in Public Subnet	Internet	EC2 → Route Table → Internet Gateway → Internet
Internet inbound to public subnet	Internet	EC2 in Public Subnet	Internet → Internet Gateway → Route Table → EC2
Private subnet to AWS Service (e.g., S3)	EC2 in Private Subnet	AWS Service (S3, DynamoDB, etc.)	EC2 → VPC Endpoint → AWS Service (Private Connection)
VPC to VPC (Peering)	VPC A	VPC B	VPC Peering Connection → Route Tables

CIDR BLOCK QUICK GUIDE

- IPv4 CIDR example: 10.0.0.0/16 (65,536 IPs)
- Common subnet sizes:

/28	/26	/24	/20	/16
16 IPs	64 IPs	256 IPs	4,096 IPs	65,536 IPs

- Each subnet must be within the VPC CIDR block and in one AZ.

4. VPC CONNECTION OPTIONS

Connection Type	Use Case	Notes
VPC Peering	Connect two VPCs (private)	Non-transitive, same or different account, same Region
AWS Transit Gateway	Connect multiple VPCs and on-premises at scale	Hub-and-spoke model, transitive routing
Site-to-Site VPN	Connect on-premises network to VPC	Uses Internet (IPsec tunnel)
AWS Direct Connect	Dedicated private connection to AWS	More secure, consistent bandwidth, lower latency
VPC Endpoints (Gateway)	Private access to S3, DynamoDB	Adds a route in route table (Gateway Endpoint)
VPC Endpoints (Interface)	Private access to other AWS services	Uses ENIs with private IPs (Interface Endpoint / PrivateLink)

5. AWS LOAD BALANCERS COMPARISON

Load Balancer Type	OSI Layer	Use Cases	Key Features
Application Load Balancer (ALB)	7	Web apps, microservices, HTTP/HTTPS	- Path-based routing - Host-based routing - WebSockets, HTTP/2
Network Load Balancer (NLB)	4	High performance, TLS, TCP/UDP traffic	- Ultra-low latency - Static IP support - Millions of requests/sec
Gateway Load Balancer (GLB)	3	Deploy, scale, and manage third-party appliances	- Transparent to traffic - Supports GENEVE tunneling
Classic Load Balancer (CLB)	4 & 7	Legacy applications (HTTP, TCP)	- Older generation - Not recommended for new applications

6. ROUTE 53 – ROUTING POLICIES

Routing Policy	Best For	Description
Simple	Simple use cases	Uses a single resource record.
Weighted	Load balancing traffic	Split traffic across multiple resources based on weight.
Latency	Reduce latency	Routes to the Region with the lowest latency for the user.
Failover	High availability	Routes to a primary resource; if unhealthy, routes to secondary.
Geolocation	Location-based content	Routes based on the geographic location of users.
Geoproximity	Proximity to resources	Routes based on user location and resource location (bias).
Multivalue Answer	High availability	Returns multiple healthy records (available IPs) to route traffic.



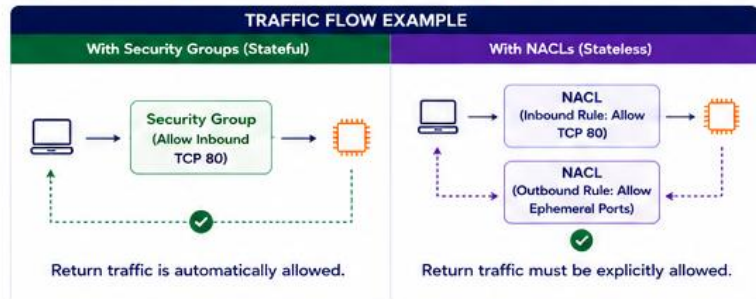
Key Takeaway: Use private subnets + NAT Gateway for outbound access, Security Groups for instance-level control, and the right load balancer for your application.



AWS SECURITY GROUPS vs NACLs - CHEAT SHEET (CLF-C02)

Control traffic to your resources at the instance level (Security Groups) and subnet level (NACLs).

FEATURE	SECURITY GROUPS (SGs)	vs.	NACLs (NETWORK ACLs)	KEY TAKEAWAY
Scope	Instance level (attached to ENI: EC2 instance, Lambda ENI, etc.)		Subnet level (associated with a subnet)	SG = Protect your instance NACL = Protect your subnet
Type	Stateful – Return traffic is automatically allowed		Stateless – Return traffic must be explicitly allowed	SGs are easier to manage. NACLs require rules for both directions.
Rule Evaluation	Rules are evaluated before allowing traffic		Rules are evaluated in order by rule number (from lowest to highest)	NACLs process rules top-down .
Allow / Deny	Only Allow rules Deny is implicit (not allowed by default)		Allow and Deny rules (Deny rules are evaluated first by rule order)	NACLs can explicitly deny traffic.
Default Behavior (Important!)	Implicit Deny (all traffic not allowed by rules is denied)		- Default NACL (created by AWS): Allows all inbound and outbound traffic. - Custom NACL (created by you): Denies all inbound and outbound traffic until rules are added.	EXAM GOTCHA Default NACL = Allows all Custom NACL = Denies all
Rule Granularity	Allow rules only		Allow/Deny rules with rule number (1–32766)	Use NACLs for more granular control.
Traffic Direction	Rules can be Inbound or Outbound		Rules are separate for Inbound and Outbound	NACLs have 2 rule lists per subnet.
Association	Attached to ENI (can be attached to multiple instances)		Associated with a subnet . Automatically applies to all resources in the subnet.	One NACL per subnet (in each AZ).
Applies To	EC2 instances, ENIs, Lambda (in VPC), RDS, etc.		All resources in the subnet (EC2, RDS, ENI, etc.)	NACLs protect everything in the subnet.
Limits	Up to 60 rules per SG (Inbound + Outbound)		Up to 20 rules per NACL (Inbound + Outbound)	NACL rule limit is lower.
Where to Use	Use for instance-level control (e.g., allow SSH to specific instances)		Use for subnet-level control (e.g., block unwanted IP ranges at subnet edge)	Use both for defense in depth.
Change Impact	Changes affect immediately		Changes are not immediate (can take a few minutes to propagate)	Plan ahead when updating NACLs.
Logging	Flow logging supported (VPC Flow Logs)		Flow logging supported (VPC Flow Logs)	Use VPC Flow Logs to troubleshoot.



NACL RULE EVALUATION EXAMPLE

Inbound Rules (Evaluated in Order)

Rule #	Type	Protocol	Source	Action
100	ALLOW	TCP 22	10.0.0.0/16	ALLOW
200	DENY	TCP	0.0.0.0/0	DENY
*	ALLOW	ALL	0.0.0.0/0	ALLOW (Default)

Outbound Rules (Evaluated in Order)

Rule #	Type	Protocol	Destination	Action
100	ALLOW	ALL	0.0.0.0/0	ALLOW
200	DENY	ALL	10.0.0.0/16	DENY
*	ALLOW	ALL	0.0.0.0/0	ALLOW (Default)

HOW TO CHOOSE?

Use Security Groups when you want to:

- ✓ Control traffic to specific instances/ENIs
- ✓ Allow traffic from specific allowed sources (e.g., office IPs)
- ✓ Simplify management (stateful)
- ✓ Follow least privilege at the instance level

Use NACLs when you want to:

- ✓ Block entire IP ranges at the subnet level
- ✓ Add an extra layer of security (defense in depth)
- ✓ Comply with security policies
- ✓ Control traffic for all resources in a subnet

BEST PRACTICES

- ✓ Use Security Groups as the primary control for instance-level security.
- ✓ Use NACLs as a secondary layer for subnet-level protection.
- ✓ Keep rules as simple as possible.
- ✓ Test changes carefully – NACL misconfigurations can block traffic.
- ✓ Use VPC Flow Logs to monitor and troubleshoot.

COMMON EXAM SCENARIOS

- 1 Allow SSH from office IP to EC2 instance → Use Security Group (inbound rule).
- 2 Block unknown IP range from accessing resources in a subnet → Use NACL (Deny rule).
- 3 Allow return traffic automatically → Security Group (stateful).
- 4 Explicitly deny outbound traffic to a range → NACL (Deny outbound rule).
- 5 Apply rules to all resources in subnet → NACL.

CLF-C02 TIP

Think: SG for instances, NACL for subnets.
Both together provide layered security.

KEY DIFFERENCES AT A GLANCE

	Stateful vs. Stateless
	Allow only vs. Allow & Deny
	Instance level vs. Subnet level



AWS VPC CONNECTION OPTIONS COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right way to connect your VPC to other AWS services, on-premises networks, or the internet.

CONNECTION OPTION	INTERNET GATEWAY (IGW)	NAT GATEWAY (NAT GW)	VPC ENDPOINTS (Interface / Gateway)	VPC PEERING CONNECTION	TRANSIT GATEWAY (TGW)	AWS SITE-TO-SITE VPN (VPN CONNECTION)	AWS DIRECT CONNECT (DX CONNECTION)
What it is	Horizontally scaled, redundant VPC component that allows communication between your VPC and the internet.	Managed service that enables instances in private subnets to connect to the internet, but prevents the internet from initiating connections.	Private connection between your VPC and supported AWS services (Interface Endpoint uses ENI; Gateway Endpoint uses route table).	Network connection between two VPCs that enables traffic between them using private IP addresses.	Hub-and-spoke network transit service that connects multiple VPCs and on-premises networks (transitive routing).	Encrypted connection over the internet between your VPC and your on-premises network.	Dedicated network connection between your on-premises network and AWS over private connectivity.
Purpose	Allow resources in public subnets to access the internet.	Allow resources in private subnets to access the internet for outbound traffic only.	Access AWS services privately without using the internet.	Allow resources in different VPCs to communicate as if they are in the same network.	Connect many VPCs and on-premises networks and enable transitive routing.	Extend your on-premises network to your VPC securely.	Provide a more consistent network experience between on-premises and AWS.
Use Cases	- Host a public website - Internet-facing applications - Allow SSH/RDP access (bastion host)	- Private instances need OS updates - Access to AWS APIs or internet (e.g., yum, npm, external APIs) - Software license activation	- Access S3, DynamoDB, CloudWatch, SSM, KMS privately - Reduce NAT gateway costs - Improve security (keep traffic off the internet)	- App tier in one VPC, DB tier in another VPC - Shared services VPC - Multi-account architecture	- Hub-and-spoke network - Centralized egress - Multiple VPCs + on-premises - Simplified routing at scale	- Hybrid cloud connectivity - Extend on-prem data center - Secure access to on-prem resources	- High bandwidth workloads - Consistent network performance - Compliance requirements - Large data transfers
Traffic Flow	VPC ↔ Internet (Bidirectional)	Private Subnet → Internet (Outbound only)	VPC ↔ AWS Services (Private network)	VPC ↔ VPC (Private network)	VPCs ↔ VPCs ↔ On-premises (Transitive)	VPC ↔ On-premises (Bidirectional)	VPC ↔ On-premises (Bidirectional)
Scope	VPC	Per Availability Zone	Per VPC (attached via route tables or ENIs)	Between two VPCs (Non-transitive)	Multiple VPCs and on-premises (Transitive routing)	Between VPC and on-premises (One VPN per connection)	Between AWS and on-premises (One DX per connection)
Attachment	Attach to VPC	In a public subnet (Elastic IP required)	Gateway Endpoint: Route Table Interface Endpoint: ENI (in subnets)	Accept connection in the peer VPC	Attach VPCs and VPNs/DX via TGW attachments	Attach to Virtual Private Gateway (VGW)	Attach to Direct Connect Gateway (DXGW)
Routing	Add 0.0.0.0/0 route to IGW in route table	Add 0.0.0.0/0 route to NAT GW in private subnet route table	Gateway: Prefix list route to endpoint Interface: Private DNS (optional)	Add routes for peer VPC CIDR in both VPC route tables (No transitive routing)	Route tables in TGW control traffic between attachments (Transitive routing)	Add routes for on-prem CIDR in route table via VGW	Add routes for on-prem CIDR in route table via DXGW
Encryption	No (use HTTPS, SSL at application layer)	No (use HTTPS, SSL at application layer)	Yes (AWS private network)	No (use encryption at application layer if needed)	No (use encryption at application layer if needed)	Yes (IPsec VPN encryption)	Yes (private network; can use MACsec encryption)
Bandwidth	Up to 5 Gbps per AZ (scales horizontally)	Up to 45 Gbps per AZ	Scales with AWS network	Up to 50 Gbps	Up to 1.25 Gbps per attachment (Scales horizontally)	Up to 1.25 Gbps per VPN tunnel (max 2 tunnels = 2.5 Gbps)	From 50 Mbps to 100 Gbps+ (Port speed dependent)
Availability	Highly available (Managed by AWS)	Highly available within AZ (Deploy in multiple AZs for HA)	Highly available (Regional service)	Highly available (Managed by AWS)	Highly available (Managed by AWS)	Highly available (use 2 VPN tunnels for redundancy)	Highly available (use redundant connections)
Cost	No charge	Hourly + Data processing	Gateway: No hourly charge (Data processing applies) Interface: Hourly + Data processing	No charge (Data transfer charges apply)	Hourly + Data processing	No hourly charge (Data transfer applies)	Port hourly charge + Data transfer (Local/Regional data charges)
Key Notes	Required for public subnet internet access.	For outbound internet access from private subnets only.	Use Gateway Endpoint for S3 and DynamoDB; Interface Endpoint for other services.	Does not support transitive peering. No overlapping CIDR blocks.	Supports transitive routing (Hub-and-spoke). No overlapping CIDRs.	Use BGP for dynamic routing (optional but recommended).	Provides more consistent performance than internet-based connections.

QUICK DECISION GUIDE

Need internet access for public resources?	Use Internet Gateway
Need outbound internet for private resources?	Use NAT Gateway
Need private access to AWS services?	Use VPC Endpoints
Need communication between two VPCs?	Use VPC Peering (non-transitive)
Need to connect many VPCs and on-prem?	Use Transit Gateway (transitive)
Need secure connection to on-prem over internet?	Use Site-to-Site VPN
Need high-performance private connection to on-prem?	Use AWS Direct Connect

VPC ENDPOINT TYPES



- Gateway Endpoint**
- Used for S3 and DynamoDB only
 - Added to route table
 - No hourly charge



- Interface Endpoint (AWS PrivateLink)**
- Used for most other AWS services
 - Creates ENI in your subnets
 - Supports Private DNS
 - Hourly charge + data processing

EXAM TRAP – TRANSIT GATEWAY vs VPC PEERING



VPC Peering = NON-TRANSITIVE

You cannot route traffic through a peered VPC (often tested!).



Transit Gateway = TRANSITIVE (HUB-AND-SPOKE)



KEY TAKEAWAYS

- ✓ IGW = for inbound internet traffic (public resources).
- ✓ NAT Gateway = for outbound internet traffic (private resources).
- ✓ VPC Endpoints = keep traffic off the internet and improve security.
- ✓ VPC Peering = non-transitive. Use for simple 1:1 VPC communication.
- ✓ Transit Gateway = transitive routing for many VPCs and on-prem.
- ✓ Site-to-Site VPN = encrypted over internet; good for most use cases.
- ✓ Direct Connect = dedicated private link; best performance & consistency.



Remember: Choose the connection method that meets your requirements for security, performance, availability, and cost.



Further Reading: AWS VPC User Guide | AWS Transit Gateway User Guide | AWS Direct Connect User Guide

AWS TRANSIT GATEWAY – CHEAT SHEET (CLF-C02)

Connect multiple VPCs and on-premises networks at scale using a central hub.



CLF-C02 EXAM TRIGGER KEYWORDS

many VPCs, transitive routing, hub-and-spoke, centralized connectivity, on-premises

WHAT IS AWS TRANSIT GATEWAY?

AWS Transit Gateway (TGW) is a network transit hub that you can use to interconnect your VPCs and on-premises networks through a central gateway.

- ✓ Enables transitive routing between connected networks
- ✓ Simplifies network architecture at scale
- ✓ Reduces management overhead and costs
- ✓ Supports high availability and scalability
- ✓ Centralized control over routing and connectivity

KEY CAPABILITIES (TESTED IN CLF-C02)

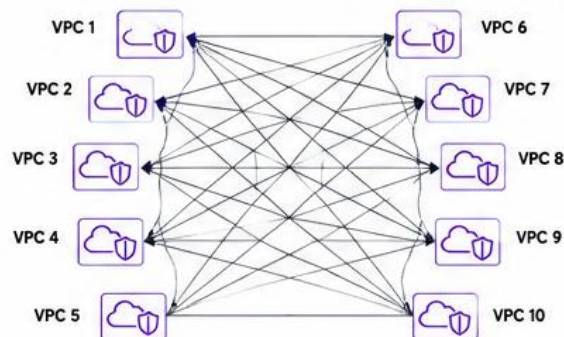
- Hub-and-spoke connectivity for VPCs and on-premises
- Transitive routing between attached networks
- Connect VPCs across accounts and Regions
- Integrates VPN and Direct Connect for hybrid access
- Route tables for fine-grained traffic control
- Scales to thousands of VPCs and on-prem networks

COMMON USE CASES

- Connect many VPCs together at scale
- Connect on-premises networks to multiple VPCs
- Centralize egress/inspection (with GWLB)
- Multi-account network connectivity
- Mergers, acquisitions, and hybrid cloud architectures
- Replace complex mesh of VPC Peering connections

TRANSIT GATEWAY VS VPC PEERING: THE SCALABILITY DIFFERENCE

BEFORE: VPC PEERING (MESH) – COMPLEX & DOES NOT SCALE



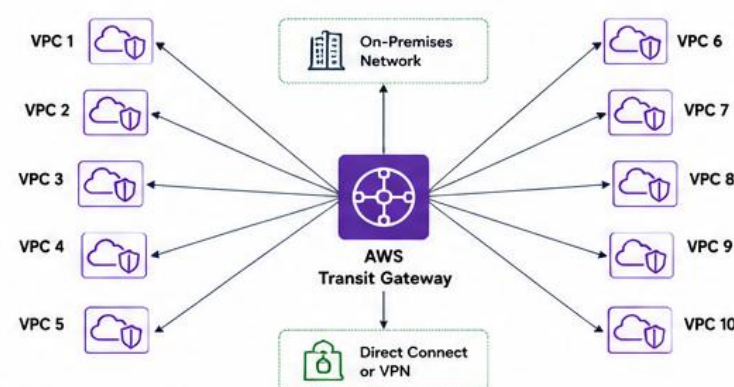
45

INDIVIDUAL CONNECTIONS

For 10 VPCs:
 $n \times (n-1) / 2$
 $= 10 \times 9 / 2$
 $= 45$ connections

- ✗ Complex to manage
- ✗ Harder to scale
- ✗ Does not support transitive routing
- ✗ Higher cost

AFTER: TRANSIT GATEWAY (HUB-AND-SPOKE) – SIMPLE & SCALABLE



10

CONNECTIONS

For 10 VPCs:
 $n = 10$ connections

- ✓ Simple to manage
- ✓ Highly scalable
- ✓ Supports transitive routing
- ✓ Lower cost

HOW IT WORKS

- Create a Transit Gateway in your account.
- Attach VPCs, VPN connections, and/or Direct Connect to the Transit Gateway.
- Create and associate Transit Gateway route tables.
- Add routes to control traffic flow between attachments.
- Traffic is routed transitively between connected networks.

ATTACHMENTS (SUPPORTED)

- VPC Attachments
- VPN Attachments (Site-to-Site VPN)
- Direct Connect Gateway Attachments
- Peering Attachments (to other TGWs)

KEY BENEFITS

- ✓ Centralized connectivity hub
- ✓ Transitive routing between connected networks
- ✓ Scales to thousands of VPCs and on-prem networks
- ✓ Reduces complexity and operational overhead
- ✓ High availability across AZs
- ✓ Fine-grained routing control with route tables

EXAM TIPS (CLF-C02)

Use Transit Gateway when the question mentions:

- ✓ Many VPCs connecting together
- ✓ Transitive routing between VPCs
- ✓ Hub-and-spoke architecture
- ✓ Centralized connectivity or network hub
- ✓ On-premises + multiple VPCs
- ✓ Reducing management complexity



NOT FOR: Connecting only 2 VPCs (use VPC Peering). TGW is for multiple connections at scale.

PRICING (CLF-C02)

Charged for:

- Transit Gateway hourly usage
- Data processing (per GB)
- Attachments (per hour)
- Cross-AZ data transfer

COMMON SCENARIOS

- Multiple VPCs across accounts → TGW
- On-premises + many VPCs → TGW + VPN/DX
- Multiple Regions → TGW Peering
- Centralized inspection → TGW + GWLB

ROUTE TABLES

Transit Gateway route tables control how traffic is routed.

- Multiple route tables supported
- Associate attachments with route tables
- Propagate routes as needed

QUICK RECALL (CLF-C02)

Best for connecting many networks?	Transit Gateway
Supports transitive routing?	Yes
Connects to on-premises?	Yes (VPN/DX)
Two VPCs only?	Use VPC Peering
Architecture pattern?	Hub-and-Spoke



AWS LOAD BALANCERS COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right load balancer for your application.

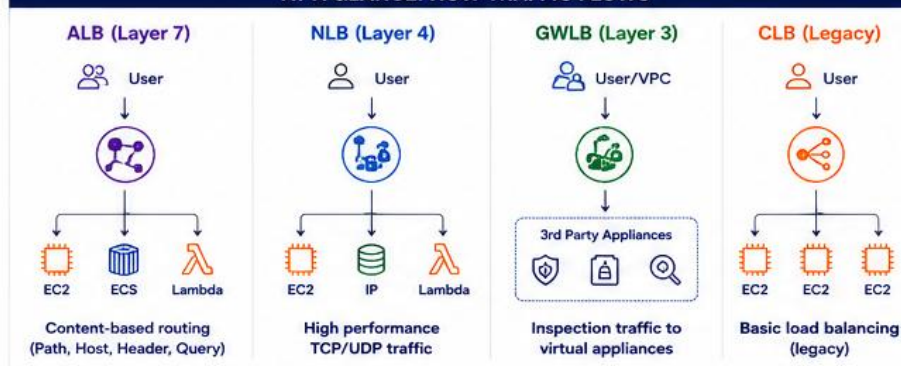


CLF-C02 SCOPE NOTE

Focus on the 4 load balancer types below.
Deep implementation details are out of scope.

FEATURE	APPLICATION LOAD BALANCER (ALB)	NETWORK LOAD BALANCER (NLB)	GATEWAY LOAD BALANCER (GWLB)	CLASSIC LOAD BALANCER (CLB) (Legacy)
OSI Layer	Layer 7 (Application Layer)	Layer 4 (Transport Layer)	Layer 3 (Network Layer)	Layer 4 (Transport Layer)
Protocol Support	HTTP, HTTPS, WebSocket	TCP, TLS, UDP, TCP_UDP	IP (GENEVE)	HTTP, HTTPS, TCP, SSL (deprecated)
Use Cases	- Web applications - Microservices - HTTP/HTTPS traffic	- High performance - Extreme scale - TCP/UDP based applications - Low latency	- Deploy, scale, and manage third-party virtual appliances - Security, compliance, monitoring	- Legacy applications - Basic load balancing needs - HTTP, HTTPS, TCP traffic
Routing	Content-based routing (Path, Host, Header, Query)	IP-based routing (Source IP hash, least outstanding requests)	Transparent routing (Uses GENEVE to forward traffic)	Basic routing (Round Robin)
Target Type	IP, Instance, Lambda, ALB, NLB	IP, Instance, Lambda	GWLB Endpoints (Appliances)	Instance Only
IP Address Type	IPv4	IPv4 and IPv6	IPv4	IPv4
Supported Features	SSL termination, Host/Path routing, WAF, Cognito auth, Fixed response, Redirect, Stickiness	TLS termination (pass-through or terminate), Static IP, Cross-Zone LB, Proxy Protocol v2	Appliance insertion, GENEVE tunneling, Health checks	Basic LB, SSL termination (old), Stickiness
Cross-Zone Load Balancing	✔ Yes (Enabled by default)	✔ Yes (Enabled by default)	✔ Yes (Enabled by default)	✘ No (Disabled by default)
Idle Timeout	1 to 4000 seconds (Default: 60s)	350 seconds (Default)	350 seconds (Default)	1 to 3600 seconds (Default: 60s)
Best For	Modern apps (HTTP/HTTPS), microservices, containers	Real-time apps, gaming, IoT, TCP/UDP, microservices, high performance	Security, compliance, and inspection appliances (firewalls, IDS/IPS, etc.)	Legacy apps (Migrate to ALB/NLB)
Pricing	LCU-based (per hour + LCU usage)	LCU-based (per hour + LCU usage)	Hour + Data processing (GB)	Hour + Data processing (GB)

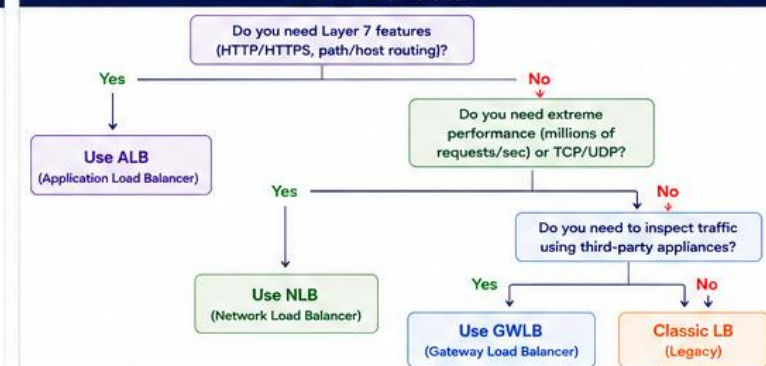
AT A GLANCE: HOW TRAFFIC FLOWS



USE CASES SUMMARY

ALB	- Multi-tenant web applications - Microservices & containers - Path-based routing (e.g., /api, /app) - Integrates with WAF, Cognito, OIDC
NLB	- High-throughput, low-latency apps - Real-time apps, gaming - IoT, TCP/UDP workloads - Static IP requirements
GWLB	- Insert third-party appliances - Security, firewall, IDS/IPS, DLP - Traffic inspection & monitoring
CLB	- Legacy applications - Migrate to ALB or NLB

HOW TO CHOOSE?



QUICK SELECTION GUIDE

ALB	Need advanced routing for web applications? → Use Application Load Balancer
NLB	Need ultra-high performance or static IP? → Use Network Load Balancer
GWLB	Need to inspect traffic with 3rd-party appliances? → Use Gateway Load Balancer
CLB	Have legacy application with basic needs? → Use Classic Load Balancer

KEY TAKEAWAYS

- ✔ ALB is best for modern HTTP/HTTPS applications.
- ✔ NLB is best for high performance, low latency, and TCP/UDP traffic.
- ✔ GWLB is for traffic inspection with third-party virtual appliances.
- ✔ CLB is legacy – use ALB or NLB for new workloads.
- ✔ All support Health Checks and High Availability across AZs.
- ✔ Choose the right LB to meet performance, cost, and feature needs.



TIPS FOR EXAM (CLF-C02)

- HTTP/HTTPS and content-based routing → ALB
- TCP/UDP, high performance, or static IP → NLB
- Firewall/IDS/inspection appliances → GWLB
- Legacy or basic requirements → CLB
- Cross-Zone LB is enabled by default for ALB, NLB, GWLB.
- CLB does not support cross-zone LB by default.

IMPORTANT NOTES

- ✔ You can't convert one load balancer type to another.
- ✔ ALB and NLB support IP targets (no need to register instances).
- ✔ NLB is ideal for extreme scale and low latency workloads.
- ✔ GWLB is used with third-party virtual appliances.
- ✔ CLB is in maintenance mode (no new features).



Remember: Match the load balancer to your application requirements for security, performance, availability, and cost.

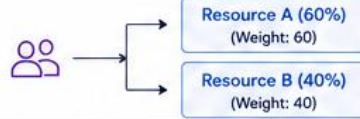


Further Reading: [AWS ALB User Guide](#) | [AWS NLB User Guide](#) | [AWS GWLB User Guide](#) | [AWS CLB User Guide](#)



AMAZON ROUTE 53 ROUTING POLICIES – CHEAT SHEET (CLF-C02)

Route traffic to the right resources for high availability, performance, and reliability.

ROUTING POLICY	DESCRIPTION	USE CASES	HOW IT WORKS	EXAMPLE	HEALTH CHECK SUPPORT
 Simple (Simple Routing)	Routes traffic to a single resource.	- Single resource - Small websites - No failover needed		example.com →  EC2	 No
 Weighted Routing	Routes traffic to multiple resources based on assigned weights.	- Load balancing - Blue/Green deployments - Canary releases		example.com → 60% →  EC2 40% →  EC2	 Yes (recommended)
 Latency Routing	Routes traffic to the AWS Region with the lowest network latency from the user.	- Improve performance - Global applications		example.com → - us-east-1 50 ms - eu-west-1 90 ms - ap-southeast-1 120 ms	 Yes (recommended)
 Failover Routing	Routes traffic to a primary resource. If it fails, traffic is routed to a secondary (resource).	- Disaster recovery - High availability		example.com → - Primary (us-east-1) - Secondary (us-west-2) (Standby)	 Yes (Required)
 Geolocation Routing	Routes traffic based on the geographic location of the user.	- Local content - Legal compliance - Regional services		example.com → - USA → us.example.com - Europe → eu.example.com - Asia → ap.example.com	 Yes (Optional)
 Geoproximity Routing (Latency-based)	Routes traffic based on geographic location and resource proximity (uses AWS Network).	- Local + performance - Regional load balancing		example.com → - us-east-1 (closest) - us-west-2 - eu-west-1	 Yes (Recommended)
 Multi-value Answer Routing	Returns up to 8 healthy records at random.	- Simple load balancing - High availability		example.com → - IP 1 - IP 2 - IP 3 (Up to 8)	 Yes (Required)

HEALTH CHECKS

Route 53 Health Checks monitor your resources and applications.



- Specify endpoint (HTTP, HTTPS, TCP)
- Check interval: 30 sec (minimum)
- Unhealthy threshold: 3 (default)
- Used by most routing policies

WHEN TO USE WHICH POLICY?

Goal	Best Policy
Single resource, simple setup	Simple Routing
Distribute traffic by ratio	Weighted Routing
Lowest latency for users	Latency Routing
Failover to backup if primary fails	Failover Routing
Route based on user location	Geolocation Routing
Route based on location + proximity	Geoproximity Routing
Return multiple IPs for simple load balancing	Multi-value Answer Routing

ROUTE 53 RECORD TYPES (COMMON)

Type	Use
A	Maps domain to IPv4 address
AAAA	Maps domain to IPv6 address
CNAME	Maps domain to another domain name
MX	Mail server records
NS	Name server records
TXT	Text records (SPF, verification, etc.)
SRV	Service location records

QUICK TIPS

- ✓ Use Health Checks with routing policies for high availability.
- ✓ Weighted Routing is great for gradual traffic shifting (Blue/Green, Canary).
- ✓ Latency and Geoproximity use AWS Network data (bandwidth + latency).
- ✓ Failover Routing requires Health Checks.
- ✓ You can combine Route 53 with CloudFront, ALB, or S3 for best results.



KEY TAKEAWAY: Route 53 routing policies help you control how DNS traffic is routed to your resources based on performance, availability, cost, and location.



Learn more: aws.amazon.com/route53



AWS CLOUDFRONT ADVANCED FEATURES (CLF-C02)

CloudFront is not just caching — it's a global edge platform for performance, security, and serverless compute.



KEY TAKEAWAY

CloudFront delivers content, runs code at the edge, and secures applications. Global Accelerator improves network performance at the transport layer (Layer 3/4). Different problems, different tools.

WHAT CLOUDFRONT REALLY DOES

Global Content Delivery	Dynamic Content Acceleration	Edge Compute (Codeless or Serverless)	Security at the Edge	Origin Offload & Optimization
				
Caches and delivers your static and dynamic content from 630+ edge locations worldwide. Benefits • Low latency • High availability • Reduced origin load • Cost effective	Accelerates dynamic requests (HTML, API, personalized content) using optimized network routes and persistent connections. Benefits • Faster TTFB • Better user experience • Origin shielding support	Run code at the edge to customize requests and responses, make real-time decisions, and integrate with AWS services. Includes • Lambda@Edge • CloudFront Functions	Protects applications from threats and attacks close to users. Includes • AWS WAF integration • DDoS protection • HTTPS/TLS • Signed URLs/Cookies • Field-level encryption	Reduces load on origins with caching, compression, connection pooling, and origin shield. Includes • Caching policies • Compression (Gzip/Brotli) • Origin Shield • Keep-alive

EDGE COMPUTE OPTIONS IN CLOUDFRONT

Lambda@Edge	CloudFront Functions
 Run Lambda functions in response to CloudFront events at edge locations. Useful for complex logic with full Node.js runtime. Event Triggers • Viewer Request • Viewer Response • Origin Request • Origin Response Use Cases • URL rewrites, redirects • Auth, A/B testing • Custom headers • Personalization	 Lightweight JavaScript functions that run at the edge with microsecond latency. Best For • Simple logic (modify request/response) • Very low latency • Lower cost than Lambda@Edge Limitations • No network calls (can't call AWS services) • Shorter runtime, smaller code size

CLOUDFRONT vs GLOBAL ACCELERATOR

Aspect	Amazon CloudFront	AWS Global Accelerator
Purpose	Deliver content and accelerate applications (Layer 7 – HTTP/HTTPS)	Improve availability and performance (Layer 3/4 – TCP/UDP)
Best For	Websites, APIs, streaming, downloads, SaaS applications	Real-time gaming, VoIP, IoT, any TCP/UDP or non-HTTP traffic
How It Works	Caches content at edge locations and routes requests to the optimal origin	Routes traffic over AWS's global network to optimal healthy endpoints
OSI Layer	Layer 7 (Application Layer)	Layer 3/4 (Network/Transport Layer)
Protocols	HTTP, HTTPS	TCP, UDP, TCP/UDP
IP Anycast	No	Yes (Static anycast IPs)
Static IPs	No	Yes (2 per accelerator)
Health Checks	At the edge (HTTP/HTTPS)	At edge (TCP/HTTP/HTTPS/Custom)
Use With	S3, ALB, API Gateway, Custom Origins, MediaStore, etc.	ALB, NLB, EC2, ECS, EIP, Service VPC endpoints
Pricing Model	Per GB data transfer + requests	Per GB data transfer + accelerator hours



When to use what?

Use CloudFront when you need:

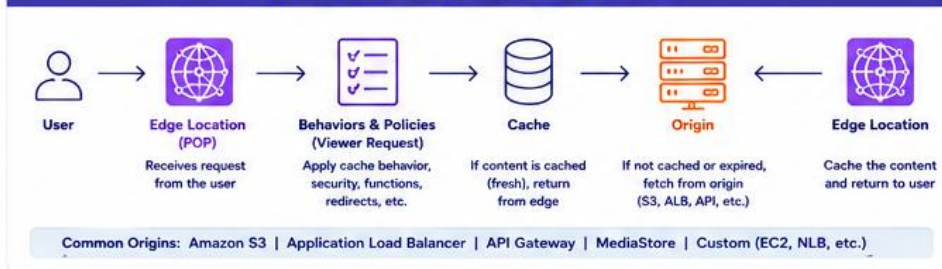
- ✓ Content caching and delivery
- ✓ Support for HTTP/HTTPS
- ✓ Edge compute (Lambda@Edge / Functions)
- ✓ Integration with WAF, Shield, signed URLs/Cookies



Use Global Accelerator when you need:

- ✓ Static anycast IPs
- ✓ Fast, reliable TCP/UDP connectivity
- ✓ Resilient routing across Regions
- ✓ Best performance for real-time apps

CLOUDFRONT ARCHITECTURE OVERVIEW



COMMON USE CASES

	Static website hosting (S3 + CloudFront)
	API acceleration & protection (ALB/API GW + CloudFront)
	Video on demand & live streaming
	Secure file downloads (signed URLs/Cookies)
	Multi-tenant SaaS with edge logic (Lambda@Edge)
	AB testing, geo redirects, bot protection

EXAM TRAPS (DON'T CONFUSE!)

✗	CloudFront does not provide static IPs. (Use Global Accelerator)
✗	CloudFront works at Layer 7 (HTTP/HTTPS), not Layer 4.
✗	Lambda@Edge ≠ Lambda (runs at edge locations, different replication and triggers).
✗	CloudFront Functions cannot call AWS services.
✗	Caching is optional. You can disable caching for dynamic content.
✗	Global Accelerator does not cache content.



KEY TAKEAWAYS

- CloudFront is more than caching: it's a global edge platform for delivery, security, and compute.
- Use Lambda@Edge for complex logic; CloudFront Functions for lightweight, low-latency logic.
- Use Global Accelerator for network-level acceleration with static IPs (Layer 3/4).

FEATURE

CLOUDFRONT

GLOBAL ACCELERATOR

Dynamic Content Acceleration	Yes (optimized HTTP/TCP routes)	Yes (TCP/UDP optimization)
Edge Compute	Yes (Lambda@Edge, CloudFront Functions)	No
Content Caching	Yes	No
Static Anycast IPs	No	Yes

IAM BEST PRACTICES CHEAT SHEET (CLF-C02)

Secure your AWS environment by following IAM best practices

1. PRINCIPLE OF LEAST PRIVILEGE

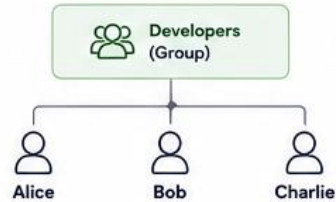
Grant only the minimum permissions necessary to perform a task.



- ✓ Use managed policies when possible
- ✓ Create custom policies for fine-grained access
- ✓ Regularly review and remove unnecessary access

2. USE GROUPS, NOT USERS

Assign permissions to groups, not individual users. Add users to groups.



- ✓ Easier to manage access
- ✓ Add/remove users without changing permissions
- ✓ Ensure consistent permissions across team

3. ENABLE MULTI-FACTOR AUTHENTICATION (MFA)

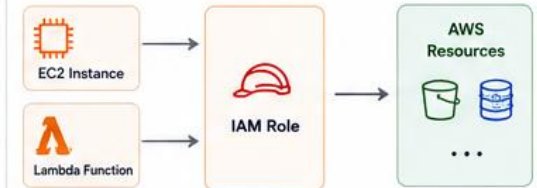
Require MFA for all users, especially those with console access and admin privileges.



- ✓ Protects against compromised passwords
- ✓ Use virtual MFA devices (e.g., Authenticator app) or hardware security keys (FIDO2)
- ✓ Make MFA mandatory for privileged users

4. USE ROLES FOR AWS ACCESS

Use IAM roles for applications, AWS services, and cross-account access.



- ✓ Avoid long-term access keys
- ✓ Roles provide temporary credentials
- ✓ Use instance profiles for EC2, task roles for ECS, etc.

5. SECURE YOUR ACCESS KEYS

	Do	Don't
	Use IAM users only for human access	Don't use access keys for applications
	Rotate access keys regularly	Don't share access keys
	Store access keys securely	Don't hard-code access keys in scripts or code
	Use IAM credentials reporting to monitor unused keys	Don't keep unused access keys active

6. USE PERMISSIONS BOUNDARIES



- ✓ Permissions boundary limits the maximum permissions a user can have
- ✓ Useful to delegate permission management safely
- ✓ Boundary + Identity-based policy = Effective permissions (intersection)

7. MONITOR & REVIEW ACCESS

-  Use AWS IAM Access Analyzer to identify unused permissions and external access.
-  Enable IAM credential report to find unused users and access keys.
-  Use AWS CloudTrail to log and monitor API activity.
-  Regularly review permissions and remove excess access.
-  Set up alerts for root user usage and security-sensitive actions.

8. ROOT USER BEST PRACTICES

- ✓ Do not use the root user for daily tasks.
- ✓ Enable MFA on the root user.
- ✓ Use a strong, unique password.
- ✓ Store root credentials securely and offline.
- ✓ Use IAM users and roles for everyday access.
- ✓ Use AWS Organizations to avoid root for member account access.

9. POLICY BEST PRACTICES

- ✓ Start with least privilege.
- ✓ Use specific actions and resources.
- ✓ Avoid "*" unless absolutely necessary.
- ✓ Use conditions to further restrict access.
- ✓ Validate policies using IAM Policy Simulator.

Example: Good Policy (Least Privilege)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:ListBucket"],
      "Resource": "arn:aws:s3:::my-bucket"
    }
  ]
}
```

10. COMMON IAM ENTITIES

Entity	Description	Use Case
 User	An identity with long-term credentials	Human users accessing AWS
 Group	Collection of users	Apply permissions to multiple users at once
 Role	Identity with temporary credentials	AWS services, applications, cross-account access
 Policy	JSON document that defines permissions	Attach to users, groups, or roles
 Permission Boundary	Sets the maximum permissions an identity can have	Control the highest level of permissions



KEY TAKEAWAY: Secure your AWS environment by following least privilege, using MFA, monitoring access, and auditing regularly.



Remember: Security is a shared responsibility. You are responsible for securing your IAM users and permissions.

AWS SECURITY SERVICES COMPARISON (CLF-C02)

Know what each service does and when to use it.

KEY TAKEAWAY
Layer these services together for defense in depth.

SERVICE	WHAT IT DOES	DETECTS / PROTECTS AGAINST	HOW IT WORKS (KEY POINTS)	TARGET RESOURCES	INTEGRATES WITH	BEST FOR / USE CASES
 GuardDuty Threat Detection	Intelligent threat detection service that continuously monitors for malicious activity and unauthorized behavior.	- Malicious IPs - Suspicious API calls - Credential compromise - Crypto mining - Data exfiltration	✓ Analyzes logs and events: - VPC Flow Logs - CloudTrail Management Events - DNS Logs - EKS Audit Logs (optional) - S3 Data Events (optional)	- AWS Accounts - Workloads (EC2, EKS, etc.) - Containers - Serverless	- CloudWatch - Security Hub - EventBridge - SNS / Lambda	Detecting active threats and anomalous behavior across your AWS environment.
 Inspector Vulnerability Management	Automated security assessment service that identifies vulnerabilities and misconfigurations.	- CVEs on instances - Missing security patches - Network exposure - Insecure configurations	✓ Scans EC2 instances, containers, ECR images, and Lambda functions against best practices and CVE databases.	- EC2 instances - ECR container images - Lambda functions - On-prem (via SSM)	- Security Hub - EventBridge - SSM - SNS / Lambda	Finding vulnerabilities and security misconfigurations before attackers exploit them.
 Security Hub Security Posture Management	Central security dashboard that aggregates, correlates, and prioritizes security findings from multiple AWS services and partners.	- Misconfigurations - Compliance violations - Vulnerabilities - Threats (via integrations)	✓ Aggregates findings into one place ✓ Normalizes and deduplicates findings ✓ Assigns severity and insights ✓ Supports custom insights & standards (CIS, PCI, etc.)	All AWS resources across accounts and Regions	GuardDuty, Inspector, Macie, IAM Access Analyzer, AWS Firewall Manager, and 50+ partners	Getting a comprehensive view of your security posture in one place.
 Macie Data Security & Privacy	Uses ML and pattern matching to discover and protect sensitive data in Amazon S3.	- PII (Personally Identifiable Information) - Financial data - Credentials / Secrets - Sensitive documents	✓ Automatically classifies S3 objects ✓ Alerts on sensitive data exposure ✓ Monitors access to sensitive data	Amazon S3 buckets and objects	- Security Hub - CloudTrail - EventBridge - SNS / Lambda	Discovering and protecting sensitive data at rest in Amazon S3.
 AWS Shield DDoS Protection	Managed DDoS protection service that safeguards applications running on AWS.	- DDoS attacks (Volumetric & Protocol) - L3, L4 attacks - Network floods	✓ Standard: Always on, automatic protection ✓ Advanced: Enhanced DDoS protection + cost protection ✓ Works at the edge of AWS network	- Amazon CloudFront - Route 53 - ELB / ALB / NLB - Global Accelerator	- AWS WAF - CloudWatch - AWS Firewall Manager - Support (Advanced)	Protecting against DDoS attacks to keep applications available.
 AWS WAF Web Application Firewall	Protects web applications from common web exploits that could affect availability, compromise security, or consume resources.	- SQL injection - XSS - Bad bots - IP reputation attacks - OWASP Top 10 threats	✓ Uses rules (managed or custom) ✓ Filters requests at Layer 7 (HTTP/S) ✓ Rate limiting & geo match ✓ Bot control and IP reputation lists	- CloudFront - API Gateway - ALB / AppSync - AWS Verified Access	- Shield - CloudWatch - AWS Firewall Manager - AWS Bot Control	Protecting web apps and APIs from application-layer attacks.
 IAM Access Analyzer Access Analysis	Analyzes IAM policies to identify unintended access and potential security risks.	- Overly permissive policies - External access - Cross-account access - Unused access	✓ Validates access to resources ✓ Checks for public and cross-account access ✓ Generates findings & policy recommendations	- IAM policies - S3 bucket policies - KMS key policies - Resource policies	- Security Hub - AWS Organizations - CloudTrail - EventBridge	Ensuring least privilege and identifying unintended access.

HOW THESE SERVICES WORK TOGETHER



SECURITY BEST PRACTICES

- ✓ Enable all these services.
- ✓ Centralize findings in Security Hub.
- ✓ Review findings and act on fixes.
- ✓ Use least privilege (IAM Access Analyzer).
- ✓ Protect at the edge (Shield + WAF).
- ✓ Encrypt & monitor sensitive data (Macie).



CLF-C02 TIP

Remember the purpose:

GuardDuty = Detect threats
Inspector = Find vulnerabilities
Security Hub = Centralize findings
Macie = Protect sensitive data
Shield = Stop DDoS attacks
WAF = Protect web applications
IAM Access Analyzer = Fix access risks

★ EXAM REMINDERS

- Know what each service does.
- Understand the inputs each service uses.
- Know the difference between threat detection vs vulnerability management vs data protection.
- Expect scenario-based questions!



AMAZON GUARDDUTY vs AMAZON INSPECTOR

Know the difference. Both are security services, but they do very different jobs.



KEY TAKEAWAY

GuardDuty finds active threats.
Inspector finds vulnerabilities.



AMAZON GUARDDUTY Threat Detection



AMAZON INSPECTOR Vulnerability Scanning

Purpose (What it does)	Intelligent threat detection service that continuously monitors for malicious activity and unauthorized behavior.	Automated vulnerability management service that scans your workloads for security vulnerabilities and misconfigurations .
Primary Focus	Detects active threats , suspicious activity, and potential compromises in your AWS environment.	Identifies vulnerabilities (CVEs) and security best practice deviations in your resources.
What It Uses (Data Sources)	Uses logs, events, and threat intelligence - CloudTrail (API activity) - VPC Flow Logs (network traffic) - DNS Logs (Route 53 Resolver DNS Firewall) - AWS CloudWatch Logs - EKS Audit Logs - Threat intelligence feeds CloudTrail VPC Flow Logs DNS Logs CloudWatch Logs	Uses agents and scans workloads directly - Agents installed on EC2, on-prem, containers - Scans operating systems, applications, and libraries - Checks for known CVEs - Evaluates security best practices - Amazon ECR image scanning (vulnerabilities in container images) EC2 / On-prem Amazon ECR
What It Detects / Finds	✓ Malicious IP communications ✓ Unusual API activity (e.g., privilege escalation) ✓ Crypto mining behavior ✓ Data exfiltration patterns ✓ Compromised credentials ✓ Reconnaissance, backdoors, and more DETECTS COMPROMISE Answers: "Is something bad happening right now?"	✓ CVEs in OS, packages, and software ✓ Unpatched vulnerabilities ✓ Security misconfigurations ✓ Network exposure issues ✓ Amazon ECR image vulnerabilities DETECTS CVEs & VULNERABILITIES Answers: "What could be exploited?"
How It Works	- Continuously analyzes data sources using ML, anomaly detection, and threat intelligence. - Generates findings when suspicious or malicious activity is detected.	- Actively scans your workloads on a schedule. - Compares findings against CVE databases and best practices. - Generates findings for vulnerabilities and issues.
Response / Outcome	Produces findings about potential threats. Helps you investigate and respond to active incidents.	Produces findings about vulnerabilities and misconfigurations. Helps you remediate issues before they are exploited.
Integrates With	Security Hub EventBridge CloudWatch SNS / Lambda	Security Hub EventBridge SSM SNS / Lambda

SIDE-BY-SIDE SUMMARY

Aspect	GuardDuty	Inspector
Main Job	Threat detection	Vulnerability scanning
Uses	Logs & events	Agents & scans
Detects	Compromise / Active threats	CVEs / Vulnerabilities
Data Sources	CloudTrail, VPC Flow Logs, DNS Logs, CloudWatch Logs, EKS Audit Logs	EC2, On-prem, Containers, ECR images
Examples	Malicious IP, Crypto mining, Unauthorized API calls	Missing patches, Outdated software, Open ports, ECR CVEs
Question to Ask	Is something bad happening now?	What could be exploited?

KEY DATA SOURCES EXPLAINED

CloudTrail	API activity & account actions
VPC Flow Logs	Network traffic going to/from ENIs
DNS Logs	DNS queries & domain lookups
ECR Scanning	Scans container images for CVEs
EC2 / On-prem	Scans OS, packages, libraries, configs

WHEN TO USE WHAT?

Use GuardDuty when you want to:

- ✓ Detect active threats and suspicious behavior
- ✓ Monitor for breaches and compromise
- ✓ Get alerted to unusual or malicious activity

Use Inspector when you want to:

- ✓ Continuously scan for vulnerabilities
- ✓ Find CVEs and misconfigurations
- ✓ Improve security posture over time

CLF-C02 EXAM TIPS

- ★ GuardDuty = Threat detection using logs/events (CloudTrail, VPC Flow Logs, DNS, etc.)
- ★ Inspector = Vulnerability scanning of workloads (EC2, on-prem, containers, ECR)
- ★ GuardDuty detects compromise. Inspector detects CVEs.
- ★ Both integrate with Security Hub.
- ★ Common exam trap: Don't confuse logs-based threat detection with workload scanning!



AWS CLF-C02 KEYWORDS CHEAT SHEET

Memorize the keywords. Recognize the intent. Pick the best AWS service.

HOW TO USE

- 1 Read the scenario
- 2 Identify the key words (intent)
- 3 Pick the best AWS service



1. COST OPTIMIZATION

Analyze / visualize costs	→	AWS Cost Explorer (cost trends, spend history, charts, breakdown)
Forecast future costs	→	AWS Cost Explorer (forecast, predict, projected cost)
Set budgets / alerts	→	AWS Budgets (budget, threshold, alert, notify)
Estimate cost before launch	→	AWS Pricing Calculator (estimate, pricing, monthly cost)
Get savings recommendations	→	AWS Trusted Advisor (optimize, best practice, save money)
Detect unused / underutilized	→	AWS Trusted Advisor (unused, idle, rightsize, waste)
Check service limits / quotas	→	AWS Trusted Advisor (service limits, quota, request increase)
Compare support plans	→	AWS Support Plans (basic, business, enterprise, TAM)
Manage multiple accounts	→	AWS Organizations (org, consolidated billing, payer)
Apply policies across accounts	→	AWS Organizations (SCPs) (SCP, OU, restrict services)
Tag for cost allocation	→	AWS Cost Allocation Tags (tag, cost allocation, billing tag)

2. PERFORMANCE & SCALABILITY

Handle increased traffic	→	Auto Scaling (scale, traffic spike, demand)
Scale stateless apps	→	Elastic Load Balancing (distribute, balance, multi-AZ)
Low latency to users	→	Amazon CloudFront (edge, CDN, global users)
High performance app	→	Amazon EC2 / EBS / ElastiCache (compute, high performance, IOPS)
Decouple components	→	Amazon SQS (decouple, queue, buffer)
Process large amounts of data	→	Amazon Kinesis (streaming, real-time data)

3. RELIABILITY & AVAILABILITY

Highly available application	→	Multi-AZ (high availability, HA, uptime)
Fault tolerant application	→	Auto Scaling + ELB (fault tolerant, resilient)
Back up data	→	AWS Backup / S3 (backup, snapshots, protect data)
Disaster recovery	→	AWS Backup / Cross-Region (DR, failover, restore, secondary region)
Check service health	→	AWS Health Dashboard (service health, maintenance)

4. SECURITY

Manage users / permissions	→	AWS IAM (user, role, policy, permission)
Temporary access / assume role	→	IAM Role (STS) (assume role, cross-account, temporary)
Encrypt data	→	AWS KMS (encrypt, key, CMK)
Protect web applications	→	AWS WAF (web protection, SQLi, XSS, IP blocking)
Network security	→	Security Group / NACL (firewall, allow, deny, ports)
Detect threats	→	Amazon GuardDuty (threat, anomaly, suspicious)
Scan vulnerabilities	→	Amazon Inspector (vulnerability, scan, CVE)
Store secrets	→	AWS Secrets Manager (secret, credential, password)

5. MONITORING & LOGGING

Monitor metrics / alarms	→	Amazon CloudWatch (metrics, alarm, dashboard, log)
Log API activity / audit	→	AWS CloudTrail (API calls, audit, activity, changes)
Log resource changes / compliance	→	AWS Config (config, compliance, change history)
Collect and store logs	→	Amazon CloudWatch Logs (log group, log stream, aggregation)
Trace requests	→	AWS X-Ray (trace, latency, service map)
Automate on events / schedule	→	Amazon EventBridge (event, rule, schedule, trigger)
Centralize application logs	→	CloudWatch Logs (centralized logging, multiple sources)
Get personal health alerts	→	AWS Health Dashboard (health alert, personal view)

6. DATA MANAGEMENT

Store unstructured data	→	Amazon S3 (object storage, files, documents)
Store relational data	→	Amazon RDS / Aurora (SQL, relational database)
Store NoSQL data	→	Amazon DynamoDB (NoSQL, key-value, fast)
Data warehouse / analytics	→	Amazon Redshift (data warehouse, analytics, BI)
Archive data (long term)	→	Amazon S3 Glacier (archive, cold storage, long term)
Backup databases	→	AWS Backup (backup, restore, snapshots)
Migrate data	→	AWS DataSync / Snowball (data migration, transfer)

7. COMPUTE

Run virtual servers	→	Amazon EC2 (server, instance, VM)
Serverless code execution	→	AWS Lambda (serverless, function, event)
Run containers	→	Amazon ECS / EKS (container, Docker, Kubernetes)
Batch processing jobs	→	AWS Batch (batch, queue, job)
High performance computing	→	Amazon EC2 (Compute Opt.) (HPC, compute optimized)

8. NETWORKING & CONTENT DELIVERY

VPC / private network	→	Amazon VPC (VPC, private, subnet, CIDR)
Connect on-premises to AWS	→	AWS Direct Connect (on-prem, private connection)
Secure site-to-site access	→	AWS VPN (VPN, site-to-site, IPsec)
Route internet traffic	→	Amazon Route 53 (DNS, domain, routing, health check)
Deliver content globally	→	Amazon CloudFront (CDN, global, static content)
Protect against DDoS	→	AWS Shield (DDoS, protection, shield)

9. MIGRATION & HYBRID

Migrate servers	→	AWS Migration Hub / MGN (server migration, lift and shift)
Migrate databases	→	AWS DMS / Schema Conversion Tool (database migration, schema)
Migrate large data	→	AWS Snowball / Snowmobile (large data transfer, offline)
Application discovery	→	AWS Application Discovery Service (discovery, inventory, assessment)

COMMON QUESTION INTENTS (WHAT THEY REALLY WANT)

Reduce cost	Start/ingr/ce, CloudWatch
Improve performance	Storage / services
Ensure availability	Migration services
Secure resources	EventBridge / Automation
Monitor / Troubleshoot	VPC / Network services
Store / Manage data	Analytics / Reporting
Migrate / Move	

KEYWORD INTENT SHORTCUTS

Analyze / Report	Cost Explorer, CloudWatch, Redshift
Alert / Notify	CloudWatch, Budgets, EventBridge
Allow / Deny / Restrict	IAM, Security Group, NACL, SCP
Backup / Restore	Backup, S3, RDS Snapshot
Connect / Link	VPC, Direct Connect, VPN
Distribute / Route	ELB, Route 53, CloudFront
Detect / Monitor	CloudWatch, GuardDuty, Inspector
Encrypt / Key	KMS, Secrets Manager
Store / Save	S3, EBS, RDS, DynamoDB
Scale / Handle Load	Auto Scaling, ELB, ECS/EKS

EXAM SUCCESS TIPS

- ☑ Read the scenario 2-3 times.
- ☑ Underline key words (intent).
- ☑ Match intent → service.
- ☑ Eliminate wrong options.
- ☑ Know the core services deeply.
- ☑ Practice scenario-based questions!

★ **REMEMBER:** Understand the requirement → Identify keywords → Apply AWS best practices → Choose the BEST AWS service!





AWS KMS vs SECRETS MANAGER vs PARAMETER STORE – CHEAT SHEET (CLF-C02)

Choose the right service to encrypt data, store secrets, and manage configuration.



CLF-C02 SCOPE NOTE
All three services are tested. Know their purpose, features, and key differences.

FEATURE	AWS KMS (Key Management Service)	AWS SECRETS MANAGER	SSM PARAMETER STORE
 Purpose	Encryption key management service. Used to encrypt data and control access to keys.	Store, manage, and automatically rotate secrets such as database passwords, API keys.	Store and manage configuration data, parameters, and basic secrets.
 What it Stores	Encryption keys (CMKs)	Secrets (DB passwords, API keys, certs, tokens, etc.)	Parameters (configuration values, AMI IDs, feature flags, connection strings, basic secrets)
 Auto-Rotation	N/A	✔ Yes (built-in rotation for supported services e.g., RDS, Redshift, DocumentDB)	✘ No (manual rotation required)
 Encryption	Provides encryption/decryption using keys. Can be used by other AWS services.	Secrets are encrypted using KMS CMK (default <code>aws/secretsmanager</code>).	SecureString parameters are encrypted using KMS CMK.
 Integration	Integrated with most AWS services (EBS, S3, RDS, Lambda, etc.)	Integrated with RDS, Redshift, DocumentDB, ElastiCache, Lambda, ECS, EC2, and more.	Integrated with EC2, Lambda, CodeBuild, CodePipeline, AppConfig, and more.
 Versioning	✔ Yes (automatic key versions)	✔ Yes (automatic)	✔ Yes
 Access Control	IAM policies + Key policies	IAM policies + Resource policies	IAM policies
 Cost Model (CLF-C02)	Per key per month (\$1.00) + API requests (encryption/decryption)	~\$0.40 per secret per month + API requests	✔ Standard parameters: Free Advanced parameters: ~\$0.05 per parameter per month + API requests
 Best For	Encrypting data and managing encryption keys. Control who can use the keys.	Storing and rotating secrets securely. If the question mentions "auto-rotation" → Use Secrets Manager.	Storing configuration values and basic secrets. If the question mentions "store configuration" → Use Parameter Store.
 Examples	Encrypt S3 objects, EBS volumes, (RDS storage, Lambda, env vars, etc.	Database passwords, API keys, OAuth tokens, third-party service credentials.	App config (e.g., log level), AMI IDs, DB endpoints, feature flags, simple credentials.

WHEN TO USE WHICH?



Use AWS KMS when you need to:

- Encrypt data (at rest or in transit)
- Manage encryption keys
- Control key usage and access
- Encrypt across many AWS services

Keywords: encrypt, decrypt, CMK, key policy, customer managed key



Use AWS Secrets Manager when you need to:

- Store secrets (passwords, API keys, tokens)
- Automatically rotate secrets
- Integrate with databases and services

Keywords: secret, password, API key, token, auto-rotation, rotate



Use SSM Parameter Store when you need to:

- Store configuration values
- Store simple or non-rotating secrets
- Use hierarchical parameters
- Keep costs low (free tier available)

Keywords: configuration, parameter, setting, value, AMI ID, feature flag, non-secret

HOW THEY WORK (AT A GLANCE)



Application / Service → AWS KMS → Encrypt / Decrypt → Data / AWS Services



Application / Service → AWS Secrets Manager → Secret (Auto-Rotated) → Database / Service



Application / Service → SSM Parameter Store → key-value → Parameter (Config Value) → Application / EC2 / Lambda

KEY EXAM TAKEAWAYS

- ✔ If the question says "auto-rotation" → Secrets Manager.
- ✔ If the question says "store configuration" → Parameter Store.
- ✔ If the question says "encrypt data/keys" → KMS.
- ✔ KMS = encryption keys, not for storing secrets.
- ✔ Secrets Manager = secrets with built-in rotation.
- ✔ Parameter Store = configuration and basic secrets.
- ✔ All three use IAM for access control.
- ✔ Secrets Manager and Parameter Store (SecureString) use KMS for encryption.

PRICING QUICK REFERENCE (CLF-C02)			QUICK RECALL (CLF-C02)		EXAM TIP (MUST REMEMBER)
AWS KMS	AWS SECRETS MANAGER	SSM PARAMETER STORE	Scenario / Requirement	Use This Service	 ✔ KMS = Keys (encrypt data) ✔ Secrets Manager = Secrets (rotate automatically) ✔ Parameter Store = Parameters (store configuration) ✔ Right tool = Security + Simplicity + Cost savings
\$1.00 per CMK per month \$0.03 per 10,000 API requests - No charge for automatic key rotation	~\$0.40 per secret per month \$0.05 per 10,000 API requests - No charge for automatic rotation	Standard Parameters (Free) 10,000 parameters 4 KB size limit Advanced Parameters (Paid) ~\$0.05 per parameter per month 8 KB size limit - Policies, expiration, more	Need to encrypt data or manage keys	AWS KMS	
			Need to store DB password with auto-rotation	Secrets Manager	
			Need to store API key (rotate automatically)	Secrets Manager	
			Need to store configuration like AMI ID or log level	SSM Parameter Store	
			Need the cheapest option for config values	SSM Parameter Store (Standard)	
			Need fine-grained access control to secrets	Secrets Manager	



AWS COMPLIANCE & SECURITY STANDARDS – CHEAT SHEET (CLF-C02)

Understand compliance frameworks, AWS compliance tools, and the shared responsibility for compliance.



CLF-C02 SCOPE NOTE
You are responsible for compliance in the cloud. AWS provides tools, reports, and services to help.

KEY COMPLIANCE FRAMEWORKS TESTED IN CLF-C02

FRAMEWORK	PURPOSE	WHO IT'S FOR	KEY FOCUS AREAS	EXAMPLE AWS SERVICES THAT HELP
HIPAA (Health Insurance Portability and Accountability Act)	Protect patient health information (PHI) in the cloud.	Healthcare providers, health plans, healthcare clearinghouses, and their business associates.	Privacy, security, access controls, encryption, auditing, and safeguards for electronic PHI (ePHI).	AWS Identity and Access Management (IAM), Amazon S3 (encryption), Amazon CloudWatch, AWS CloudTrail, AWS KMS, AWS Config.
PCI DSS (Payment Card Industry Data Security Standard)	Protect payment card data and reduce credit card fraud.	Organizations that store, process, or transmit cardholder data.	Network security, encryption, access control, vulnerability management, monitoring, and policy management.	Amazon VPC, AWS WAF, AWS Shield, AWS KMS, Amazon GuardDuty, AWS Security Hub, AWS CloudTrail, AWS Config.
SOC 1, SOC 2, SOC 3 (Service Organization Controls)	Assure customers about the controls at a service organization.	Businesses using cloud services (their auditors may request these).	Security, availability, processing integrity, confidentiality, privacy (SOC 2), and financial reporting (SOC 1).	AWS Artifact (download reports), AWS Audit Manager (evidence collection), AWS Organizations (control governance).
ISO 27001 (Information Security)	International standard for information security management systems (ISMS).	Any organization seeking information security certification.	Risk management, information security policies, access control, cryptography, physical security, and incident management.	AWS KMS, AWS IAM, AWS Config, AWS CloudTrail, Amazon Macie, AWS Security Hub, AWS Audit Manager.
GDPR (General Data Protection Regulation)	Protect personal data and privacy of EU residents.	Organizations that collect or process personal data of individuals in the EU.	Lawful processing, data minimization, consent, access rights, data portability, breach notification.	Amazon S3 (encryption), AWS KMS, Amazon Macie, AWS CloudTrail, AWS Data Lifecycle Manager, AWS Control Tower.

AWS COMPLIANCE TOOLS (TESTED IN CLF-C02)

AWS ARTIFACT



On-demand access to AWS compliance reports and agreements.

WHAT YOU CAN DO

- ✓ Download AWS security and compliance reports.
- ✓ Access business associate addendums (BAA) for HIPAA.
- ✓ Review ISO, SOC, PCI, FedRAMP, and more.
- ✓ Get AWS service-specific compliance information.

EXAM TIP
AWS Artifact = Download compliance reports and agreements on demand.

AWS AUDIT MANAGER



Automate audit evidence collection and simplify compliance at scale.

WHAT YOU CAN DO

- ✓ Automate collection of evidence from AWS services.
- ✓ Map controls to frameworks (ISO, SOC, PCI, etc.).
- ✓ Continuously monitor and assess compliance.
- ✓ Generate audit reports and reduce manual effort.

EXAM TIP
AWS Audit Manager = Automate ongoing evidence collection and compliance assessments.



EXAM TIP

You are responsible for your own compliance in the cloud. AWS is responsible for security OF the cloud; you are responsible for security IN the cloud and for your data and workloads. AWS provides the reports and tools to prove their part (e.g., via AWS Artifact).

SHARED RESPONSIBILITY MODEL FOR COMPLIANCE



AWS (RESPONSIBLE FOR)
Security OF the Cloud

- Global infrastructure (Regions, AZs, Edge)
- Physical security of data centers
- Hardware, networking, and virtualization
- Managed services availability and security
- Compliance certifications and attestations (You can download via AWS Artifact)



YOU (CUSTOMER) (RESPONSIBLE FOR)
Security IN the Cloud

- Your data classification and protection
- IAM users, roles, policies, and permissions
- Your applications, instances, and workloads
- Network configuration and security groups
- Logging, monitoring, and threat detection
- Compliance with laws and regulations

Compliance is a shared effort — know your responsibilities.

COMPLIANCE LIFECYCLE WITH AWS



1. PLAN
Identify regulatory requirements (HIPAA, PCI, SOC, ISO, GDPR, etc.)



2. DESIGN
Design your AWS architecture and controls to meet requirements.



3. IMPLEMENT
Use AWS services and security best practices to implement controls.



4. MONITOR
Continuously monitor and collect evidence with AWS services and Audit Manager.



5. REPORT
Validate and report compliance. Use AWS Artifact for reports.

EXAMPLE FRAMEWORK MAPPING

Framework	Common AWS Services Used
HIPAA	IAM, KMS, CloudTrail, CloudWatch, Config, S3, GuardDuty, Audit Manager
PCI DSS	VPC, WAF, Shield, KMS, GuardDuty, Security Hub, CloudTrail, Audit Manager
SOC 2	CloudTrail, Config, Security Hub, CloudWatch, Organization, Audit Manager
ISO 27001	IAM, KMS, CloudTrail, Config, Macie, Security Hub, Audit Manager
GDPR	S3 (encryption), KMS, Macie, CloudTrail, Data Lifecycle Manager, Control Tower

WHY COMPLIANCE MATTERS (BUSINESS VALUE)

- ✓ Build customer trust and confidence
- ✓ Meet legal and regulatory requirements
- ✓ Reduce risk of fines and penalties
- ✓ Enable market access and industry requirements
- ✓ Improve security posture and governance
- ✓ Support business growth and global expansion
- ✓ Improve operational efficiency with automation (Audit Manager)



Compliance is not just about avoiding penalties — it's about building secure, trustworthy, and resilient business.

QUICK RECALL (CLF-C02)



Where do you download AWS compliance reports? **AWS Artifact**



Which service helps automate audit evidence collection? **AWS Audit Manager**



Who is responsible for compliance in AWS? **The Customer**



AWS responsibility? **Security OF the Cloud**



Customer responsibility? **Security IN the Cloud**

HIGH-FREQUENCY EXAM TRAPS

- ✗ AWS is compliant. Incorrect. You are responsible for your compliance.
- ✗ AWS Artifact is for monitoring resources. Incorrect. It is for compliance reports & agreements.
- ✗ SOC reports are for AWS internal use only. Incorrect. Customers/auditors can request them.
- ✗ HIPAA automatically enabled in AWS. Incorrect. You must architect for HIPAA compliance.
- ✗ Compliance is the same as security. Incorrect. Compliance = meeting requirements.

KEY TAKEAWAYS

- ★ Know the compliance frameworks: HIPAA, PCI DSS, SOC 1/2/3, ISO 27001, GDPR.
- ★ Use AWS Artifact to download compliance reports and BAAs.
- ★ Use AWS Audit Manager to automate and track compliance continuously.
- ★ You are responsible for your compliance in the cloud.
- ★ Compliance reduces risk, builds trust, and enables business growth.



FURTHER READING: AWS Artifact User Guide | AWS Audit Manager User Guide | AWS Compliance Programs | AWS Shared Responsibility Model | AWS Well-Architected Framework – Security & Governance Pillars



AWS IDENTITY SERVICES – CHEAT SHEET (CLF-C02)

Authentication, authorization, and directory services on AWS.

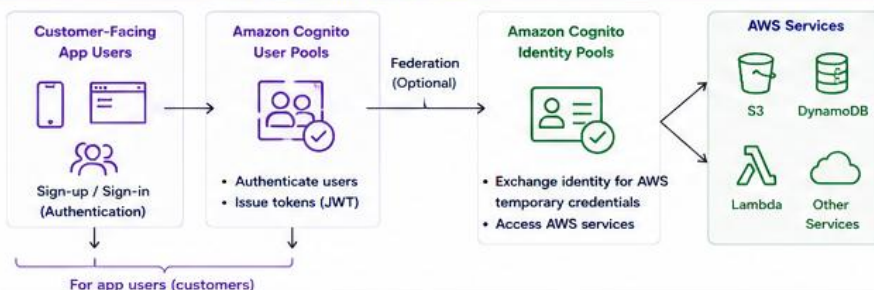


CLF-C02 SCOPE NOTE

Focus: Cognito, AWS Directory Service, IAM Identity Center (SSO). Deep implementation of AD is out of scope.

SERVICE	AMAZON COGNITO USER POOLS	AMAZON COGNITO IDENTITY POOLS	AWS DIRECTORY SERVICE (Managed Microsoft AD)	IAM IDENTITY CENTER (successor to AWS SSO)
Purpose	User authentication — sign-up, sign-in, issue tokens.	Authorize users to access AWS resources with temporary AWS credentials.	Managed Microsoft Active Directory in AWS or connected to on-prem AD.	Single sign-on (SSO) to access AWS accounts and cloud applications.
Who It's For	Customer-facing app users (Internet / Mobile applications)	App users who need AWS access to call AWS services	Enterprise IT admins (Workforce / Organizations)	Employees / Workforce (Internal users)
What It Does	Manages user directory and authentication. Provides tokens (JWT) for your app.	Maps identities (from Cognito, social, SAML, or OIDC) to IAM roles for AWS access.	Provides managed AD, domain join, group policies, and LDAP/Kerberos.	Provides SSO to AWS accounts and thousands of cloud apps.
Key Features	• Sign-up / sign-in / sign-out • Multi-factor authentication (MFA) • Social IdP (Google, Facebook, Apple, etc.) • Custom branding (hosted UI) • JWT / OAuth 2.0 / OpenID Connect	• Get temporary AWS credentials • Supports unauthenticated identities • Access AWS services (S3, DynamoDB, etc.) • Fine-grained access via IAM roles	• Managed Microsoft AD (Simple AD or AD Connector) • Domain join EC2 instances • Group Policy support • LDAP / Kerberos authentication • Integrate with on-prem AD	• SSO to multiple AWS accounts • SSO to cloud applications (SAML 2.0) • Centralized user and permission management • Permission sets and account assignments • MFA and multi-account access
Integrates With	Your applications, API Gateway, ALB, SAML / OIDC IdPs	Cognito User Pools, social IdPs, SAML / OIDC, Developer-authenticated identities	EC2, AWS Managed Microsoft AD, RDS (SQL Server), WorkSpaces, AWS apps	AWS Organizations, AWS accounts, SaaS applications, SAML apps
Tokens / Access	ID Token, Access Token, Refresh Token (JWT)	AWS STS temporary credentials (Access Key ID, Secret, Session Token)	Kerberos tickets, NTLM, LDAP authentication	Short-term credentials via AWS STS, SAML assertions.
Pricing Model	Per MAU (Monthly Active Users) + optional features	No charge for identities. Pay for AWS service usage.	Hourly charge based on directory size/type + AWS infrastructure usage	No additional charge for SSO. Pay for AWS service usage.
Exam Keywords	Sign-up, sign-in, authentication, JWT, user directory, hosted UI	Authorize, temporary credentials, IAM roles, unauthenticated identities	Managed AD, domain join, LDAP, Kerberos, Group Policy	SSO, permission sets, AWS accounts, SAML, cloud apps

HOW THEY WORK TOGETHER



KEY DISTINCTIONS (EXAM IMPORTANT)

Cognito User Pools	Customer-facing applications. Handles user authentication and issues tokens.
Cognito Identity Pools	Provides AWS access (temporary credentials) to authenticated or unauthenticated identities.
AWS Directory Service	Enterprise directory (AD) for workforce access to AWS resources and applications.
IAM Identity Center	Single sign-on for employees to access AWS accounts and cloud applications.

COMMON USE CASES

Mobile/Web App Authentication	Use Cognito User Pools.
App Users Accessing AWS Services	Use Cognito Identity Pools.
Extend Microsoft Active Directory to AWS	Use AWS Directory Service (Simple AD or AD Connector).
Employee Access to AWS & SaaS Apps	Use IAM Identity Center.

EXAM TIP (MUST REMEMBER)



- ✓ Cognito = Customer identities for your applications.
- ✓ IAM Identity Center = Workforce identities for employees.
- ✓ Identity Pools give AWS access (credentials), not authentication.
- ✓ You are responsible for your users, authentication, and authorization.
- ✓ AWS provides the services and tools to help.

QUICK RECALL (CLF-C02)

Question Clue	Use This Service
Sign-up / sign-in for app users	Cognito User Pools
Get AWS credentials for app users	Cognito Identity Pools
Managed Microsoft Active Directory	AWS Directory Service
SSO to AWS accounts and SaaS apps	IAM Identity Center

KEY TAKEAWAYS

- ★ Cognito User Pools authenticate users and issue tokens.
- ★ Cognito Identity Pools authorize identities to access AWS services.
- ★ Directory Service provides managed AD for enterprise use.
- ★ IAM Identity Center provides SSO across accounts and cloud apps.
- ★ Choose the right service based on who the users are and what access they need.

AWS COMPLIANCE & REPORTING

Use **AWS Artifact** to download compliance reports (SOC, ISO, PCI, HIPAA, etc.).
Use **AWS Audit Manager** to automate audit evidence collection.



AWS Artifact
Compliance reports & agreements



AWS Audit Manager
Continuous audit evidence collection



FURTHER READING: [Amazon Cognito Developer Guide](#) | [Amazon Cognito Identity Pools Developer Guide](#) | [AWS Directory Service User Guide](#) | [IAM Identity Center User Guide](#) | [AWS Artifact User Guide](#) | [AWS Audit Manager User Guide](#)

AWS SYSTEMS MANAGER (SSM) – CHEAT SHEET (CLF-C02)

Operations management, secure access, and configuration management at scale.

CLF-C02 SCOPE NOTE
SSM is a core service for managing EC2 and resources at scale. Learn the 3 must-know capabilities below.

OVERVIEW



AWS Systems Manager (SSM)

Unified interface to view and operate AWS resources at scale.

- No servers to manage
- Works through AWS APIs and IAM (no inbound ports)
- Pay-as-you-go pricing
- Integrates with CloudWatch, IAM, EC2, and more

KEY CAPABILITIES (TESTED IN CLF-C02)

SESSION MANAGER

Secure access to instances

Replace SSH/RDP. Connect to EC2 instances securely without opening ports or using a bastion host.

Key Benefits

- ✓ No inbound ports (22/3389) required
- ✓ No bastion host needed
- ✓ Encrypted using AWS KMS
- ✓ Access control with IAM
- ✓ Full audit logs to CloudTrail and S3/CloudWatch Logs
- ✓ Works over HTTPS (port 443 outbound only)

Common Use

How do you manage EC2 instances at scale without opening port 22?
Answer: Use SSM Session Manager. Connect from the AWS Console or AWS CLI.



PARAMETER STORE

Store configuration & secrets

Store and manage configuration data, passwords, API keys, and other secrets.

Key Benefits

- ✓ Hierarchical structure
- ✓ SecureString (encrypted with KMS)
- ✓ Versioning and labels
- ✓ Access control with IAM
- ✓ Cheaper than Secrets Manager for non-rotating secrets
- ✓ Integrates with Lambda, EC2, and more

Common Use

Store database passwords, API keys, application settings, feature flags.

String
(plain text)
/app/dev/db/host

SecureString
(encrypted)
/app/dev/db/password



PATCH MANAGER

Automate OS patching

Automate the process of patching operating systems and applications at scale.

Key Benefits

- ✓ Automated patch compliance
- ✓ Schedule patch windows
- ✓ Supports Amazon Linux, Windows, RHEL, Ubuntu, and more
- ✓ Maintenance windows
- ✓ Patch baselines (approve/deny patches)
- ✓ Patch reports and compliance status

Common Use

Keep hundreds or thousands of EC2 instances fully patched and compliant.

 Define Maintenance Window

 Create Patch Baseline

 Run Patch Operation

 Review Compliance Reports

OTHER IMPORTANT SSM FEATURES

- Run Command**
Run scripts or commands on one or more instances (e.g., configuration, troubleshooting).
- State Manager**
Automate configuration tasks and enforce the desired state of your resources.
- Automation**
Create runbooks to automate common operational and deployment tasks.
- Inventory**
Collect and view detailed inventory of your AWS resources.
- Documents**
Store and share documents (SSM documents) for automation and command execution.
- Maintenance Windows**
Schedule and centralize operational tasks during approved time windows.

SSM SESSION MANAGER: HOW IT WORKS



- No inbound access to the instance.
- SSM Agent (pre-installed on Amazon Linux 2, Windows AMI, or installed on others) maintains an outbound connection to SSM.
- Traffic is encrypted and logged.

SSM AGENT REQUIREMENTS

SSM Agent must be running on the instance.

Amazon Linux 2 / 2023	Pre-installed
Windows Server (2016 and later)	Pre-installed
Amazon Linux / RHEL / Ubuntu	Install SSM Agent
Other OS	Install SSM Agent

IAM Permissions Required

Instance profile/role must allow:

- ssm:*, ssmmessages:*, ec2messages:*, logs:CreateLogStream, logs:PutLogEvents

COMPARISON: ACCESS METHODS

Method	Inbound Ports	Bastion Host	IAM Control	Audit Logging	Recommendation
SSH (Traditional)	Port 22	Often Needed	Limited	Limited	Avoid
RDP (Traditional)	Port 3389	Often Needed	Limited	Limited	Avoid
SSM Session Manager	None	Not Needed	Strong	Full (CloudTrail)	Recommended

PRICING (CLF-C02)

All core SSM capabilities are distributed under the "Systems Manager" pricing.

Pay-as-you-go based on:

- Number of managed instances
- Number of API calls
- Data transferred / stored

EXAM TIPS (CLF-C02)

- ✓ Use SSM Session Manager to access EC2 instances without opening ports.
- ✓ Use Parameter Store for configuration and non-rotating secrets (cheaper than Secrets Manager).
- ✓ Use Patch Manager to automate and track OS patching at scale.
- ✓ Make sure the SSM Agent is installed and the instance role has required permissions.

QUICK RECALL (CLF-C02)

- Access EC2 without SSH/RDP? → SSM Session Manager
- Store app config / secrets? → SSM Parameter Store
- Automate OS patching? → SSM Patch Manager
- Run script/command on instances? → SSM Run Command
- Automate operational tasks? → SSM Automation

REMEMBER: SSM helps you manage resources securely, at scale, and with no inbound access. It's a key service for operational excellence and security.

FURTHER READING: AWS SSM User Guide | SSM Session Manager Guide | SSM Parameter Store Guide | SSM Patch Manager Guide

EVENT-DRIVEN AWS ARCHITECTURE (CLF-C02)

Build decoupled, scalable, and resilient systems that respond to events in real time.



KEY TAKEAWAY

EventBridge = The central event router.
SNS = Fan-out to many. SQS = Buffer & decouple.
Lambda = Compute. Step Functions = Orchestrate.

SERVICE	AMAZON EVENTBRIDGE	AMAZON SNS	AMAZON SQS	AWS LAMBDA	AWS STEP FUNCTIONS
WHAT IT IS	Serverless event bus service that routes events from AWS services, SaaS apps, and custom sources.	Fully managed pub/sub messaging service for fan-out notifications to multiple endpoints.	Fully managed message queue service for decoupling and buffering between systems.	Serverless compute service that runs code in response to events without managing servers.	Serverless workflow orchestration service that coordinates multiple steps in a state machine.
PRIMARY PURPOSE	Route and transform events to the right targets based on rules.	Broadcast messages to many subscribers (fan-out).	Decouple producers and consumers with durable message queues.	Run code to process events and perform business logic.	Orchestrate and coordinate complex workflows.
KEY FEATURES	- Built-in bus (default) & custom buses - Rules with filtering & pattern matching - Event enrichment & transformation - Scheduler (cron / one-time / rate) - Cross-account & cross-region events - Archive & Replay events	- Topics (standard & FIFO) - Multiple protocols (HTTP/S, Email, SMS, Lambda, SQS, mobile push) - Message filtering (attributes) - Fan-out to many subscribers - Cross-account subscriptions	- Standard & FIFO queues - Long polling - Visibility timeout - Dead-letter queues (DLQ) - Message retention - High durability & availability	- Automatic scaling - Pay per request / duration - Multiple runtimes - Event source integrations - IAM roles & fine-grained access - Concurrency controls	- Visual workflow designer - Choice, Parallel, Map, Wait, Retry - Error handling & timeouts - Service integrations (200+ AWS services, APIs, Lambda) - Standard & Express workflows
DELIVERY MODEL	At-least-once delivery (near real-time)	At-least-once delivery (fan-out)	At-least-once delivery (queued until consumed)	At-least-once invocation (with retries + DLQ options)	At-least-once execution (with retries + error handling)
BEST FOR / USE CASES	Event-driven architectures, event routing, SaaS integrations, scheduled tasks, cross-account event buses.	Notifications, alerts, mobile push, email/SMS, broadcasting updates to multiple endpoints.	Buffering, decoupling, load leveling, microservices communication, reliable delivery.	Real-time compute, data processing, file processing, API backends, integrations.	Multi-step workflows, approvals, ETL pipelines, saga patterns, complex orchestration.
INTEGRATES WITH (EXAMPLES)	AWS Services SaaS Apps Custom Apps	Email SMS SQS Lambda HTTP/S	Lambda EC2 ECS Batch On-prem	EventBridge S3 DynamoDB API Gateway SQS	Lambda AWS Services API Calls SQS / SNS

TYPICAL EVENT-DRIVEN FLOW



COMMON USE CASES

Real-time Processing	React to changes in data (e.g., S3 upload → process image).
Microservices Communication	Decouple services using SQS or SNS for reliable messaging.
Scheduled Workflows	Use EventBridge Scheduler to run tasks at intervals or cron.
Notifications & Alerts	Fan-out alerts via SNS to email, SMS, Slack, or webhooks.
Data Pipelines	Orchestrate ETL jobs with Step Functions and Lambda.
Cross-Account / SaaS Integrations	Route events across accounts or from SaaS apps via EventBridge.

WHEN TO USE WHAT?

EventBridge	When you need to route events from multiple sources using rules, patterns, or schedules.
SNS	When you need to broadcast a message to many endpoints (fan-out).
SQS	When you need to buffer messages and decouple producers from consumers.
Lambda	When you need to run code in response to an event without managing servers.
Step Functions	When you need to orchestrate multiple steps with logic, retries, and error handling.

EVENTBRIDGE PATTERN EXAMPLES

Event Filtering	Match events by source, detail-type, or custom patterns.
Event Transformation	Reshape or enrich events before sending to targets.
Event Routing	Send different events to different targets.
Scheduled Events	Run tasks at a specific time (e.g., daily report).
Cross-Account / Cross-Region	Share events across accounts/regions securely.
Archive & Replay	Store events and replay them for debugging or reprocessing.

BEST PRACTICES

- ✓ Use EventBridge as the central event router.
- ✓ Use SNS for fan-out, SQS for decoupling.
- ✓ Use DLQs for SQS and Lambda for failed messages.
- ✓ Keep Lambda functions small and focused.
- ✓ Use Step Functions for complex workflows, not for simple logic.
- ✓ Use least-privilege IAM roles for all services.
- ✓ Monitor with CloudWatch metrics and alarms.
- ✓ Enable retries and backoff for reliability.



CLF-C02 EXAM TIPS

- EventBridge can receive events from AWS services, SaaS partners, and custom apps.
- SNS = fan-out (many subscribers).
- SQS = queue (one consumer at a time).
- Lambda = compute.
- Step Functions = orchestrate multiple steps.
- Know delivery models (at-least-once) and use cases.



REMEMBER: Decouple components, design for failure, and build systems that scale automatically.



AWS ANALYTICS QUERY SERVICES (CLF-C02)

Choose the right service to collect, prepare, store, process, query, and visualize your data.



KEY TAKEAWAY

Athena + Glue + S3 is the serverless analytics power trio. Add Redshift for data warehousing, EMR for advanced processing, Kinesis for streaming, and QuickSight for visualization.

SERVICE	AMAZON ATHENA	AWS GLUE	AMAZON REDSHIFT	AMAZON EMR	AMAZON KINESIS	AMAZON QUICKSIGHT
	 Serverless Interactive Query	 Data Catalog & ETL	 Data Warehouse	 Big Data Processing	 Real-time Data Streaming	 Business Intelligence & Visualization
WHAT IT IS	Serverless SQL query service built for data in Amazon S3.	Serverless data integration service with a data catalog.	Fully managed, petabyte-scale data warehouse.	Managed big data framework for processing large datasets.	Real-time streaming service for ingesting and processing data streams.	Serverless BI service for creating interactive dashboards and visualizations.
PRIMARY PURPOSE	Run ad-hoc SQL queries on data in S3 without managing servers.	Discover, catalog, clean, and prepare data for analytics.	Store and analyze structured data at scale with high performance.	Process large-scale data using open-source frameworks (Spark, Hive, Presto, etc.).	Ingest, process, and analyze streaming data in real time.	Visualize data and share insights through dashboards and reports.
KEY FEATURES	- Standard SQL - Serverless (pay per query) - Works with S3, Glue Catalog - Supports CSV, JSON, Parquet, ORC, Avro	- Data Catalog (metadata store) - Crawlers to discover schema - ETL jobs (extract, transform, load) - Data quality & schema registry	- Columnar storage - MPP architecture for speed - SQL analytics - Materialized views - Concurrency scaling	- Managed Hadoop/Spark - Supports Hive, Presto, HBase, Kafka, Flink, etc. - Auto Scaling - Spot Instances	- Kinesis Data Streams - Kinesis Data Firehose - Kinesis Data Analytics - Real-time ingestion - Milliseconds latency	- Interactive dashboards - SPICE in-memory engine - Scheduled reports & alerts - Row-level security - ML insights & Q
DATA SOURCE	Amazon S3 (and other sources via connectors)	Amazon S3, RDS, JDBC, DynamoDB, Salesforce, etc.	Data loaded from S3, RDS, DynamoDB, Kinesis, applications, etc.	Amazon S3, HDFS, DBs, Kinesis, DynamoDB, Kafka, etc.	Applications, devices, logs, IoT devices, websites, Amazon services	Athena, Redshift, RDS, S3, SPICE, Salesforce, etc.
HOW IT WORKS	Athena reads data from S3, uses Glue Data Catalog for metadata, and runs SQL queries. Results returned instantly.	Crawlers scan data sources and create metadata in the Data Catalog. ETL jobs transform data and store it in targets like S3.	Data is loaded into Redshift. Queries run on columnar, compressed data with massively parallel execution.	EMR launches a cluster and runs frameworks (Spark, Hive, etc.) to process big data stored in S3 or HDFS.	Data is streamed in real time. Services process, transform, and deliver to targets like S3, Redshift, OpenSearch, or Lambda.	Connect to data sources, build visualizations, and share dashboards. Uses SPICE for fast, interactive performance.
BEST FOR / USE CASES	Ad-hoc queries, log analysis, data exploration, quick insights on data lakes.	Data discovery, cataloging, ETL pipelines, schema management, data preparation.	Data warehousing, BI reporting, dashboards at scale, complex analytics on structured data.	Big data analytics, machine learning, data processing at scale, custom frameworks.	Real-time analytics, IoT data, clickstream data, log ingestion, fraud detection.	Dashboards, visual analytics, KPIs, business reporting, embedding analytics in apps.
INTEGRATES WITH	S3, Glue, Lake Formation, QuickSight, IAM	S3, Athena, Redshift, EMR, Lake Formation, Step Functions, CloudWatch	S3, Glue, QuickSight, Kinesis, DynamoDB, IAM, SageMaker	S3, Glue, Kinesis, Redshift, Hive Metastore, Step Functions	S3, Redshift, OpenSearch, Lambda, EMR, SageMaker, QuickSight	Athena, Redshift, S3, RDS, IAM, SSO (IAM Identity Center)

THE POWER PAIRING: ATHENA + GLUE + S3



Serverless, scalable, and cost-effective analytics on your data lake.

WHEN TO USE WHAT?

Athena	You need to run ad-hoc SQL queries on data in S3 quickly without provisioning infrastructure.
Glue	You need to catalog data, run ETL jobs, and prepare data for analytics.
Redshift	You need high-performance analytics on structured data at scale with complex queries and reporting.
EMR	You need to process large-scale data using open-source frameworks or custom workloads.
Kinesis	You need to ingest and process real-time streaming data with low latency.
QuickSight	You need to visualize data and build dashboards for business users.

KEY TAKEAWAYS

- ✓ Athena + Glue + S3 = the foundation of serverless analytics.
- ✓ Use Redshift for data warehousing and complex BI workloads.
- ✓ Use EMR for big data processing and advanced analytics.
- ✓ Use Kinesis for real-time data streaming and analytics.
- ✓ Use QuickSight to visualize and share insights.



EXAM TIP:

They may give you a scenario. Match the requirement (ad-hoc query, real-time, ETL, visualization, etc.) to the right service.



AWS AI SERVICES COMPARISON (CLF-C02)

Choose the right AI service for the right use case. Each service solves a specific problem.



KEY TAKEAWAY

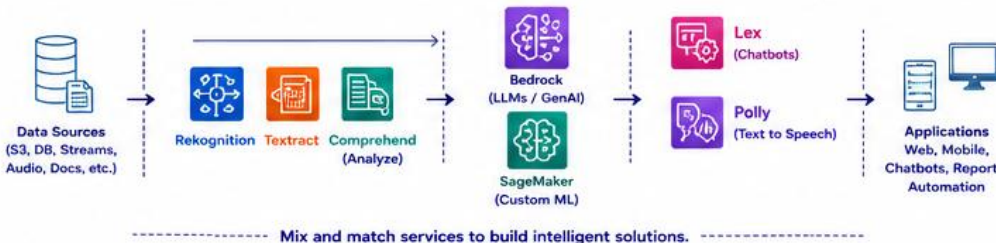
Bedrock = Foundation models (LLMs). SageMaker = Build & train models.
Other services = Pre-trained AI APIs for specific tasks.

SERVICE	AMAZON BEDROCK (Foundation Models)	AMAZON SAGEMAKER (ML Platform)	AMAZON REKOGNITION (Image & Video Analysis)	AMAZON Textract (Document Analysis)	AMAZON COMPREHEND (NLP – Text Analysis)	AMAZON LEX (Chatbots)	AMAZON POLLY (Text to Speech)	AMAZON TRANSCRIBE (Speech to Text)
WHAT IT IS	Managed service to access foundation models (FMs) from leading AI companies through an API.	End-to-end machine learning platform to build, train, and deploy custom ML models.	Image and video analysis service using deep learning to detect objects, people, text, and more.	Automatically extracts printed and handwritten text, data, and tables from documents.	Natural language processing (NLP) to extract insights and meaning from text.	Build conversational interfaces (chatbots) with speech and text interaction.	Converts text into natural-sounding human speech.	Converts speech into accurate text (batch or real-time).
PRIMARY PURPOSE	Use FMs (LLMs, Titan, Claude, Llama, etc.) without managing infrastructure.	Build, train, and deploy custom machine learning models at scale.	Analyze images and videos to identify and detect objects, faces, and text.	Extract text, key-value pairs, and tables from scanned documents.	Understand text (sentiment, entities, PII, key phrases, etc.).	Create chatbots and voice assistants with AI.	Generate lifelike speech from text.	Convert speech to text.
KEY FEATURES	- Access to multiple FMs - Serverless - Customization (fine-tuning, RAG with Knowledge Bases) - Guardrails for safety - Pay for tokens used	- Data labeling - Model training - AutoML - Model hosting - Pipelines & MLOps - End-to-end ML lifecycle	- Object & scene detection - Face recognition - Celebrity recognition - Text in image (OCR) - Moderation (unsafe content) - Video analysis	- OCR (printed & handwritten) - Key-value extraction - Table detection - Form extraction - Query documents (Analyze Document)	- Sentiment analysis - Entity recognition - Key phrase extraction - Language detection - PII detection - Syntax analysis - Topic modeling	- Built-in intents & slots - Multi-turn conversations - Integrates with Lambda, Kinesis, Polly - Supports speech & text	- Neural TTS voices - SSML support - Multiple languages - Customize speech rate, pitch, volume	- Batch & real-time - Speaker diarization - PII redaction - Vocabulary filtering - Multi-language - Timestamps
INPUT	Text prompts, images (vision models), documents	Your data (any format)	Images, videos	Scanned documents, images (PDF, JPG, PNG, etc.)	Text	Text or Speech (voice or chat)	Text (SSML supported)	Audio (voice)
OUTPUT	Generated text, code, answers, summaries, image outputs, etc.	Predictions, insights, deployed models	Labels, bounding boxes, faces, text, moderation results, metadata	Extracted text, key-value pairs, tables, forms	Sentiment, entities, key phrases, PII, topics, language, syntax	Intent detected, slots/values, response (text or speech)	Audio file (MP3, PCM)	Transcribed text, speaker labels, timestamps
BEST FOR / USE CASES	- GenAI apps (chatbots, Q&A) - Content generation - Code generation - Summarization - RAG (chat with your data)	- Predictive modeling - Recommendation engines - Fraud detection - Forecasting - Custom ML solutions	- Face search/verification - Image moderation - Object counting - Video indexing - ID verification (KYC)	- Invoice & receipt processing - Form data extraction - Contract analysis - Table extraction - Document digitization	- Customer feedback analysis - Sentiment monitoring - PII detection in text - Topic discovery - Language analytics	- Customer support bots - IVR (interactive voice response) - Order tracking - Appointment booking	- Voice apps - E-learning - Accessibility - PIV prompts - Announcements	- Call center analytics - Meeting notes - Interviews - Voice commands - Captions/subtitles
INTEGRATES WITH	Knowledge Bases (RAG), Lambda, S3, KMS, CloudWatch, Agents, Guardrails	S3, Redshift, Lambda, ECR, CloudWatch, EventBridge, Pipelines	S3, Kinesis Video, Lambda, DynamoDB, SNS, SQS	S3, Lambda, Step Functions, Comprehend, A2I	S3, Lambda, Kinesis, Redshift, QuickSight	Lambda, Kinesis, DynamoDB, Polly, CloudWatch, API Gateway	S3, Lambda, Lex, CloudWatch	S3, Lambda, Kinesis, Comprehend, Redshift, Transcribe Streaming SDK
FULLY MANAGED?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YOU MANAGE	Prompts, data, permissions, model choice, fine-tuning, security, integration	Data, features, models, training jobs, tuning, deployment, monitoring	Images/videos, configuration, permissions	Documents, queries, configuration, permissions	Text data, configuration, IAM permissions	Intents, slots, bot logic, fulfillment, permissions, integration	Text input, voice selection, permissions	Audio input, settings, vocabulary, permissions

QUICK DECISION GUIDE

- Need a powerful LLM to generate content? → Amazon Bedrock
- Need to build & deploy a custom ML model? → Amazon SageMaker
- Need to analyze images or videos? → Amazon Rekognition
- Need to extract text/data from documents? → Amazon Textract
- Need to understand text (sentiment, entities, PII)? → Amazon Comprehend
- Need a chatbot or voice assistant? → Amazon Lex
- Need to convert text to speech? → Amazon Polly
- Need to convert speech to text? → Amazon Transcribe

SERVICE RELATIONSHIPS



Mix and match services to build intelligent solutions.

EXAM TRAPS (AVOID THESE MISTAKES!)

- ✗ Bedrock is NOT for training custom models.
- ✗ SageMaker is NOT a pre-trained API for image/text/speech.
- ✗ Textract is NOT for understanding sentiment.
- ✗ Rekognition is NOT for speech or text transcription.
- ✗ Polly does NOT convert speech to text (That's Transcribe).
- ✗ Lex is NOT a general-purpose LLM.



EXAM TIPS

- ✓ Know the primary purpose of each service.
- ✓ Focus on input/output and common use cases.
- ✓ Understand integrations and what you manage vs AWS.

AWS MIGRATION SERVICES COMPARISON (CLF-C02)

Choose the right AWS migration service for your migration strategy and workload.

SERVICE	 AWS Migration Hub	 AWS Application Migration Service (MGN)	 AWS Database Migration Service (DMS)	 AWS Server Migration Service (SMS)	 AWS Transfer Family	 AWS Snow Family
Purpose	Track and manage migrations across AWS and partners	Lift-and-shift (rehost) to AWS with minimal changes	Migrate databases to AWS with minimal downtime	Migrate on-prem VMware workloads to AWS	Move files securely into and out of AWS	Move large amounts of data physically to AWS
What it migrates	Applications, servers, databases, and data across AWS & 3rd parties	Servers (physical, virtual) and applications	Databases (homogeneous or heterogeneous)	VMware VMs	Files (SFTP, FTPS, FTP, and AS2 protocols)	Large datasets (TBs to PBs)
Source Environment	Any (on-prem, multi-cloud, SaaS, etc.)	On-prem, VMware, Hyper-V, AWS, Azure, GCP	On-prem, AWS, other clouds, databases, S3, etc.	On-prem VMware vSphere	On-prem, partner, or cloud systems	On-prem, edge locations, offline environments
Target Environment	AWS	Amazon EC2, Amazon EBS	Amazon RDS, Aurora, Redshift, DynamoDB, S3, etc.	Amazon EC2	Amazon S3, EFS, Amazon FSx, or on-prem systems	Amazon S3, Amazon EBS, FSx, Glacier, etc.
Downtime	Varies (depends on migration approach)	Low to no downtime with continuous replication	Very low (near zero) with CDC	Low (short cutover window for final sync)	Not applicable (file transfer)	Not applicable (offline transfer)
Migration Type	Planning & tracking (not a migration service itself)	Rehost (lift-and-shift)	Database migration & replication	Rehost (VMware to EC2)	Data transfer	Data transfer (offline)
Best For	Centralized visibility and tracking of migrations	Quickly move servers and applications to AWS	Migrate databases with minimal downtime	Migrate VMware workloads to AWS	Secure managed file transfer workflows	Very large datasets where network transfer is not practical
Key Features	Dashboard, migration tracking, discovery, progress, integrations	- Continuous replication - Automated orchestration - Cutover with minimal downtime	- Ongoing replication (CDC) - Schema conversion - Multiple migration types	- Automated conversion - Agentless migration - Retain VMware tools	- Fully managed - Multiple protocols - Audit & logging	- Snowcone, Snowball Edge, Snowball, Snowmobile - Encryption, tracking
Pricing Model	Free (pay for integrated services you use)	Pay for replication server hours + data transfer + target resources	Pay for replication instance hours + storage + data transfer	Pay for replication server hours + data transfer + target resources	Pay for data transfer + connector (hourly)	Pay for device & shipping + data transfer + storage

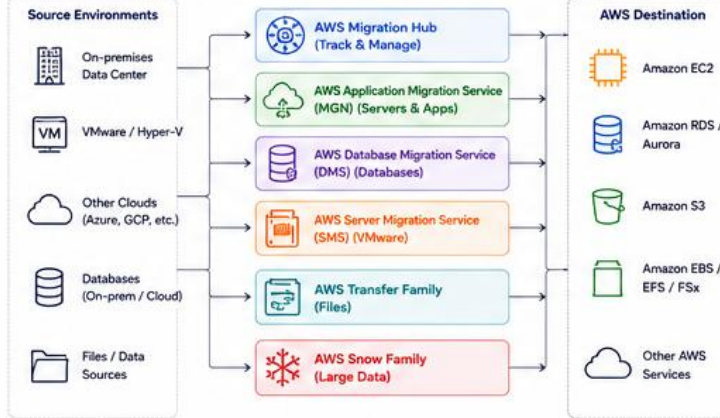
AWS MIGRATION JOURNEY



HOW TO CHOOSE THE RIGHT SERVICE?

-  Need to track and manage multiple migrations? → Use AWS Migration Hub
-  Moving servers to EC2 with minimal changes? → Use AWS Application Migration Service (MGN)
-  Migrating databases to AWS with minimal downtime? → Use AWS Database Migration Service (DMS)
-  Migrating VMware workloads to EC2? → Use AWS Server Migration Service (SMS)
-  Need to transfer files securely? → Use AWS Transfer Family
-  Moving large amounts of data offline? → Use AWS Snow Family

MIGRATION SERVICE MAP



BEST PRACTICES

- ✓ Start with a well-defined migration strategy and business case.
- ✓ Inventory and assess workloads using AWS Migration Hub or AWS Application Discovery Service.
- ✓ Choose the right migration service based on workload type and business requirements.
- ✓ Perform test migrations to validate the approach.
- ✓ Plan for security, IAM roles, and network connectivity.
- ✓ Monitor migration progress and optimize post-migration.
- ✓ Leverage AWS Well-Architected Framework for guidance.

KEY TAKEAWAY

-  Use the right AWS migration service based on what you are migrating. Plan, test, and optimize to ensure a successful and cost-effective migration to AWS.



Remember: There is no one-size-fits-all migration approach. Match the AWS migration service to your workload, downtime tolerance, data size, and business goals.



AWS MIGRATION & HYBRID SERVICES ARCHITECTURE

Migrate, extend, and modernize your workloads to AWS with a secure, reliable, and scalable hybrid architecture.



1. ASSESS & DISCOVER

Discover and assess your on-premises environment and workloads.



2. PLAN & DESIGN

Define migration strategy, target architecture, and operating model.



3. MIGRATE & TRANSFER

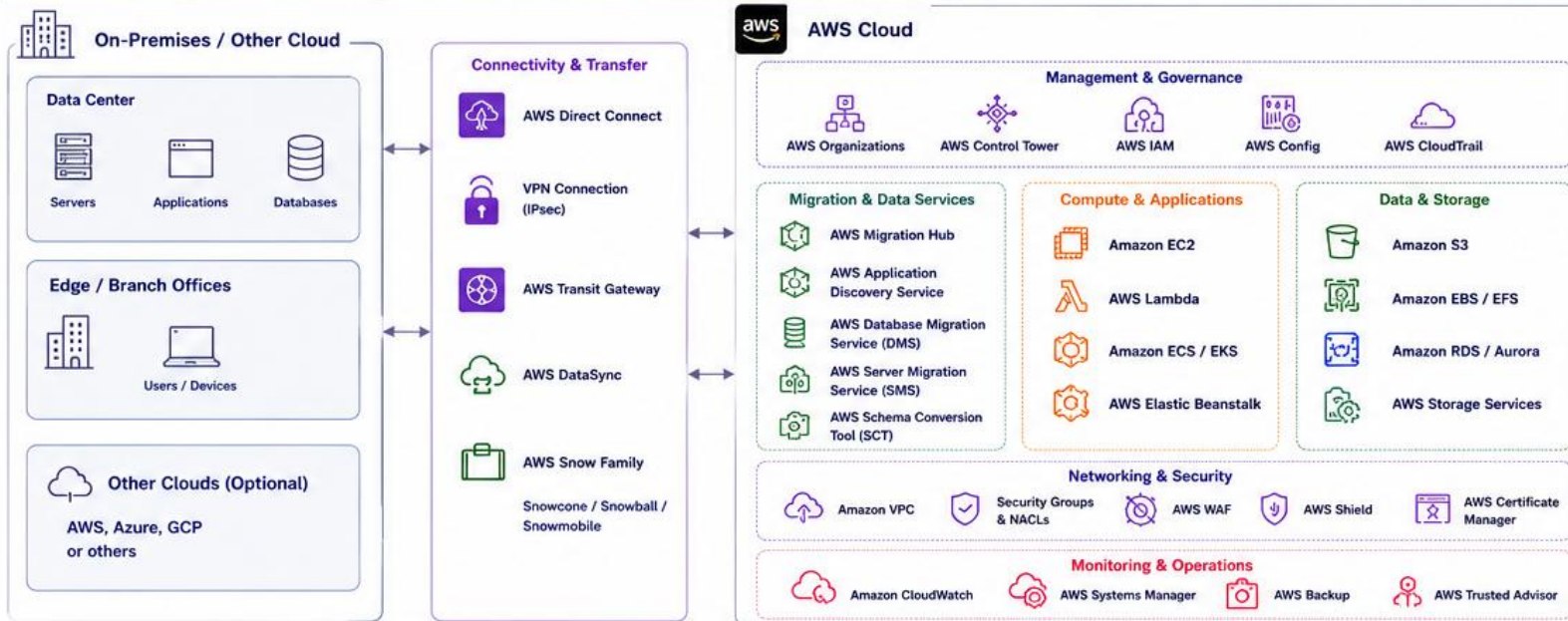
Move data and workloads securely and efficiently to AWS.



4. OPERATE & OPTIMIZE

Run, optimize, and evolve workloads using AWS services.

HYBRID ARCHITECTURE OVERVIEW



KEY AWS HYBRID & MIGRATION SERVICES

	AWS Application Discovery Service	Automated discovery and inventory of on-premises applications.
	AWS Migration Hub	Centralized dashboard to track and manage migration progress.
	AWS Database Migration Service (DMS)	Migrate databases to AWS with minimal downtime.
	AWS Server Migration Service (SMS)	Agentless migration of servers to AWS.
	AWS DataSync	Fast and automated data transfer over the network.
	AWS Snow Family	Large-scale data transfer using physical devices.
	AWS Direct Connect	Dedicated network connection from on-prem to AWS.
	AWS Storage Gateway	Hybrid storage integration with on-prem applications.
	AWS Outposts	Run AWS infrastructure and services on-premises.
	VMware Cloud on AWS	Run VMware SDDC natively on AWS.
	AWS WAN	Simplify network connectivity across sites and regions.

MIGRATION STRATEGIES – THE 7 R's

1 REHOST (Lift and Shift)	2 REPLATFORM (Lift, Tinker and Shift)	3 REFACTOR / RE-ARCHITECT	4 REPURCHASE (Replace)	5 RETIRE (Decommission)	6 RETAIN (Revise Later)	7 RELOCATE (Move)
Move applications as-is with minimal changes.	Make some optimizations to leverage AWS managed services.	Redesign applications to improve scalability, performance, and cost-efficiency.	Replace with a different product (SaaS) that meets business needs.	Remove applications that are no longer needed.	Keep applications on-prem for now and re-evaluate later.	Move to a different infrastructure without refactoring.
Example: Move EC2 instance to AWS	Example: Migrate to RDS from self-managed DB	Example: Break monolith into microservices	Example: Use Salesforce instead of custom CRM	Example: Decommission old internal applications	Example: Legacy app kept on-prem temporarily	Example: VMware Cloud on AWS (migrate VMware)

HYBRID USE CASES

	Data Center Extension Extend data center to AWS for burst capacity or DR.		Modernize Applications Gradually modernize applications on AWS.
	Disaster Recovery Replicate workloads to AWS for DR and failover.		Edge & IoT Workloads Process data closer to edge, integrate with AWS.
	Cloud Bursting Scale to AWS for seasonal or unpredictable demand.		
	Mergers & Acquisitions Integrate and operate workloads across environments.		
	Data Migration		

BEST PRACTICES

- ✓ Assess workloads and dependencies thoroughly.
- ✓ Start with non-critical or low-risk workloads.
- ✓ Use automation and AWS Migration Tools.
- ✓ Design for security, scalability, and high availability.
- ✓ Validate and test before cutover.
- ✓ Optimize, monitor, and continuously improve.



CLF-C02 TIP

Understand the purpose and features of each migration and hybrid service. Know the 7 R's and when to use each strategy.



USEFUL REFERENCES

- AWS Migration Hub
 - AWS Prescriptive Guidance
 - AWS Well-Architected Framework
 - AWS Cloud Adoption Framework
- ✓ Reduce risk and cost of migration
✓ Leverage AWS innovation and scalability
✓ Maintain business continuity.

★ REMEMBER: The 7 R's help you choose the right path for every workload. | Right strategy → Faster migration → Lower risk → Greater business value.



AWS SNOW FAMILY – CHEAT SHEET (CLF-C02)

Physical data transfer and edge computing solutions when large data transfers over the internet are impractical.



CLF-C02 EXAM KEYWORD TRIGGER
"large data transfer", "limited bandwidth",
"offline migration", "physical data transport"
→ Use Snowball Edge

ONLY 2 ACTIVE DEVICES (Updated November 2024)

i The AWS Snow Family has been significantly simplified. As of November 2024, only the two Snowball Edge models below are active.

❌ DISCONTINUED DEVICES – DO NOT USE

- **Snowcone** – discontinued November 12, 2024
- **Snowmobile** – discontinued April 2024
- All previous **Snowball Edge** models (80 TB Storage Optimized, 52 vCPU Compute Optimized, GPU variant) – discontinued November 12, 2024



EXAM TIP

Remove all knowledge of Snowcone, Snowmobile, and legacy Snowball Edge models.
Only know the two active Snowball Edge models.

DEVICE	SNOWBALL EDGE STORAGE OPTIMIZED	SNOWBALL EDGE COMPUTE OPTIMIZED	KEY FACTS (TESTED IN CLF-C02)
Purpose	Large-scale data migration when internet transfer is impractical	Edge computing and data processing before transferring to AWS	Use Snowball Edge when transferring large amounts of data and internet bandwidth is insufficient or would take too long.
Storage Capacity	210 TB NVMe (100 TB tier also available)	Lower storage capacity (varies by configuration)	Data is encrypted at rest (AES-256) and in transit (TLS 1.2).
Compute	Limited compute (light management tasks only)	104 vCPU (x86-based)	You ship the device to AWS using the prepaid shipping label. AWS ships it back after data transfer.
Memory	~ 16 GB	Up to 416 GB	Free to use the device for 10 days. Daily charges apply after that.
Best For	• One-time or large-scale data transfers • Backups and archives • Data migration from remote locations	• Edge ML inference • Local data processing and analytics • Pre-processing before transfer to AWS	Data is imported into Amazon S3 or Amazon EBS.
Connection	10/25 GbE network ports	10/25 GbE network ports	Supports AWS Direct Connect (on return shipping) for faster import to S3.
Typical Use Case	You have hundreds of TBs of data and the network would take weeks or months to transfer. Ship the device, copy data, ship back.	You need to run compute workloads at the edge (e.g., video analytics, IoT processing) and then transfer results or data to AWS.	AWS OpsHub is the GUI tool to manage Snow devices.

HOW IT WORKS (AT A GLANCE)



AWS OPSHUB (DEVICE MANAGEMENT)



AWS OpsHub is the web-based interface to:

- Unlock and activate your device
- Monitor device status and usage
- Create and manage jobs
- Configure network and security
- Track data transfer and logs
- Prepare device for return

PRICING (CLF-C02)

- 10 days of device usage is free.
- After 10 days: daily charges apply (varies by model and region).
- You pay for:
 - Device usage (after free period)
 - Data transfer out of AWS
 - S3 storage, if applicable
- Shipping is prepaid by AWS.

WHEN TO USE SNOWBALL EDGE

- ✓ Moving tens, hundreds, or thousands of TBs of data.
- ✓ The internet connection is too slow, expensive, or unavailable.
- ✓ One-time or periodic large data migrations.
- ✓ Edge locations need local compute before sending data to AWS.

WHEN NOT TO USE

- ❌ Small data transfers (< few TBs). Use AWS DataSync or direct transfer.
- ❌ Continuous data sync. Use AWS Direct Connect or Transfer Family.
- ❌ You need persistent compute in the cloud (use EC2, ECS, or Outposts).
- ❌ Real-time or near real-time data streaming requirements.

QUICK RECALL (CLF-C02)

Active devices (Nov 2024)?	2 (Snowball Edge: Storage Optimized, Compute Optimized)
Discontinued devices?	Snowcone (Nov 12, 2024), Snowmobile (Apr 2024), Old Snowball Edge models (Nov 12, 2024)
Best for large data transfer?	Snowball Edge Storage Optimized
Best for edge compute?	Snowball Edge Compute Optimized
Manage devices with?	AWS OpsHub
Free usage period?	10 days
Data encrypted?	Yes (at rest & in transit)

★ **REMEMBER:** If the question says "large data transfer", "limited bandwidth", "offline migration", or "physical data transport" → Think Snowball Edge.



FURTHER READING: [AWS Snow Family Overview](#) | [AWS OpsHub User Guide](#)



AWS MONITORING & LOGGING SERVICES COMPARISON – CHEAT SHEET (CLF-C02)

Choose the right AWS service to gain visibility, collect logs, set alarms, and respond to issues quickly.

FEATURE	AMAZON CLOUDWATCH (Monitoring & Observability)	AMAZON CLOUDTRAIL (API Activity & Governance)	AMAZON VPC FLOW LOGS (Network Monitoring)	AMAZON CLOUDWATCH LOGS (Log Management)	AMAZON X-RAY (Distributed Tracing)	AWS HEALTH (AWS Service Health)	AWS CONFIG (Resource Compliance)
What it is	Monitoring and observability service for AWS resources, applications, and infrastructure.	Tracks API calls and user activity across your AWS account.	Captures information about IP traffic going to and from network interfaces in your VPC.	Centralized log management service for storing, searching, and analyzing logs.	Service for analyzing and debugging distributed applications.	Provides information about the health of AWS services and your resources.	Tracks resource configurations and changes over time for governance and compliance.
Primary Purpose	Monitor metrics, set alarms, visualize dashboards, collect logs, and automate response.	Auditing, security analysis, compliance, and troubleshooting.	Monitor network traffic, troubleshoot connectivity and security issues.	Collect, store, and analyze logs from AWS services, applications, and custom logs.	Trace requests across distributed systems to identify bottlenecks and errors.	Stay informed about AWS service events and plan for interruptions.	Assess resource compliance, track changes, and evaluate configurations.
What it Monitors / Tracks	Metrics, logs, events, alarms, dashboards, Insights.	API calls, user logins, AWS Console activity, and resource changes via API.	IP traffic metadata (source/dest, ports, protocol, bytes, action, accepted/rejected).	Log data from applications, AWS services, OS, and custom sources.	Requests, traces, segments, latency, errors across microservices.	AWS service health, events, maintenance, and your resource health.	Resource configuration changes, relationships, and compliance status.
Data Source	AWS services, EC2, RDS, Lambda, custom apps, logs, events.	All AWS account activity via AWS API.	Network interfaces (ENI), subnets, NAT Gateways, etc.	EC2, Lambda, CloudTrail, VPC Flow Logs, apps, OS, containers, etc.	Applications using AWS SDKs, ALB, API Gateway, Lambda, ECS, EKS, etc.	AWS infrastructure and service telemetry.	AWS resources in your account (supported types).
Key Features	- Metrics & custom metrics - Alarms & notifications - Dashboards - Logs & Logs Insights - Events & EventBridge - Anomaly detection	- Record API activity - Deliver logs to S3, CloudWatch Logs, or CloudTrail Lake - Event history & Insights - Compliance reports	- Capture network traffic - Accepted & rejected traffic - Send to CloudWatch Logs or S3 - Real-time or near real-time	- Centralized log storage - Logs Insights (query) - Log groups & streams - Retention policies - Metric filters & alarms	- Service map visualization - Request tracing - Fault & latency analysis - Annotations & metadata - Integration with CloudWatch	- Service health dashboard - Personal Health Dashboard - Event notifications - Affected resources - Trusted Advisor integration	- Resource inventory - Configuration history - Change tracking - Compliance rules - Config snapshots
Retention	Metrics: 15 months Logs: As per retention policy (1 day – 10 years)	Indefinitely (if delivered to S3 and managed accordingly)	Depends on destination: - CloudWatch Logs: as per retention policy - S3: as per lifecycle rules	1 day – 10 years (configurable per log group)	Up to 30 days	Event history: past 7 days (ongoing events shown)	Config history: 7 years (default) Snapshots: as configured
Integration	All AWS services, SNS, EventBridge, Auto Scaling, Lambda, IAM, etc.	S3, CloudWatch Logs, CloudTrail Lake, Athena, IAM.	CloudWatch Logs, S3, Athena, QuickSight.	CloudWatch, Lambda, Kinesis, OpenSearch, S3, SNS, etc.	CloudWatch, ALB, API Gateway, Lambda, ECS, EKS, EC2 SDKs.	CloudWatch, SNS, EventBridge, Trusted Advisor.	CloudWatch, SNS, Lambda, S3, Athena, Security Hub, etc.
Best For	Operational monitoring, alerting, dashboards, and observability.	Security auditing, compliance, forensics, and troubleshooting.	Network troubleshooting, security analysis, and traffic visibility.	Centralized logging, search, and log analysis.	Microservices monitoring, performance analysis, and troubleshooting.	Staying informed about AWS service issues and planned changes.	Governance, compliance audits, and configuration change tracking.
Pricing Model	Pay for data ingested, metrics, alarms actions, dashboards.	Management events: Free Data events & Insights: Charged (CloudTrail Lake: charged)	Pay for data processed and data stored.	Pay for data ingested, stored, and scanned.	Free tier (first 100K traces/month) Then pay per additional trace and data.	Free	Pay for configuration items recorded and rules evaluated. (First 100K free per month)

CLOUDTRAIL vs CLOUDTRAIL LAKE

CloudTrail = Records events.
CloudTrail Lake = Stores + queries events with SQL.

RPO & RTO VISUAL EXPLAINER

RPO = How much data you can afford to lose (point in time)



RPO (Recovery Point Objective)

The maximum acceptable amount of data loss measured in time.

Example: RPO = 15 minutes
You can lose up to 15 minutes of data.

RTO (Recovery Time Objective)

The maximum acceptable duration of time to restore your system after a disaster.

Example: RTO = 1 hour
You must be back online within 1 hour.

KEY CONSIDERATIONS

- Identify critical workloads and data.
- Define business impact and recovery objectives (RTO & RPO).
- Choose the right DR strategy based on cost, complexity, and objectives.
- Design for automation and test regularly.
- Document your DR plan and runbooks.
- Review and improve periodically.

ARCHITECTURE EXAMPLES



AWS WELL-ARCHITECTED PILLARS



HOW THEY WORK TOGETHER



WHEN TO USE WHAT?

- CloudWatch: Monitor performance, set alarms, create dashboards, automate response.
- CloudTrail: Audit who did what, when, and via which API. Security & compliance.
- VPC Flow Logs: Troubleshoot network connectivity and security groups/NACL issues.
- CloudWatch Logs: Collect and query logs from applications and AWS services.
- X-Ray: Debug slow app performance and errors in microservices.
- AWS Health: Check AWS service status and get notified of events.
- AWS Config: Track configuration changes and ensure compliance.

QUICK DECISION GUIDE

Need	Use
Set alarms on CPU, memory, disk, custom metrics	Amazon CloudWatch
See logs from EC2, Lambda, or application	Amazon CloudWatch Logs
Find network connectivity or security issues	Amazon VPC Flow Logs
Know who called which API and when	Amazon CloudTrail
Understand app performance and trace requests	AWS X-Ray
Check AWS service disruptions or events	AWS Health
Track resource configuration changes	AWS Config

KEY TAKEAWAYS

- CloudWatch = Monitor everything.
- CloudTrail = Audit everything.
- VPC Flow Logs = Monitor network traffic.
- CloudWatch Logs = Store and query logs.
- X-Ray = Trace and debug applications.
- AWS Health = Know about AWS issues.
- AWS Config = Track and assess configurations.

CLF-C02 TIP: Combine multiple services for full visibility. For example: Use CloudTrail for audits, CloudWatch for alarms, and CloudWatch Logs for log analysis.

Further Reading: AWS CloudWatch User Guide | AWS CloudTrail User Guide | AWS X-Ray Developer Guide | AWS Config User Guide

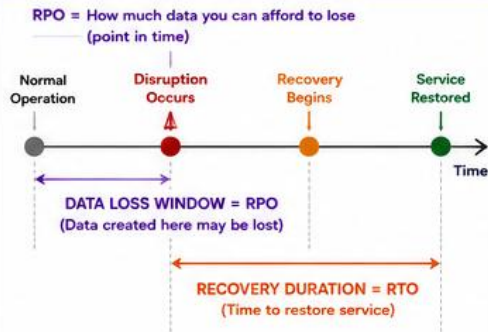


AWS DISASTER RECOVERY STRATEGIES – CHEAT SHEET (CLF-C02)

Protect your applications and data, minimize downtime, and recover quickly.

STRATEGY	BACKUP & RESTORE (Lowest Cost)	PILOT LIGHT (Low Cost)	WARM STANDBY (Moderate Cost)	MULTI-SITE ACTIVE / ACTIVE (Highest Cost)
Overview	Restore your workloads from backups after a disruption.	Keep a minimal version of your environment running in AWS for faster recovery.	Maintain a scaled-down but fully functional version of your environment in AWS.	Run your application in multiple Regions simultaneously. No downtime.
RTO (Recovery Time Objective)	Hours to Days	Minutes to Hours	Minutes	Seconds
RPO (Recovery Point Objective)	Hours to Days (depends on backup frequency)	Minutes to Hours	Minutes	Near Zero
Use Case	Non-critical workloads, data, or applications that can tolerate longer downtime.	Applications that need faster recovery than backup & restore but can accept some downtime.	Business-critical applications that require quick recovery with minimal data loss.	Mission-critical applications that require maximum availability and zero downtime.
How it Works	Restore data and infrastructure from backups in AWS.	Maintain core infrastructure (e.g., VPC, subnets, bastion host, minimal app) in AWS. Scale up and restore data when needed.	Keep a scaled-down environment in AWS. Replicate data continuously. Scale up resources to full capacity during failover.	Traffic is served from multiple Regions. Data is synchronously replicated. If one Region fails, traffic is routed to others.
Pros	- Lowest cost - Simple to implement - Good for less critical workloads	- Lower cost than always-on - Faster recovery than backup & restore - Good balance of cost and recovery	- Low RTO and RPO - Automatic failover possible - Production-like environment	- Highest availability - Zero downtime - Near zero data loss
Cons	- Highest downtime - Not suitable for mission-critical apps	- Some manual steps to scale up - Requires periodic testing	- Higher cost than Pilot Light - Requires more resources running	- Highest cost - Complex architecture and operations
AWS Services Used	AWS Backup, Amazon S3, EBS Snapshots, RDS Snapshots, AWS DataSync	Amazon EC2, Amazon RDS (small), Amazon S3, AWS Backup, Route 53, Auto Scaling (optional)	Amazon EC2, Amazon RDS (Multi-AZ or Read Replica), Amazon S3 (CRR), AWS Global Accelerator, Route 53 Failover Routing	Amazon Route 53 (Latency), AWS Global Accelerator, Amazon Aurora Global Database, DynamoDB Global Tables, S3 CRR
Cost	\$ (Lowest)	\$\$ (Low)	\$\$\$ (Moderate)	\$\$\$\$ (Highest)
Complexity	Low	Medium	High	Very High

RPO & RTO VISUAL EXPLAINER



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KEY CONSIDERATIONS

- Identify critical workloads and data.
- Define business impact and recovery objectives (RTO & RPO).
- Choose the right DR strategy based on cost, complexity, and objectives.
- Design for automation and test regularly.
- Document your DR plan and runbooks.
- Review and improve periodically.

AWS WELL-ARCHITECTED PILLARS



ARCHITECTURE EXAMPLES



AWS SERVICES FOR DR

- AWS Backup**
Centralized backup service for AWS services and on-premises.
- Amazon S3 Cross-Region Replication (CRR)**
Automatically replicate S3 data across Regions.
- Amazon EBS Snapshots**
Create point-in-time backups of EBS volumes.
- Amazon RDS Multi-AZ & Read Replicas**
Improve availability and enable faster recovery.
- AWS Elastic Disaster Recovery (DRS)**
Replicate servers to AWS and quickly recover workloads.
- Amazon Route 53**
Route traffic to the healthiest endpoint or fail over to another Region.
- AWS Global Accelerator**
Improve availability and performance for global applications.
- AWS CloudFormation**
Recreate infrastructure quickly using templates.
- Amazon DynamoDB Global Tables**
Multi-Region, multi-active database.
- AWS DataSync**
Move large datasets to AWS quickly and securely.

DR TESTING CHECKLIST

- Review DR plan and runbooks.
- Validate backups and restore procedures.
- Test failover to DR environment.
- Verify data integrity and application functionality.
- Measure RTO and RPO.
- Document results and lessons learned.
- Update plan based on findings.



CLF-C02 TIP

Choose the simplest strategy that meets your business requirements. You can always start small and improve over time.

Remember:

Lower Cost → Higher Downtime
Higher Cost → Lower Downtime

★ **Remember:** There is no one-size-fits-all DR solution. Align your strategy with your business needs, RTO, RPO, and budget.

📖 **Further Reading:** AWS Well-Architected Framework – Reliability Pillar | AWS Disaster Recovery Whitepaper

AWS BILLING & PRICING TOOLS – CHEAT SHEET (CLF-C02)

Understand, estimate, optimize, and control your AWS costs.

TOOL	PURPOSE	WHAT IT DOES	KEY FEATURES	BEST FOR	COST
 AWS Pricing Calculator	Estimate costs before you build	Estimate the cost of AWS services based on your expected usage.	- Service selection and configuration - Region selection - Save and share estimates	Planning new workloads, budgets, and architectural decisions	Free
 AWS Cost Explorer	Analyze and understand costs	Visualize, understand, and manage your AWS costs and usage over time.	- Interactive charts and reports - Filter by service, account, region, tags - Save custom reports	Cost analysis, identifying trends, anomaly detection	Free
 AWS Budgets	Set budgets and get alerts	Set custom budgets and receive alerts when costs or usage exceed thresholds.	- Budget alerts via email and SNS - Track cost, usage, RI/SP utilization - Budget actions (stop, terminate via SSM)*	Cost control, preventing over-spend	Free
 AWS Cost & Usage Report (CUR)	Access detailed billing data	Provides the most granular data about your AWS costs and usage.	- Hourly line-item data - Delivered to S3 (CSV or Parquet) - Includes resource IDs and tags	Advanced reporting, showback/chargeback, custom analytics	Free (Data transfer charges may apply)
 AWS Cost Categories	Organize costs by your business	Group and categorize costs using your own definitions (rules based on tags, account, service, region, etc.).	- Create custom cost categories - Allocate costs across business units - Works with Cost Explorer and CUR	Cost allocation, showback/chargeback, internal reporting	Free
 Savings Plans	Reduce costs (up to 72%)	Commit to a consistent amount of usage (\$/hour) in exchange for lower rates.	- Compute Savings Plans (EC2, Fargate, Lambda) - EC2 Instance Savings Plans (specific family) - Flexible across instance families, OS, regions*	Steady-state usage across services and regions	Discounted (Up to 72% savings)
 Reserved Instances (RIs)	Reduce costs (up to 72%)	Commit to specific instance configurations for 1 or 3 years for lower costs.	- Standard RIs (upfront, partial upfront, no upfront) - Convertible RIs (change instance family, OS, etc.) - Regional and Zonal RIs (Zonal = higher discount)	Steady, predictable usage of specific instances	Discounted (Up to 72% savings)
 AWS Marketplace	Buy third-party software	Purchase and deploy third-party software solutions on AWS.	- Pay-as-you-go or subscribe - Consolidated billing - Wide range of software solutions	Third-party software, SaaS, licensing on AWS	Varies by product
 AWS Trusted Advisor (Cost Optimization)	Get cost optimization recommendations	Provides real-time guidance to help you reduce costs and improve performance.	- Cost optimization checks - RI/SP recommendations - Idle resource detection	Find underutilized resources, right-size, save money	Free

PRICING MODELS	HOW COST DATA FLOWS	COST OPTIMIZATION BEST PRACTICES	KEY TAKEAWAYS
 On-Demand Pay for compute capacity by the hour or second with no long-term commitments.  Savings Plans Commit to a consistent amount of usage (\$/hour) for 1 or 3 years. Save up to 72%.  Reserved Instances (RIs) Commit to specific instance configurations for 1 or 3 years. Save up to 72%.  Spot Instances Bid on unused EC2 capacity. Save up to 90%. Great for fault-tolerant workloads.  Dedicated Hosts Physical servers dedicated for your use. Useful for compliance and licensing.		✔ Use Cost Explorer regularly to understand spending trends. ✔ Set up Budgets and alerts to avoid surprises. ✔ Tag all resources and use Cost Categories for allocation. ✔ Right-size resources (use Trusted Advisor recommendations). ✔ Use Savings Plans or Reserved Instances for predictable workloads. ✔ Use Spot Instances for interruptible, flexible workloads. ✔ Turn off or delete unused resources. ✔ Use lifecycle policies (e.g., S3, EBS snapshots) to reduce storage costs. ✔ Review CUR data for deeper insights and chargeback/showback.	★ Use the right tool for the right job: Plan → Pricing Calculator Analyze → Cost Explorer Control → Budgets Optimize → Savings Plans / RIs ★ CUR provides the most detailed data. ★ Tags + Cost Categories = better visibility and allocation. ★ Consistent monitoring and optimization lead to long-term savings.
 CLF-C02 TIP Know the purpose of each tool and when to use it. Understand pricing models and how they impact cost savings. COMMON EXAM SCENARIOS ❓ Which tool should you use to estimate costs before launching a workload? → Pricing Calculator ❓ Which service helps you set budgets and get alerts? → AWS Budgets ❓ Which tool provides detailed, line-item billing data? → Cost & Usage Report (CUR) ❓ Which option provides the highest savings for steady usage over 1–3 years? → Savings Plans or RIs IMPORTANT NOTES ✔ Prices vary by service, region, and usage. ✔ S3 storage and data transfer charges may apply (e.g., CUR to S3, Marketplace). ✔ Always check the AWS Pricing Page for the latest pricing.			

AWS COST OPTIMIZATION SERVICES (CLF-C02)

Reduce costs, improve efficiency, and maximize value with the right AWS cost optimization services.



KEY TAKEAWAY

Use the right service at the right time:
Understand → Analyze → Optimize → Save → Monitor

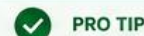
SERVICE	COST EXPLORER	BUDGETS	COMPUTE OPTIMIZER	TRUSTED ADVISOR	SAVINGS PLANS	RESERVED INSTANCES (RI)	SPOT INSTANCES
WHAT IT IS	Interactive visualization service that helps you analyze your AWS costs and usage.	Set custom budgets and get alerts when actual costs or usage exceed (or forecast to exceed) your budget.	Recommends optimal AWS compute resources for better performance and lower cost using ML.	Provides real-time guidance to help you provision your resources following AWS best practices.	Commit to a consistent amount of compute usage (\$/hour) in exchange for lower rates.	Commit to specific instance types for 1 or 3 years for significant discounts.	Use spare EC2 capacity at a discount of up to 90% compared to On-Demand.
PRIMARY PURPOSE	Understand where your money is going.	Control costs and avoid surprises.	Right-size resources and improve performance.	Optimize cost, performance, security, and fault tolerance.	Reduce costs with flexible commitment across services and regions.	Reduce costs for steady, predictable workloads.	Reduce costs for interruptible, fault-tolerant workloads.
KEY FEATURES	- Visualize costs over time - Filter by service, account, region, tags, etc. - Forecast future costs - Identify cost drivers - Save custom reports	- Cost, usage, RI utilization, Savings Plans utilization budgets - Alerts via email / SNS - Daily, weekly, monthly budgets - Budget actions via AWS Budgets Actions	- Rightsize EC2 instances - Recommend EC2, EBS, RDS, Lambda, and Auto Scaling optimizations - Performance improvement recommendations - Powered by ML using CloudWatch metrics	- Cost optimization checks - Performance improvement - Security best practices - Fault tolerance checks - Service limits - Trusted Advisor dashboard with recommendations	- Flexible across instance families, sizes, AZs, OS, tenancies, and regions - Applies to EC2, Fargate, and Lambda 1 or 3 year terms - Automatic savings	- Standard RIs - Convertible RIs 1 or 3 year terms - Payment options: All Upfront, Partial Upfront, No Upfront - Regional or Zonal	- Up to 90% discount - Spot Instance requests - Spot Fleet for large scale - Persistent / capacity-optimized fleets
BEST FOR	Everyone – finance, ops, and business teams.	Cost owners, project managers, operations teams.	Developers, architects, operators looking to rightsizing resources.	All AWS users – regular checks for best practices and savings.	Flexible workloads with steady usage across services/regions.	Predictable, steady-state workloads on specific instance families.	Fault-tolerant, batch, CI/CD, big data, containers, stateless workloads.
HOW IT HELPS SAVE COSTS	Find cost drivers and trends to make better spending decisions.	Prevent overspending with alerts and automated actions.	Rightsize over-provisioned resources and stop underutilization.	Identify underutilized resources and other cost optimization opportunities.	Lower rates than On-Demand with maximum flexibility across services and regions.	Biggest discounts for predictable workloads (Up to 75%).	Massive savings for interruptible workloads (Up to 90%).
DATA SOURCE	AWS Cost & Usage Report (CUR)	AWS Cost & Usage Report (CUR)	CloudWatch metrics, usage history, and ML analysis	AWS service metadata, usage patterns, and best practice rules	Usage tracking across EC2, Fargate, Lambda	Usage tracking for matching RI purchases	EC2 Spot market capacity and pricing

WHEN TO USE WHAT?

Understand & Analyze	→	Cost Explorer	Use when you want to understand spending patterns, identify cost drivers, and forecast costs.
Set Controls & Alerts	→	Budgets	Use when you need to control costs, set limits, and get alerted before you go over budget.
Right-size & Improve	→	Compute Optimizer	Use when you want recommendations to rightsize resources and improve performance.
Get Best Practice Guidance	→	Trusted Advisor	Use regularly to discover cost optimization opportunities and follow best practices.
Commit for Discounts	→	Savings Plans	Use when you want flexibility across instance types, services (EC2, Fargate, Lambda) and regions.
		Reserved Instances	Use when you have predictable, steady usage of specific instance families in specific regions/AZs.
Use Spare Capacity	→	Spot Instances	Use for interruptible, fault-tolerant, flexible-start, or batch workloads to get the highest discount.

DISCOUNT LEVEL (RELATIVE)

On-Demand	★☆☆☆☆	Highest cost, no commitment
Savings Plans (1 or 3 yr)	★★★★☆	Up to ~66% off On-Demand (Most flexible)
Reserved Instances (1 or 3 yr)	★★★★★	Up to ~75% off On-Demand (Most savings for steady use)
Spot Instances	★★★★★	Up to ~90% off On-Demand (Interruptible only)



PRO TIP

Combine these services for maximum savings!

Analyze (Cost Explorer) → Optimize (Compute Optimizer)
→ Commit (Savings Plans / RI) → Automate (Budgets)
→ Review (Trusted Advisor)

EXAM TIPS (CLF-C02)

- ✓ Compute Optimizer uses ML to recommend right-sizing.
- ✓ Savings Plans are more flexible than RI (applies to EC2, Fargate, Lambda).
- ✓ RI gives the highest discount for specific instance families in a region.
- ✓ Spot Instances are best for interruptible, fault-tolerant workloads.
- ✓ Budgets help you set alerts and take actions to prevent overspending.
- ✓ Trusted Advisor provides recommendations across cost, performance, security, and fault tolerance.



COMMON EXAM GOTCHAS

- Savings Plans ≠ Reserved Instances (flexibility vs. locked to instance family/region).
- Spot Instances can be interrupted with a 2-minute notice.
- Trusted Advisor does not make changes; it provides recommendations.
- Budgets uses the Cost & Usage Report (CUR) to track actuals.



REMEMBER: Optimize continuously: Monitor → Analyze → Act → Review. Cost optimization is an ongoing process, not a one-time task.

AWS SUPPORT PLANS COMPARISON (CLF-C02)

Choose the right AWS Support Plan to get the help you need, when you need it.

EXAM GOTCHA!
There are 5 support plans, not 4.
Enterprise On-Ramp sits between Business and Enterprise.

FEATURE	BASIC SUPPORT Included	DEVELOPER SUPPORT From \$29/month	BUSINESS SUPPORT From \$100/month	ENTERPRISE SUPPORT From \$15,000/month	ENTERPRISE ON-RAMP Custom Pricing
 Cost	Included	From \$29/month	From \$100/month	From \$15,000/month	Custom Pricing
 Response Time (Initial Response)	Best effort	12–24 hours	1 hour	15 minutes	Same as Enterprise
 Case Severity Availability	General guidance only	Business hours (Mon–Fri)	24/7	24/7	24/7
 Trusted Advisor Checks	Limited	Full access	Full access	Full access + Proactive guidance	Full access + Proactive guidance
 Use Case Guidance	Documentation, whitepapers, re:Post	Yes	Yes	Yes + Architecture best practices	Yes + Architecture best practices
 Technical Support	AWS re:Post, docs, whitepapers	Account & billing, service limits	Unlimited cases, all services	Unlimited cases, all services	Unlimited cases, all services
 Well-Architected Review	No	No	Yes (Basic)	Yes (Full with Recommendations)	Yes (Full with Recommendations)
 Technical Account Manager (TAM)	No	No	No	Yes	Yes
 Infrastructure Event Management	No	No	No	Yes	Yes
 Training & Best Practices	Self-service resources	General best practices	Guidance & resources	Proactive guidance & training	Proactive guidance & training
 Service Quotas (Increase Requests)	Standard	Standard	Increased limits	High priority	High priority
 Support API Access	No	No	Yes	Yes	Yes
 Security Bulletins & Alerts	Yes	Yes	Yes	Yes + Personal Health Dashboard	Yes + Personal Health Dashboard

PLAN RECOMMENDATION GUIDE	
 Basic	Great for getting started, learning AWS, and accessing self-service resources.
 Developer	Best for developers and small teams needing occasional technical support.
 Business	Ideal for production workloads that need reliable, 24/7 support and faster response.
 Enterprise On-Ramp	For large organizations that need a phased approach to Enterprise-level support.
 Enterprise	For business-critical applications that require proactive support and expert guidance.

WHAT YOU GET WITH ALL PAID PLANS	
 Access to AWS Trusted Advisor (full set of checks)	 Security bulletins & notifications
 AWS Support Center (case management)	 Architecture best practices guidance
 AWS Knowledge Center & documentation	 Access to AWS experts and re-Post
 Service quota increase requests	 Billing & account support
 Personal Health Dashboard (Business and above)	 Training & events discounts

QUICK NOTES	
✓	Basic Support is included with all AWS accounts at no additional cost.
✓	You can upgrade or downgrade your plan at any time.
✓	Pricing is based on your monthly AWS usage (excluding Marketplace & some services).
✓	Enterprise plans include a Technical Account Manager (TAM) who acts as your advocate.
✓	Use the AWS Pricing Calculator to estimate your support plan cost.
✓	For organizations with multiple accounts, consolidated billing may reduce Support plan costs.



KEY TAKEAWAY: Select a support plan based on your workload criticality, response time needs, and access to AWS experts.



EXAM TIP: CLF-C02 tests all 5 support plans. Don't forget Enterprise On-Ramp!



Learn more: aws.amazon.com/premiumsupport



HIGH-FREQUENCY EXAM TRAPS (CLF-C02)

Small differences. Big impact. Know these to avoid losing easy points.



KEY TAKEAWAY

The CLF-C02 exam tests AWS in details.
Don't just memorize services—understand the differences.

HIGH-FREQUENCY CONCEPT TRAPS

Security Group (SG) = Stateful Allows return traffic automatically.	≠	Network ACL (NACL) = Stateless Return traffic must be explicitly allowed.
Read Replica ≠ Multi-AZ Read Replica = for read scaling (asynchronous).	≠	Multi-AZ = for high availability (synchronous replication with automatic failover).
CloudFront = Caches Content Improves performance and reduces origin load.	≠	Global Accelerator = Routes TCP/UDP Traffic Improves availability and performance using AWS global network.
S3 = Regional Service Data is stored in a specific Region.	≠	Edge Locations ≠ Regions Edge locations are used by CloudFront (not the same as Regions).
Athena Queries S3 Directly No data is loaded into Athena. Pay for data scanned.	≠	Redshift Stores Data Data is loaded into Redshift. Pay for storage + compute.
DynamoDB = Serverless NoSQL Fully managed, scalable key-value and document database.	≠	Aurora = Managed Relational DB Supports SQL (MySQL/PostgreSQL compatible).
Lambda = Serverless Compute No servers to manage. Pay per request + duration.	≠	EC2 = Serverful Compute You manage the OS, patching, and scaling.
IAM Role ≠ IAM User Role = for AWS resources to access AWS services.	≠	User = for people to access AWS Management Console or CLI.
EBS = Block Storage Attach to EC2 in the same AZ.	≠	S3 = Object Storage Highly durable, used over the network.
On-Demand = Pay more, no commitment Most flexible.	≠	Savings Plans / RI = Commit upfront, pay less 1 or 3 years.
Spot Instances = Interruptible, cheap Can be terminated anytime.	≠	On-Demand Instances = Not interruptible Higher cost.
CloudFront for HTTP/HTTPS (Layer 7) Web content, API, downloads.	≠	Global Accelerator for TCP/UDP (Layer 4) Gaming, IoT, custom apps.

STORAGE TRAPS

S3 Standard General purpose 99.99% availability	≠	S3 IA (Infrequent Access) Lower cost, higher retrieval fees.
S3 Glacier For archiving (minutes to hours retrieval).	≠	S3 Glacier Deep Archive Cheapest storage, retrieval takes hours.
EFS = File Storage Shared (Linux) Multiple AZ.	≠	FSx = Managed File System For specific workloads (Windows, Lustre, etc.).
EBS = AZ-bound Cannot attach to EC2 in another AZ.	≠	S3 = Region-wide Accessible from any AZ in the Region.

NETWORKING TRAPS

Private IP = Internal (RFC1918) Not routable on the internet.	≠	Public IP = Internet-routable Assigned to reach resources.
Internet Gateway For resources in public subnets.	≠	NAT Gateway For resources in private subnets to access internet.
Route 53 = DNS (Domain names) Resolves names to IP addresses.	≠	VPC = Networking (Isolated Virtual Cloud) Your virtual network in AWS.
Security Group Stateful (Allow rules) Evaluated at instance level.	≠	NACL Stateless (Allow/Deny) Evaluated at subnet level.

QUICK REMINDERS

- ✓ SG = Stateful
- ✓ NACL = Stateless
- ✓ Read Replica ≠ Multi-AZ
- ✓ CloudFront = Caching (HTTP/HTTPS)
- ✓ Global Accelerator = TCP/UDP Routing
- ✓ S3 = Regional Service
- ✓ Edge Locations ≠ Regions
- ✓ Athena = Queries S3 Directly
- ✓ DynamoDB = Serverless NoSQL
- ✓ Lambda = Serverless Compute
- ✓ EBS = AZ-bound Block Storage
- ✓ S3 = Object Storage
- ✓ On-Demand = No Commitment
- ✓ Savings Plans / RI = Commit & Save
- ✓ Spot = Interruptible, Cheap
- ✓ IAM Role ≠ IAM User
- ✓ Private IP ≠ Public IP
- ✓ IGW ≠ NAT Gateway
- ✓ Route 53 (DNS) ≠ VPC (Networking)
- ✓ Rekognition ≠ Textract
- ✓ Comprehend ≠ Lex
- ✓ Polly ≠ Transcribe

DATABASE TRAPS

RDS Multi-AZ High availability (automatic failover).	≠	RDS Read Replica Read scaling (not for HA).
DynamoDB On-demand / Provisioned capacity.	≠	S3 Does not support SQL or joins.
DynamoDB = NoSQL (Key-Value) Not relational.	≠	RDS/Aurora = Relational (SQL) Tables, joins, ACID.

AI / ANALYTICS TRAPS

Rekognition Image/Video analysis.	≠	Textract OCR & document text extraction.
Comprehend NLP: Sentiment, Entities, PII.	≠	Lex Chatbots (Conversational AI).
Polly Text to Speech (TTS).	≠	Transcribe Speech to Text (STT).

COMMON SCENARIO TRAPS

Need low latency for global users (web app)	→	Use CloudFront (caching) or Global Accelerator (TCP/UDP)
Need HA for database	→	Use RDS Multi-AZ
Need read scaling	→	Use Read Replica
Need cheapest archive storage	→	Use S3 Glacier Deep Archive
Need to run code without managing servers	→	Use Lambda
Need full control of OS	→	Use EC2
Resources in private subnet need internet access	→	Use NAT Gateway
Need to find text in image / scanned doc	→	Use Textract (not Rekognition)

EXAM TRAP CHECKLIST

- ✓ Understand the difference, not just the definition.
- ✓ Look for keywords: low latency, cost, scalability, HA, durability, performance.
- ✓ Read every option. AWS loves similar but wrong answers.
- ✓ Eliminate wrong answers (they are usually false in one small detail).
- ✓ When two answers seem right, pick the most AWS-native / managed option.
- ✓ Remember: "Serverless" usually means AWS manages the infrastructure.
- ✓ Focus on: "Best", "Most cost-effective", "MOST scalable", "HIGHLY available".
- ✓ Know the purpose of the service and how it works.
- ✓ Practice real exam-style scenarios regularly.

EXAM TRAP ALERTS

- ✗ Don't choose Read Replica when HA is required.
- ✗ Don't choose Multi-AZ when read scaling is required.
- ✗ Don't use CloudFront for non-HTTP/HTTPS protocols.
- ✗ Don't use EC2 when serverless is required.
- ✗ Don't forget S3 is NOT for online transactions.
- ✗ Don't confuse regions with edge locations.



FINAL TAKEAWAY

- ✓ CLF-C02 is detail-heavy.
- ✓ These "traps" are tested constantly.
- ✓ Master the differences.
- ✓ Read carefully.
- ✓ Eliminate wrong answers.
- ✓ Choose the BEST answer.



YOUR GOAL: UNDERSTAND, NOT MEMORIZE. THE CLOUD REWARDS THOSE WHO TRULY UNDERSTAND IT.





MOST COMMON CLF-C02 SCENARIO QUESTIONS MAP

Identify the need in the question → Match the keywords → Choose the best AWS service

HOW TO USE

- 1 Read the scenario
- 2 Identify the key need (intent)
- 3 Pick the best AWS service



1. COST OPTIMIZATION

Analyze spending / view cost trends	→ AWS Cost Explorer
Set budgets / get alerts	→ AWS Budgets
Estimate cost before launch	→ AWS Pricing Calculator
Find savings / best practice recommendations	→ AWS Trusted Advisor
Detect unused / underutilized resources	→ AWS Trusted Advisor
Tag resources for cost allocation	→ AWS Cost Allocation Tags
Compare support plans	→ AWS Support Plans

💡 Think: cost, spend, budget, forecast, savings, optimize

2. PERFORMANCE & SCALABILITY

Handle increased traffic	→ Auto Scaling
Scale stateless applications	→ Elastic Load Balancing
Improve performance / lower latency	→ Amazon CloudFront
High performance database / cache	→ Amazon ElastiCache
Decouple components / services	→ Amazon SQS
Process large amounts of data	→ Amazon Kinesis
Run many containers at scale	→ Amazon ECS / EKS

💡 Think: scale, traffic, performance, latency, throughput

3. RELIABILITY & AVAILABILITY

Highly available application	→ Multi-AZ
Fault tolerant application	→ Auto Scaling + ELB
Back up data	→ AWS Backup / Amazon S3
Disaster recovery (DR)	→ AWS Backup / Cross-Region
Check service health	→ AWS Health Dashboard
Ensure resilience	→ Well-Architected Best Practices

💡 Think: availability, uptime, failover, backup, DR

4. SECURITY

Manage users / permissions	→ AWS IAM
Temporary access / assume role	→ IAM Role (STS)
Encrypt data	→ AWS KMS
Protect web applications	→ AWS WAF
Network security	→ Security Groups / NACL
Detect threats	→ Amazon GuardDuty
Scan for vulnerabilities	→ Amazon Inspector
Store secrets	→ AWS Secrets Manager

💡 Think: secure, protect, access, encrypt, threat, vulnerability

5. MONITORING & LOGGING

Monitor metrics / create alarms	→ Amazon CloudWatch
Log API activity / audit	→ AWS CloudTrail
Log resource changes / compliance	→ AWS Config
Collect and store logs	→ Amazon CloudWatch Logs
Trace requests / troubleshoot	→ AWS X-Ray
Automate on events / schedule	→ Amazon EventBridge
Personal health alerts	→ AWS Health Dashboard

💡 Think: monitor, log, alert, audit, track, troubleshoot

6. DATA MANAGEMENT

Store files / objects	→ Amazon S3
Store relational data	→ Amazon RDS / Aurora
Store NoSQL data	→ Amazon DynamoDB
Data warehouse / analytics	→ Amazon Redshift
Archive data (long term)	→ Amazon S3 Glacier
Back up databases	→ AWS Backup
Migrate data	→ AWS DataSync / Snowball

💡 Think: store, save, backup, archive, database, migrate

7. COMPUTE

Run virtual servers	→ Amazon EC2
Run serverless code	→ AWS Lambda
Run containers	→ Amazon ECS / EKS
Batch processing jobs	→ AWS Batch
High performance computing	→ Amazon EC2 (HPC)

💡 Think: compute, server, instance, run, process, execute

8. NETWORKING & CONTENT DELIVERY

Isolate resources / private network	→ Amazon VPC
Connect on-premises to AWS	→ AWS Direct Connect
Secure site-to-site access	→ VPN (Site-to-Site)
Route internet traffic	→ Amazon Route 53
Deliver content globally	→ Amazon CloudFront
Protect against DDoS	→ AWS Shield

💡 Think: network, connect, route, private, internet, secure

9. MIGRATION & HYBRID

Migrate servers	→ AWS Migration Hub / MGN
Migrate databases	→ AWS DMS / Schema Conversion Tool
Migrate large amounts of data	→ AWS Snowball / Snowmobile
Application discovery	→ AWS Application Discovery Service

💡 Think: migrate, move, transfer, on-prem, hybrid

COMMON QUESTION INTENTS (WHAT THEY REALLY WANT)

- 💰 Reduce cost
- 📈 Improve performance
- 🔒 Ensure availability
- 🔒 Secure resources
- 🔍 Monitor / troubleshoot
- 💾 Store / manage data
- 🔄 Migrate / move
- ⚙️ Automate processes
- 🌐 Connect / communicate
- 📊 Analyze / gain insights

KEYWORD INTENT SHORTCUTS

Reduce	Cost optimization
Scale	Handle growth / traffic
Secure	Protect / restrict access
Monitor	Observe / alert
Store	Save / persist data
Analyze	Gain insights
Migrate	Move / transition
Automate	Run without manual effort
Connect	Network / connectivity
High Availability	Minimize downtime
Backup / Restore	Data protection
Encrypt	Data security

CLF-C02 SUCCESS TIPS

- ✅ Read the scenario carefully.
- ✅ Find the main need (intent).
- ✅ Ignore extra details that don't change the need.
- ✅ Choose the AWS managed service that best fits.
- ✅ Follow AWS Well-Architected principles.

QUICK SERVICE REFERENCE (MOST USED)



★ REMEMBER: Understand the requirement → Identify keywords → Apply best practices → Choose the **BEST** AWS service!



AWS GOVERNANCE & MULTI-ACCOUNT SERVICES (CLF-C02)

Secure, govern, and manage your AWS environment at scale.

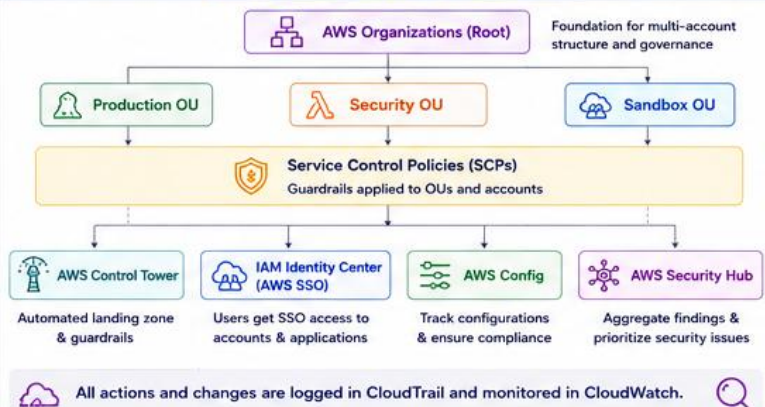


KEY TAKEAWAY

These services help you build a secure, consistent, and well-governed multi-account AWS environment.

SERVICE	AWS ORGANIZATIONS	SERVICE CONTROL POLICIES (SCPs)	AWS CONTROL TOWER	IAM IDENTITY CENTER (AWS SSO)	AWS CONFIG	AWS SECURITY HUB
WHAT IT IS	 A service to centrally manage multiple AWS accounts.	 Permissions guardrails applied at the organization level.	 The easiest way to set up and govern a secure, multi-account AWS environment.	 Centrally manage workforce identities and access to AWS accounts and cloud apps.	 A service that enables you to assess, audit, and evaluate the configurations of your AWS resources.	 A central security dashboard that aggregates and prioritizes security findings across AWS and partners.
PRIMARY PURPOSE	Enable governance, management, and consolidation of billing across accounts.	Define the maximum permissions an account can have (guardrails).	Automate account provisioning and apply guardrails based on best practices.	Provide secure, single sign-on (SSO) access to AWS accounts and applications.	Track resource configurations and ensure compliance with policies and best practices.	Provide a comprehensive view of your security posture and prioritize what to fix.
KEY FEATURES	- Create organizational structure (root, OUs, accounts) - Consolidated billing - Policy-based management - Account lifecycle management	- Applied to OUs or accounts - Sets permission boundaries - Deny or allow actions - IAM policies still apply within the guardrail	- Automated landing zone setup - Best-practice blueprints - Guardrails (preventive & detective) - Account vending (self-service) - Dashboard & monitoring	- SSO to AWS accounts - SSO to cloud applications - Permission sets (job-based access) - Directory integration (AD, etc.) - Multi-account access management	- Configuration recording - Config rules (compliance checks) - Remediation actions - Change history & timeline - Aggregators for multi-account	- Aggregates findings from AWS services and partners - Automatic severity scoring - Security standards & benchmarks - Insights & trends
HOW IT WORKS	You invite or create AWS accounts and organize them into Organizational Units (OUs) to apply policies and controls.	SCPs act as a permissions boundary. They do not grant permissions; they limit what IAM users/roles in the account can do.	Control Tower uses AWS Organizations, SCPs, and other services to set up a landing zone and enforce guardrails across accounts.	Users authenticate once and get assigned permission sets that grant access to accounts and applications.	Config records configuration changes and evaluates resources against rules. Non-compliant resources can trigger remediation.	Collects findings from integrated services (e.g., GuardDuty, Inspector, Macie, Config) and partners, normalizes, and prioritizes them.
TARGET USERS	Account administrators, cloud governance teams	Security, governance, platform teams	Cloud administrators, platform teams	End users, IT teams, security teams	Compliance officers, DevOps, security teams	Security teams, cloud administrators
INTEGRATES WITH	SCPs, AWS Control Tower, AWS IAM Identity Center, AWS Config, Security Hub, CloudTrail, Billing	AWS Organizations, IAM, Control Tower	AWS Organizations, SCPs, IAM Identity Center, AWS Config, Security Hub, CloudTrail, CloudWatch	AWS Organizations, Control Tower, IAM, AD Connector, SAML 2.0 apps, CloudTrail	Security Hub, CloudTrail, CloudWatch, Lambda, SNS, Systems Manager	GuardDuty, Inspector, Macie, AWS Config, IAM Access Analyzer, Firewall Manager, and 50+ partners
BEST FOR / USE CASES	Managing multiple accounts, applying guardrails, and centralizing billing.	Preventing risky actions, enforcing compliance, and applying least-privilege guardrails.	Quickly establishing a secure multi-account environment with governance at scale.	Centralized access management and SSO for users across multiple accounts and applications.	Compliance monitoring, change tracking, and automated remediation.	Centralized security posture management and prioritization of security issues.

HOW THESE SERVICES WORK TOGETHER



KEY BENEFITS

- Consistent Guardrails**
Enforce policies across all accounts.
- Reduced Risk**
Limit permissions and prevent unsafe actions.
- Faster Setup**
Control Tower automates secure landing zone deployment.
- Centralized Access**
SSO simplifies and secures user access across accounts and apps.
- Continuous Compliance**
Monitor configurations and remediate non-compliant resources.
- Better Visibility**
Aggregate and prioritize security findings in one place.

COMMON EXAM SCENARIOS

- You need to prevent developers from deleting S3 buckets across all accounts.
→ Use SCP (Deny s3:DeleteBucket).
- You need to give users access to multiple accounts with a single sign-on.
→ Use IAM Identity Center (SSO).
- You need to automatically set up a secure multi-account environment.
→ Use AWS Control Tower.
- You need to track who changed a security group and when.
→ Use AWS Config + CloudTrail.
- You need a centralized view of all security findings and prioritize them.
→ Use AWS Security Hub.
- You need to ensure only approved regions are used across accounts.
→ Use SCPs to restrict regions.



CLF-C02 TIPS

- Organizations = Structure & governance for multi-account.
- SCPs = Set the maximum permissions (permissions guardrails).
- Control Tower = Automates landing zone & guardrails.
- IAM Identity Center = SSO for AWS accounts & cloud apps.
- Config = Track configurations & ensure compliance.
- Security Hub = Central security dashboard & prioritization.



REMEMBER:

Organizations provides the structure, SCPs provide guardrails, Control Tower automates governance, IAM Identity Center provides access, Config ensures compliance, and Security Hub gives visibility.

These services work together to build a secure, scalable, and well-governed AWS environment.